22AIE303: Database Management Systems

Lecture 3

Entity-Relationship Model

Database Design

The ***entity-relationship (E-R) data model*** consists of a collection of basic objects, called **entities,** and of relationships among these objects.

It is used to

- develop an initial database design.
- a phase called ***conceptual database design***.

Database Design

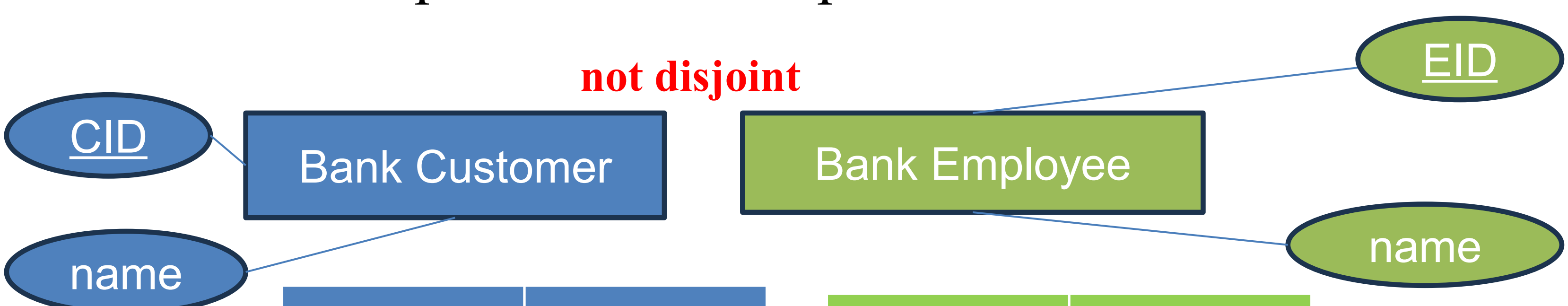
Database Design

Entity: Object in the real world that is distinguishable from another object.

Entity-set: A collection of similar entities
Entity **sets need not be disjoint**;

Attribute: An entity is described using a set of attributes

Domain: A unique set of values permitted for an attribute



CID	name
C1	Ali
C2	Puja
C3	Nila

CID	name
E1	Ali
E2	Kamal

Database Design

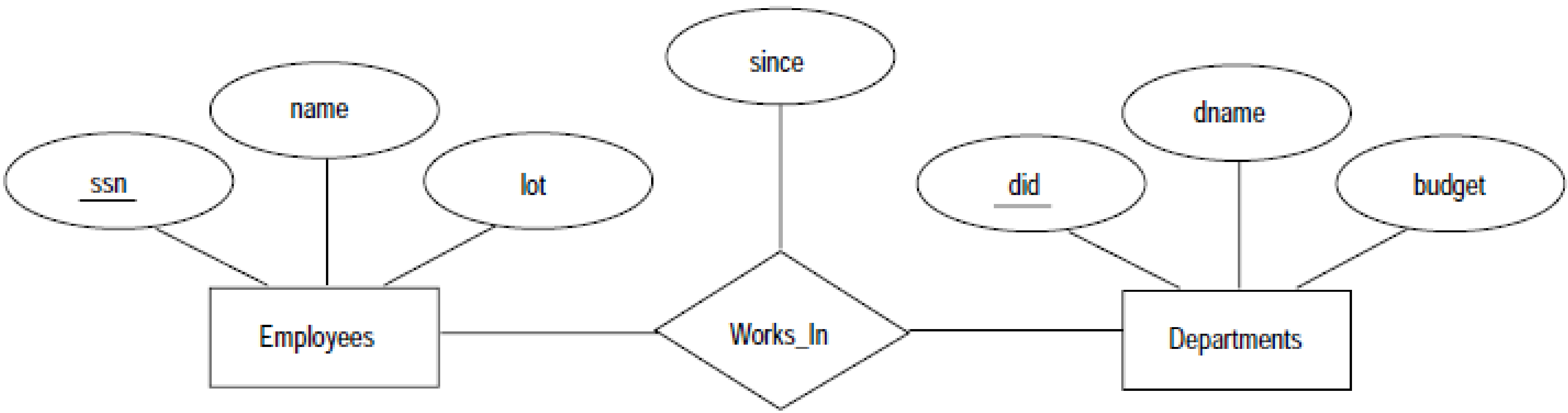
Relationship: An association between two or more entities.

Relationship set: A set of similar relationships

Descriptive attribute: Use to record information about relationship/relationship-set

E-R Diagram

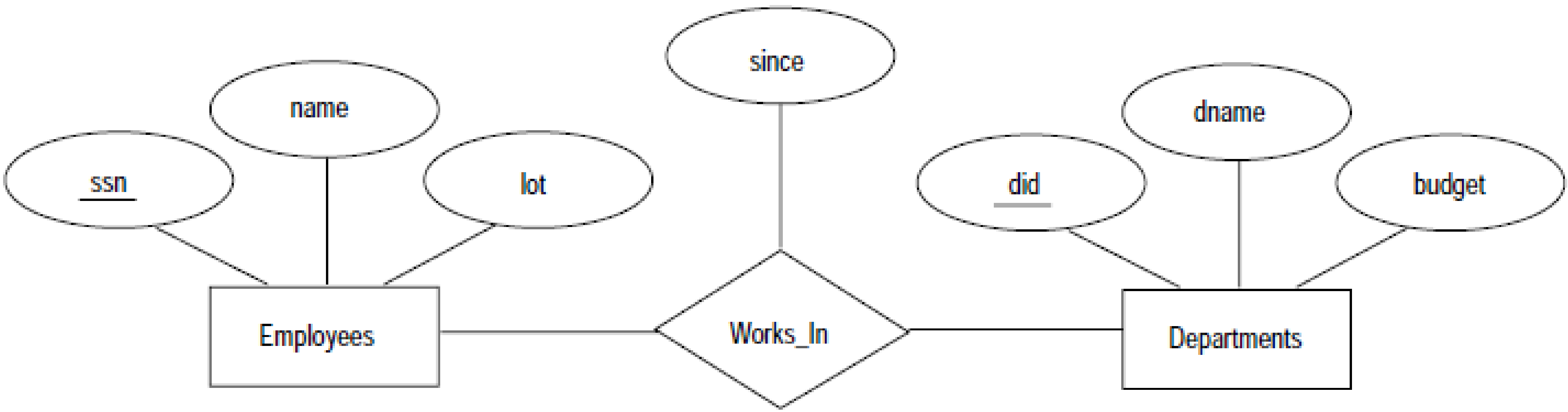
Entity-set, Relationship-set and attributes



Figures	Symbols	Represents
Rectangle		Entities in ER Model
Ellipse		Attributes in ER Model
Diamond		Relationships among Entities

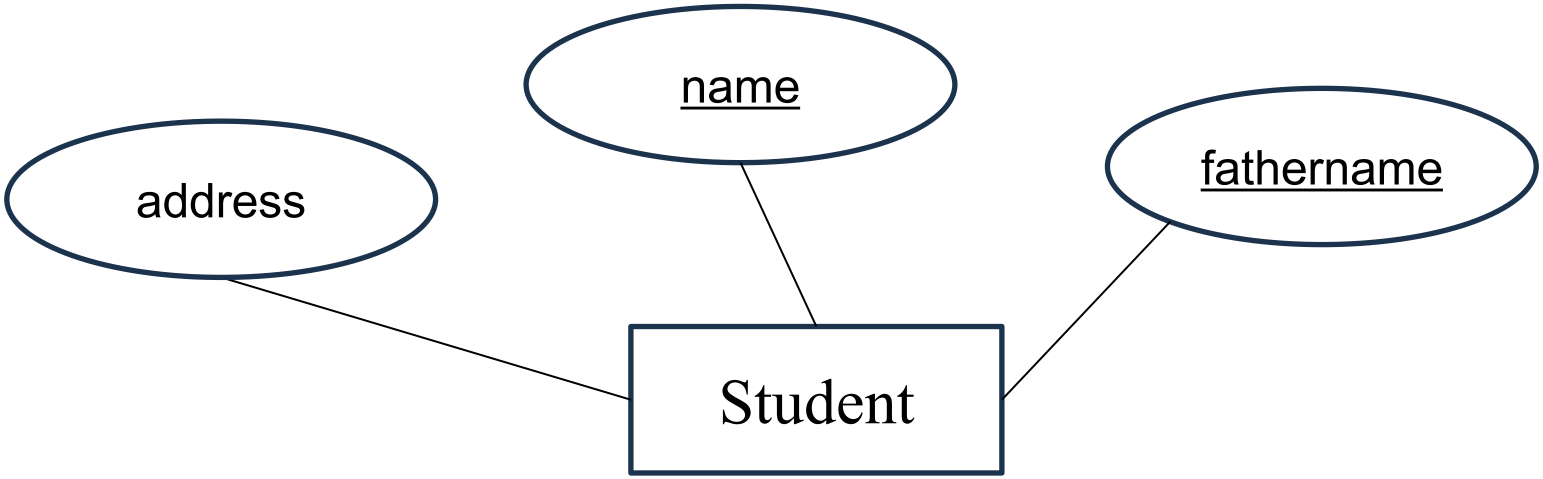
E-R Diagram

Key: <u>An attribute</u> or <u>a set of a set of attributes</u> whose **value can uniquely identify an entity in** an entity-set



E-R Diagram

Example: Key{multiple attribute}



E-R Diagram
(Identify entity-set in an English sentence)

To identify entities in a sentence, we need to look for nouns or noun phrases that represent things, people, places, concepts, or events.

- Identify nouns/noun phrases
    - These are potential candidates for the entity set
- Check for general categories or classes of things
    - An entity set is typically a **collection of similar entities** (e.g., "Students" is an entity set containing individual students).
- Naturally become **a table** in a relational database
    - where each **row = an instance**, and each **column = an attribute**.

E-R Diagram
(Identify entity-set in an English sentence)

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Example:
“John enrolled in the DBMS”

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(Identify entity-set in an English sentence)

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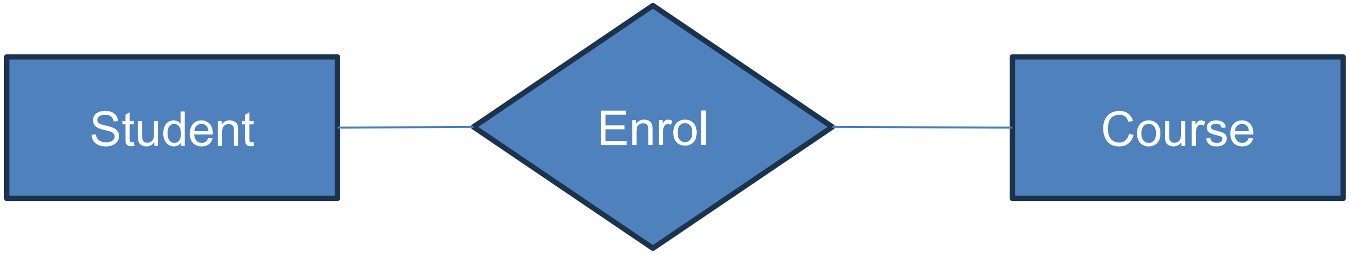
Example:
"**John** enrolled in the **DBMS**"

E-R Diagram
(Identify entity-set in an English sentence)

To identify entities in a sentence, we need to look for nouns or noun phrases that represent things, people, places, concepts, or events.

Example:
"**John** enrolled in the **DBMS**"

1. John (Student)
2. DBMS(Course)



E-R Diagram
(Identify entity-set in an English sentence)

Students work on research projects under guidance of an instructor

E-R Diagram
(Identify entity-set in an English sentence)

Students work on **research projects** under guidance of an **instructor**

E-R Diagram
(Identify entity-set in an English sentence)

Students work on research projects under the guidance of an instructor

1. Student
2. Research Project
3. Instructor

E-R Diagram
(Draw a basic E-R diagram)

Construct an ER diagram of **a university database** containing information about professors and courses. There should be a key attribute for each entity set.

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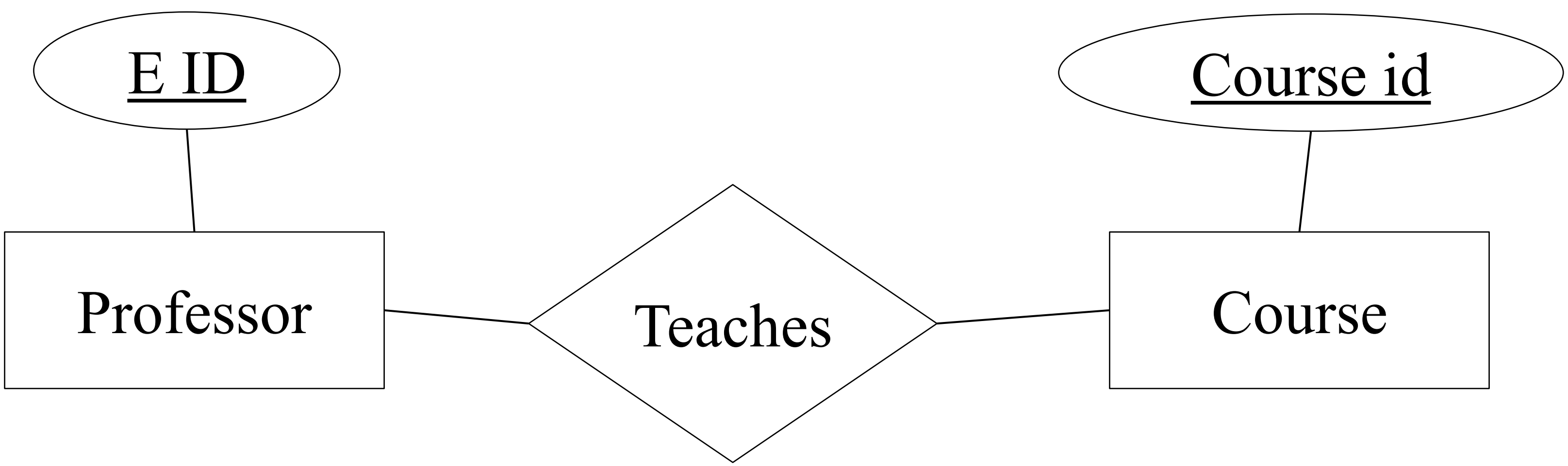
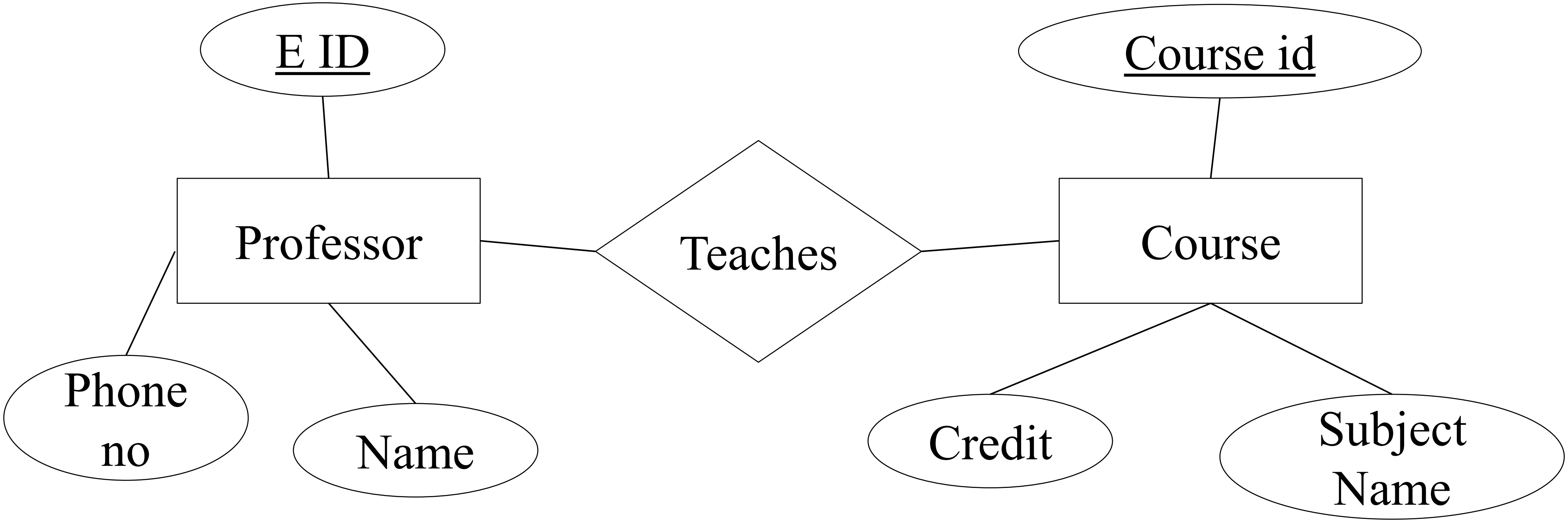


E-R Diagram
(Draw a basic E-R diagram)



E-R Diagram
(Draw a basic E-R diagram)

The National Hockey League (NHL) database contains team and players. Each team has a registration code unique to a team, a name, a city, a coach, a captain. Each player has a name, ID, a position, and a skill level.

E-R Diagram
(Draw a basic E-R diagram)

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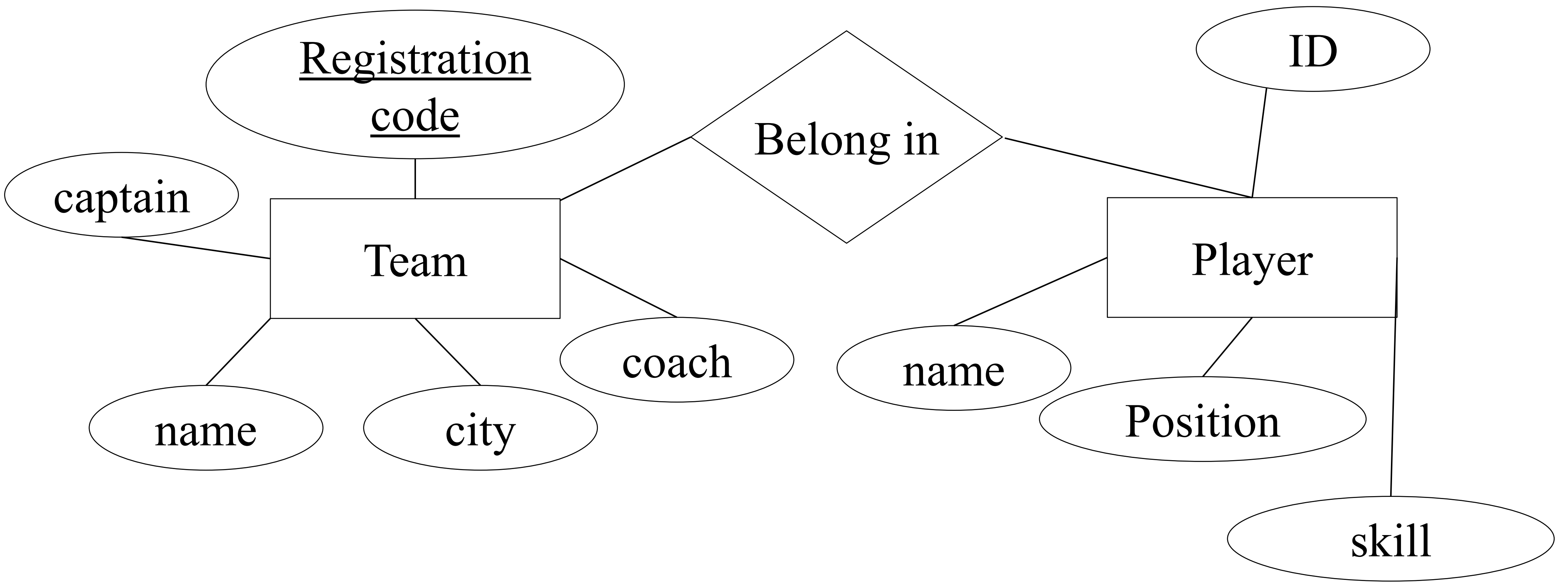

E-R Diagram
(Type of Attribute)

- Attribute types:
  - **Simple** and **Composite** attribute

Simple: An attribute that cannot be further subdivided into
Example: gender of a customer

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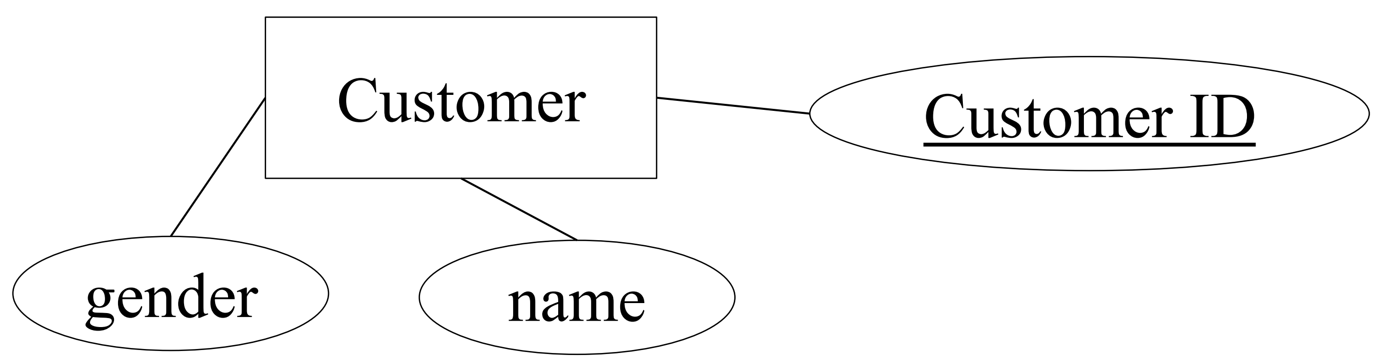
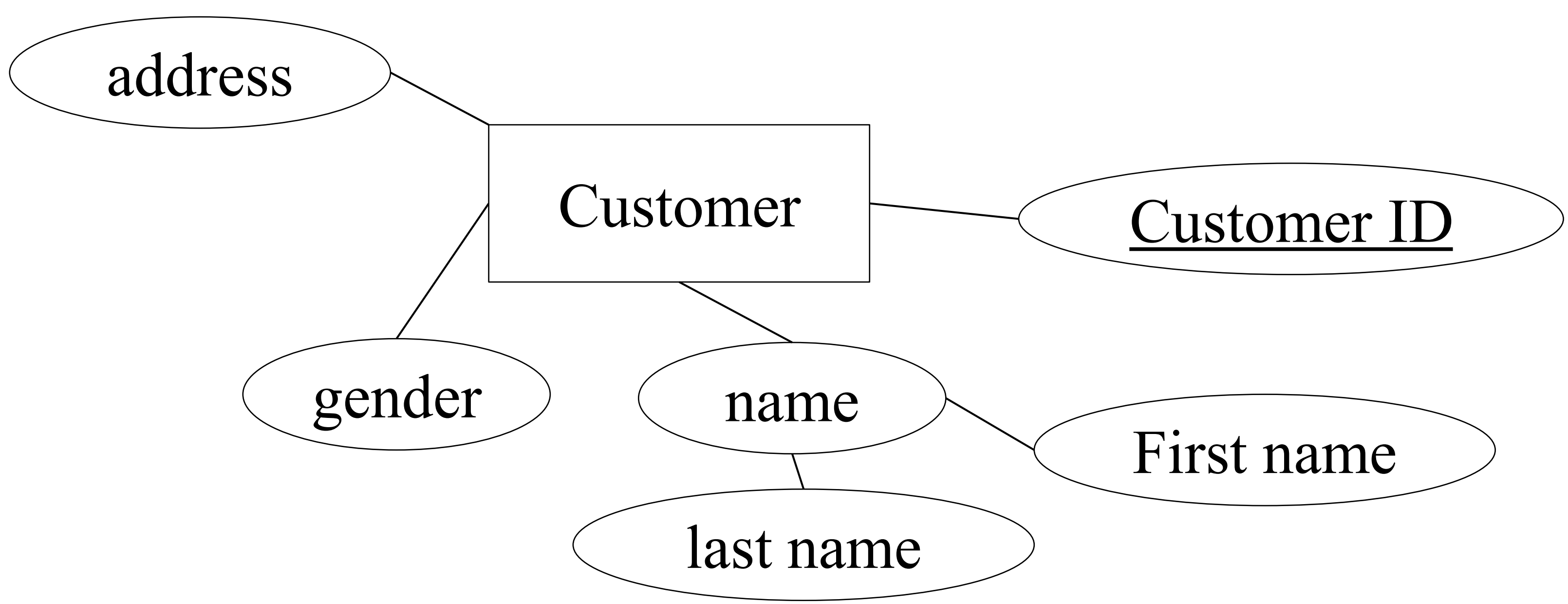


E-R Diagram (Type of Attribute)

- Attribute types:
  - **Simple** and **Composite** attribute

Composite: An attribute that can be split into components is a composite attribute.

Example: name can be split into first name, last name



E-R Diagram
(Type of Attribute)

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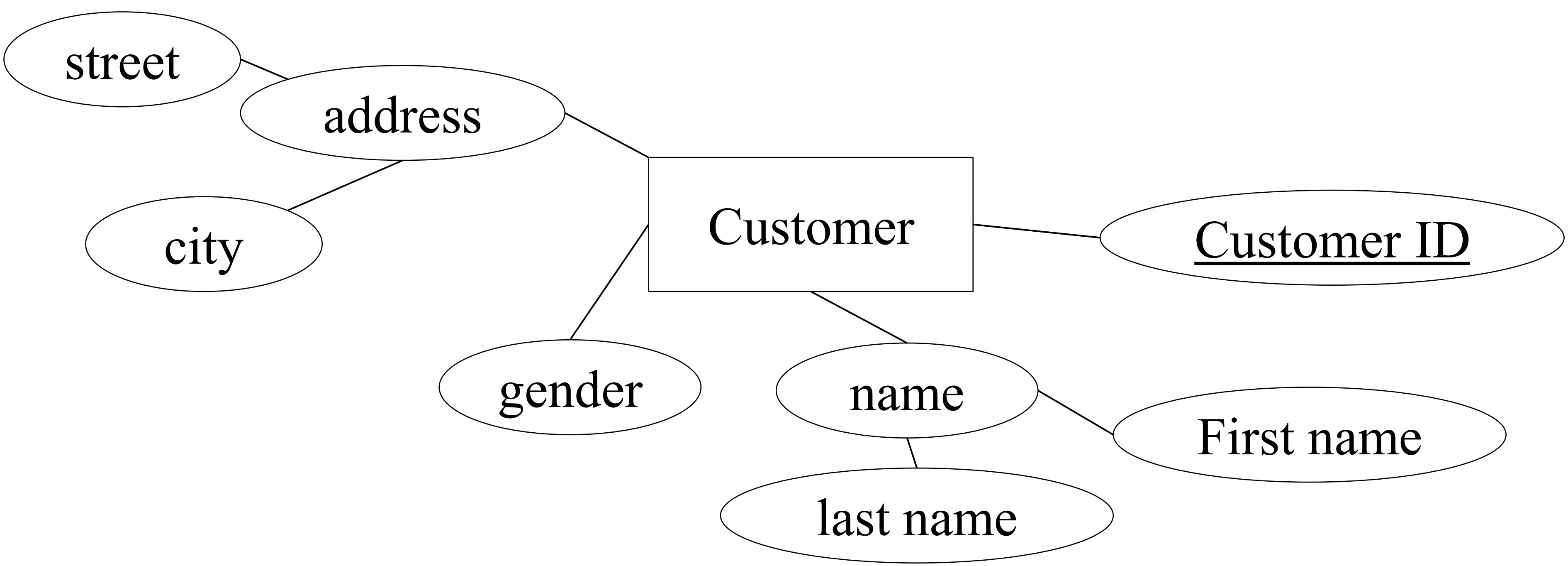
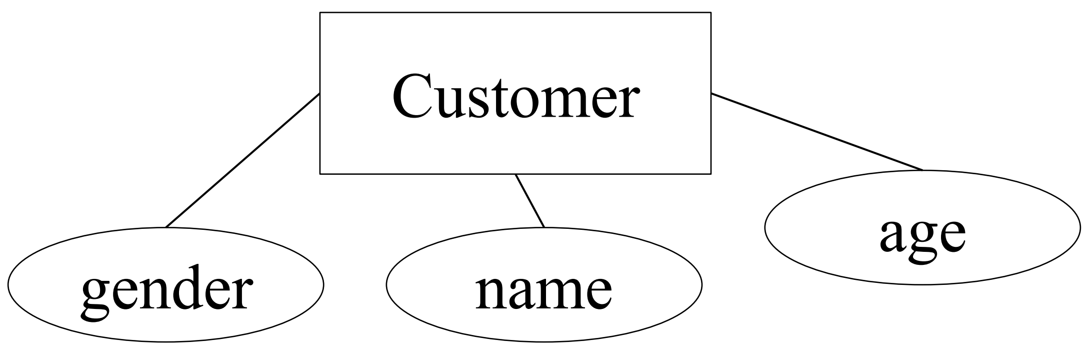
Example: address can be split into street, city



E-R Diagram
(Type of Attribute)

- Attribute types:
  - **single-valued** and **multi-valued** attribute

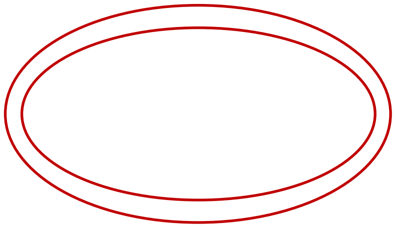
Single-valued: The attribute which takes up only a single value for each entity instance is single-valued.
Example: A entity of an entity-set can only have one gender (single value), same name is single value

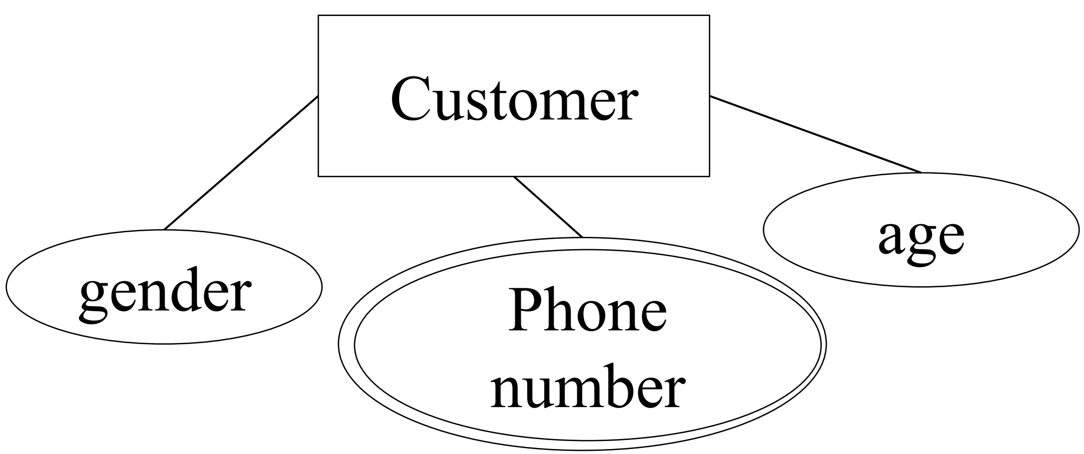


E-R Diagram
(Type of Attribute)

- Attribute types:
  - **single-valued** and **multi-valued** attribute

Multi-valued: The attribute which takes up more than a single value for each entity instance is multi-valued. It is represented by a double oval shape.





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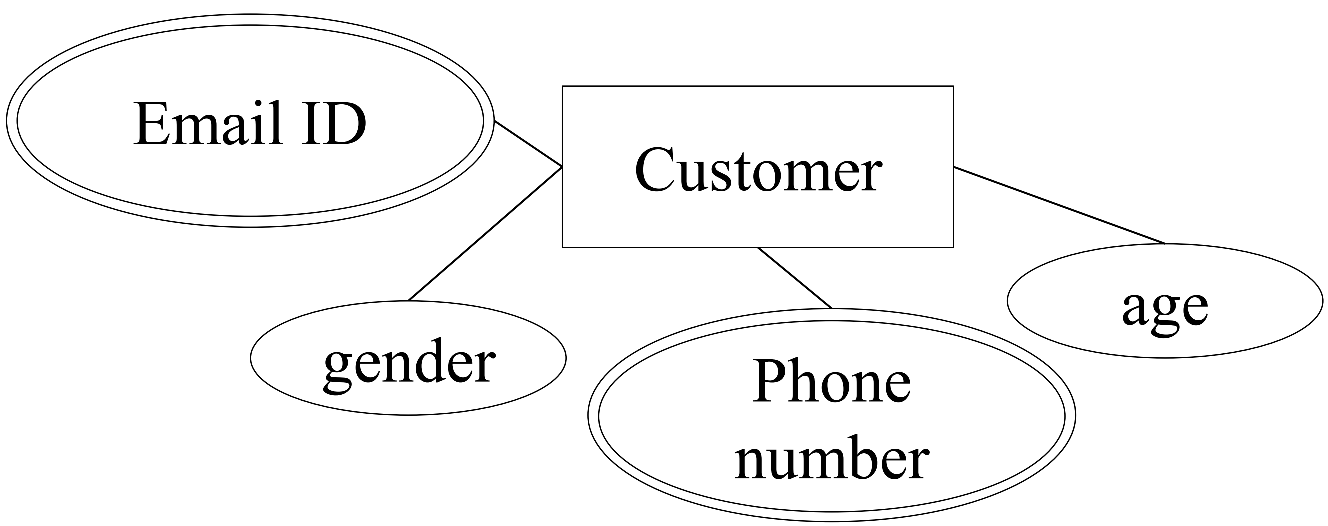

E-R Diagram
(Type of Attribute)

- Attribute types:
  - **Complex** attribute

Complex: Those attributes, which can be formed by the nesting of composite and multi-valued attributes, are called "*Complex Attributes*". These attributes are rarely used in DBMS.

E-R Diagram (Type of Attribute)

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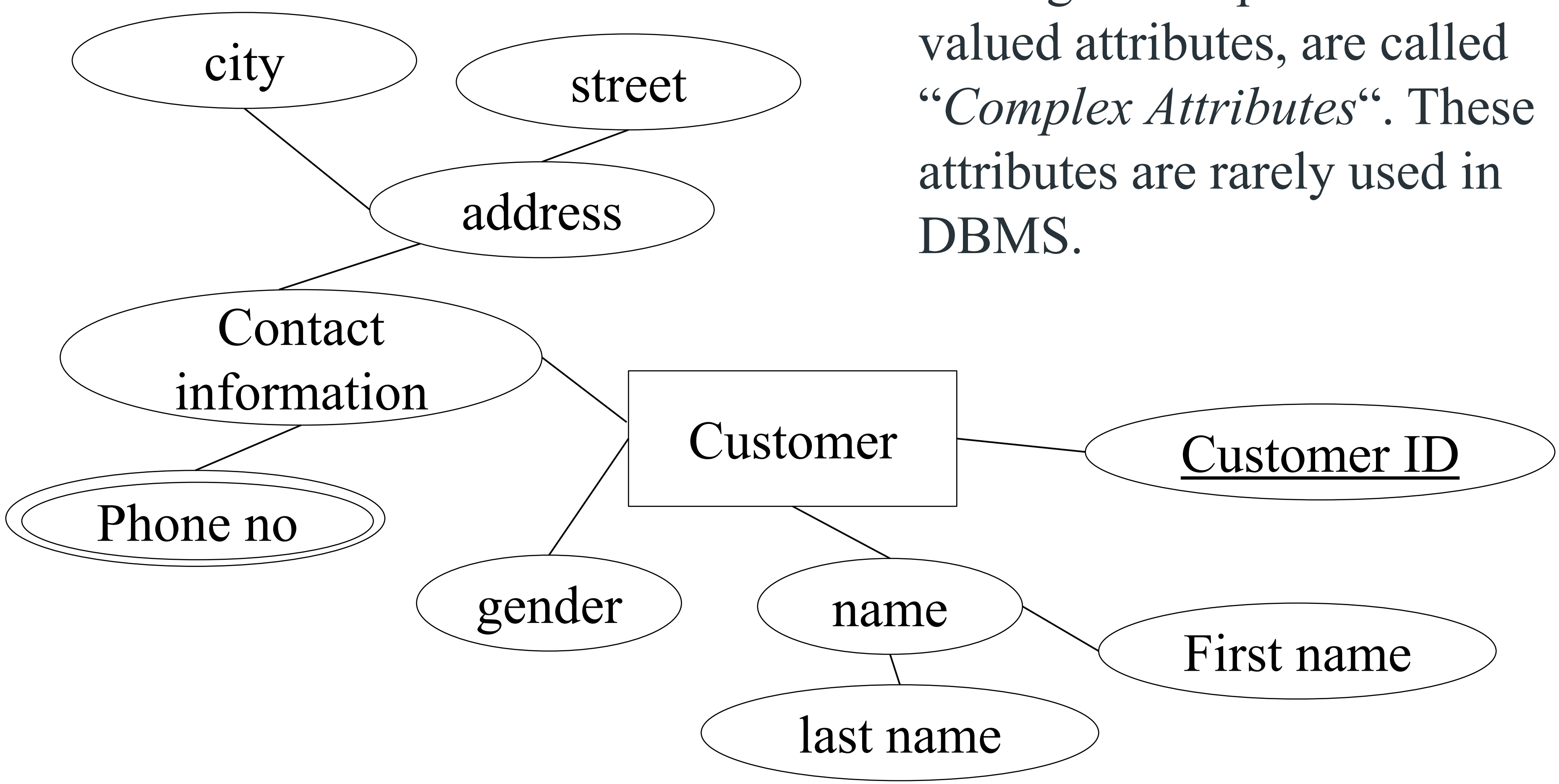
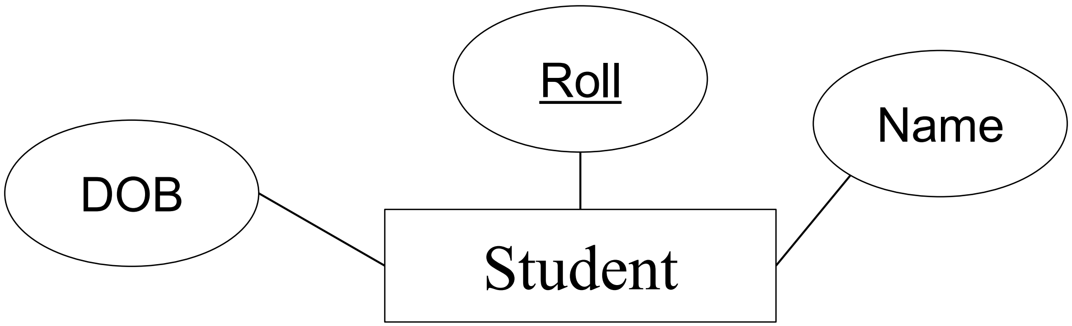


E-R Diagram
(Type of Attribute)

- Attribute types:
  - **Stored** attribute and **Derived** attribute

Stored: The stored attributes are those attributes which doesn’t require any type of further update since they are stored in the database.

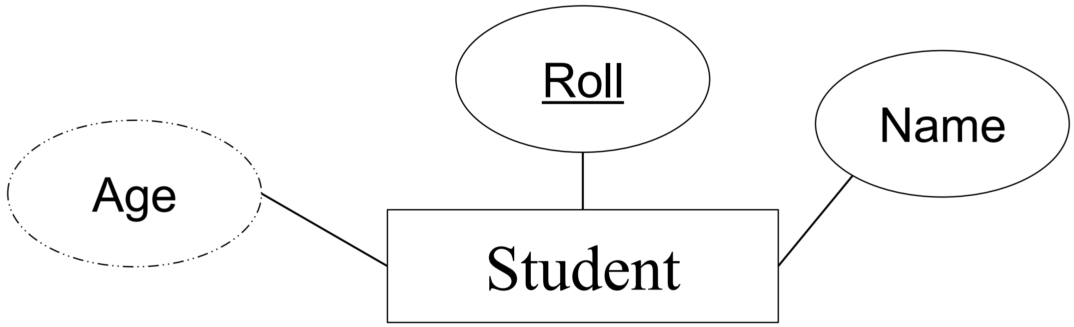
Example: DOB(Date of birth) is the stored attribute.

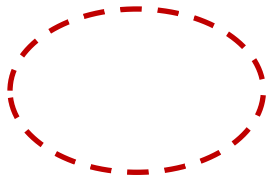


E-R Diagram
(Type of Attribute)

- Attribute types:
  - **Stored** attribute and **Derived** attribute

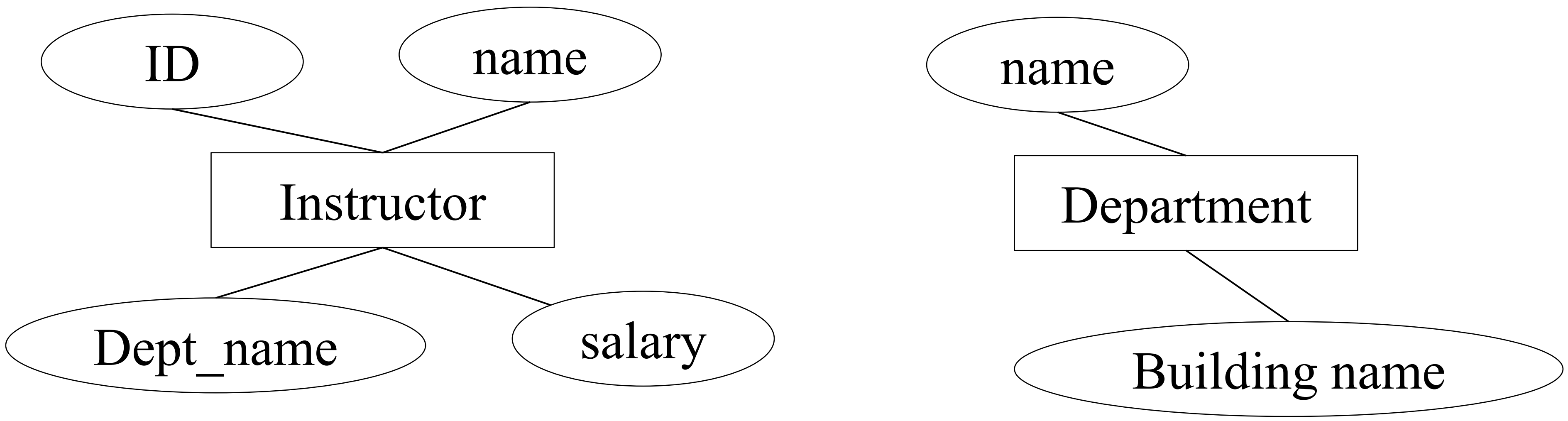
Derived: An attribute that can be derived from other attributes is derived attributes. And it is represented by dotted oval shape.
Example: Age is the stored attribute.





E-R Diagram
(Type of Attribute)

Redundant Attribute



E-R Diagram
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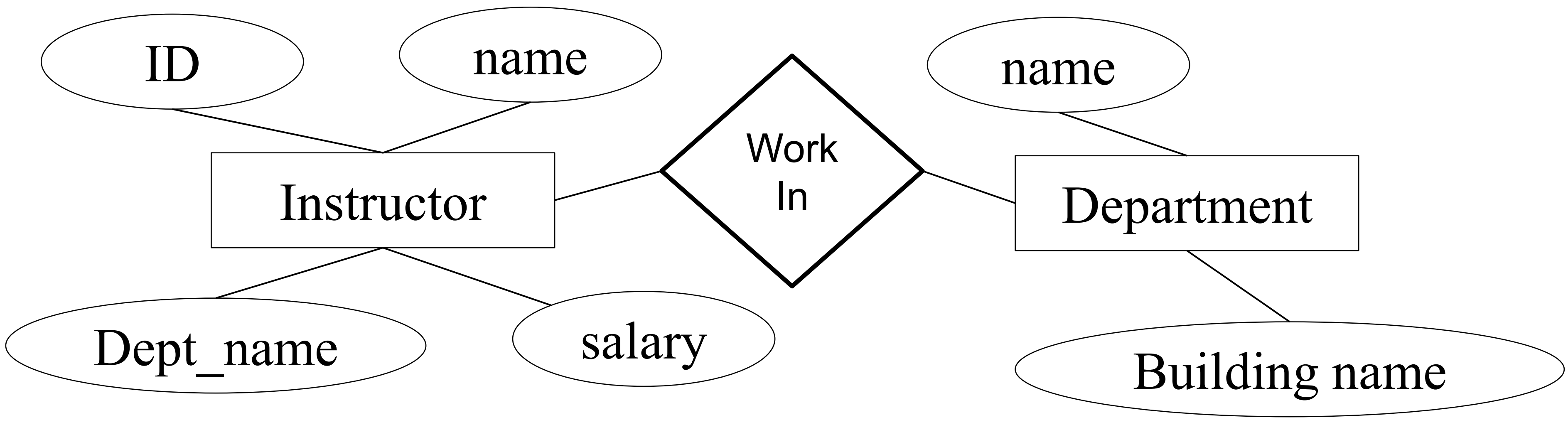

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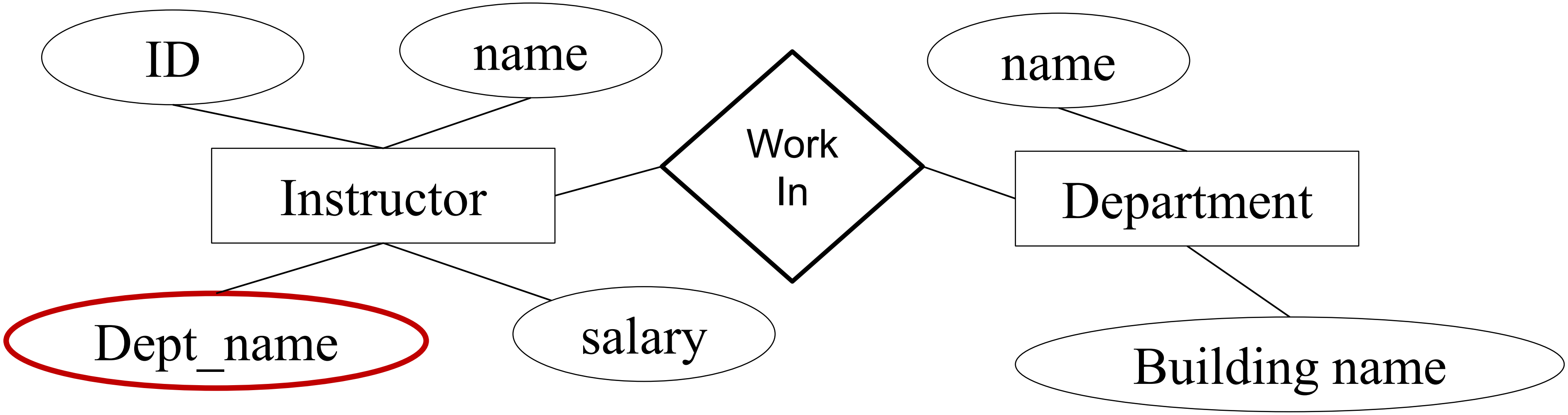

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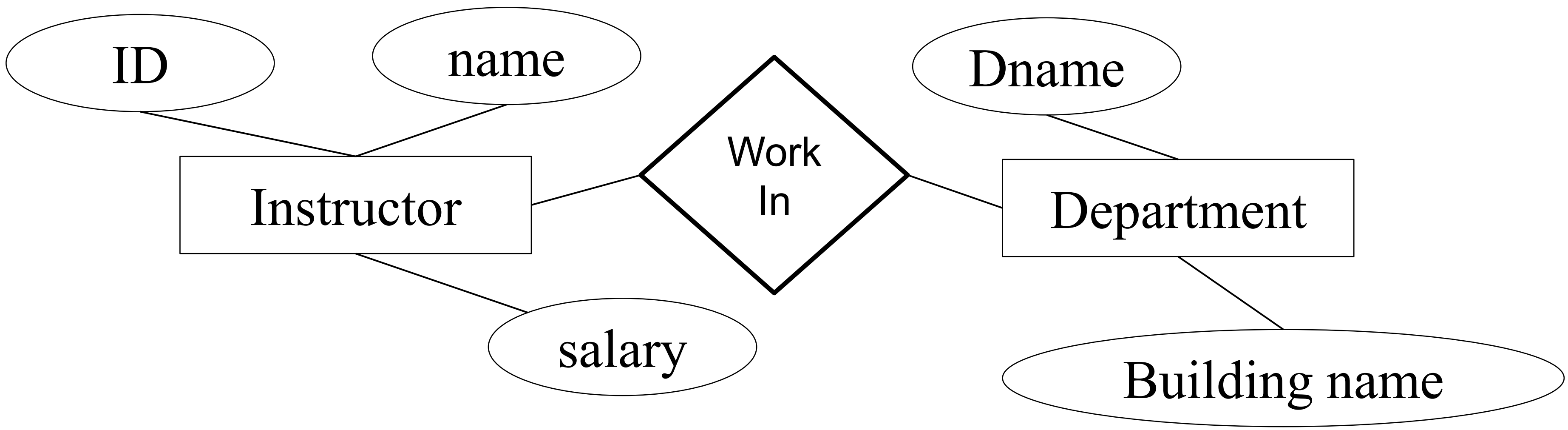

E-R Diagram
(Type of Entity Sets)

Weak entity

- An entity set that does not have a primary key is referred to as a **weak entity set**.
- The existence of a weak entity set depends on the existence of ***identifying (ow owner) entity set***

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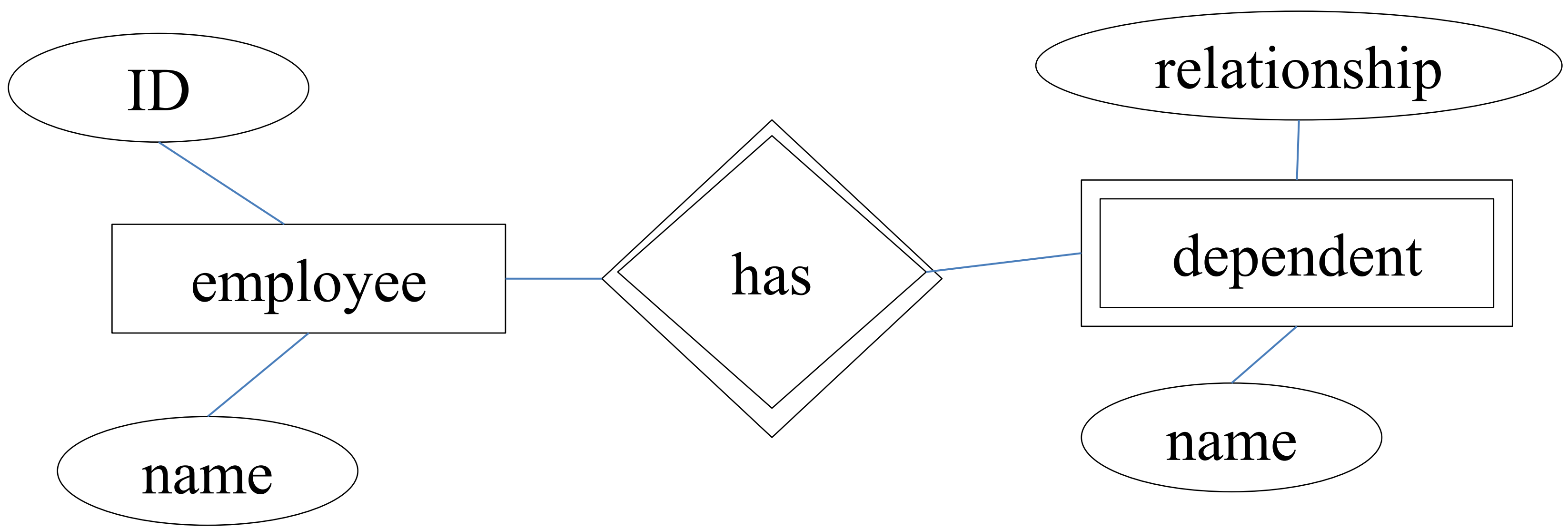
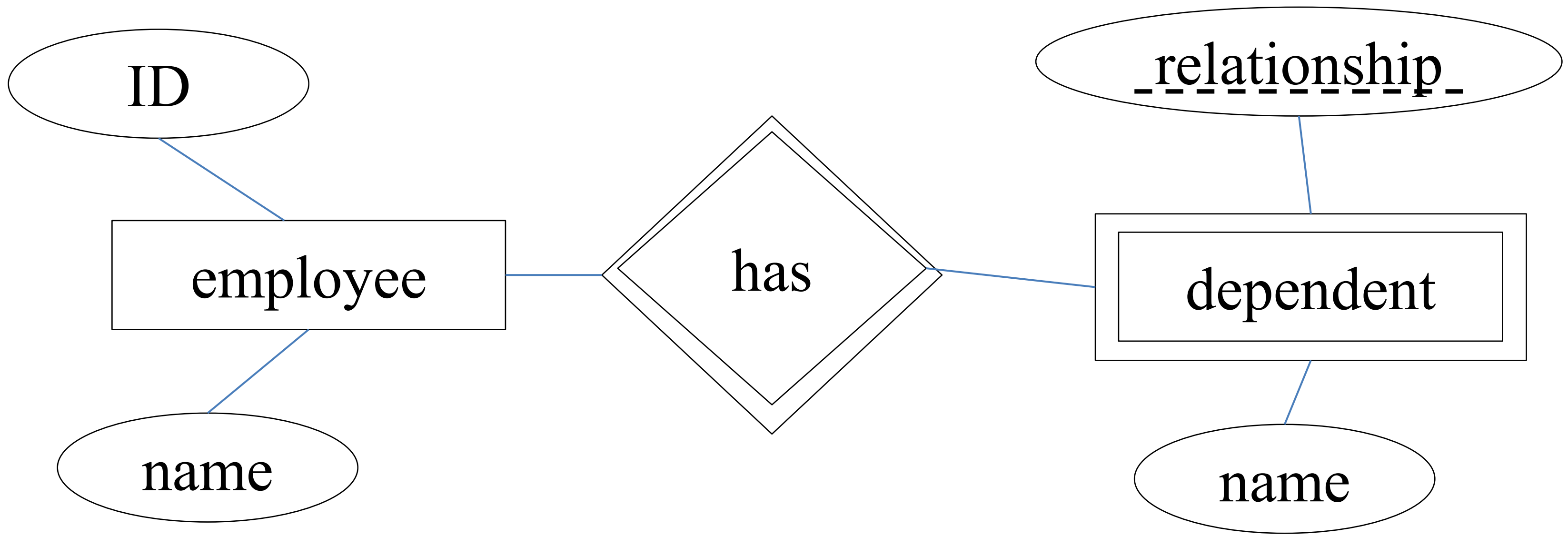


E-R Diagram
(Type of Entity Sets)

Week entity

- An entity set that does not have a primary key is referred to as a **weak entity set**.
- The existence of a weak entity set depends on the existence of ***identifying (ow owner) entity set***
- Discriminator key: the set of attributes / an attribute that distinguishes among all the entities of a weak entity set



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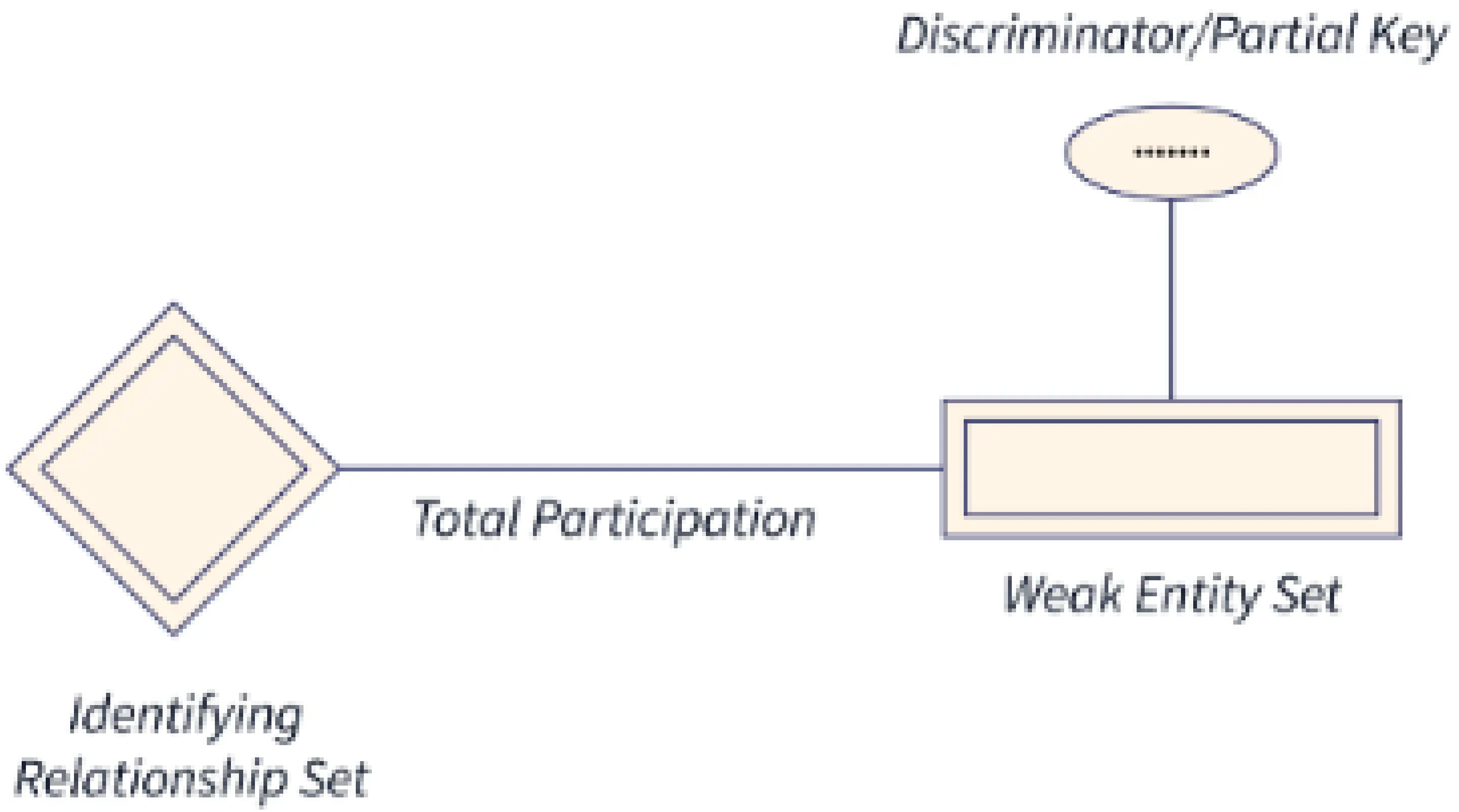
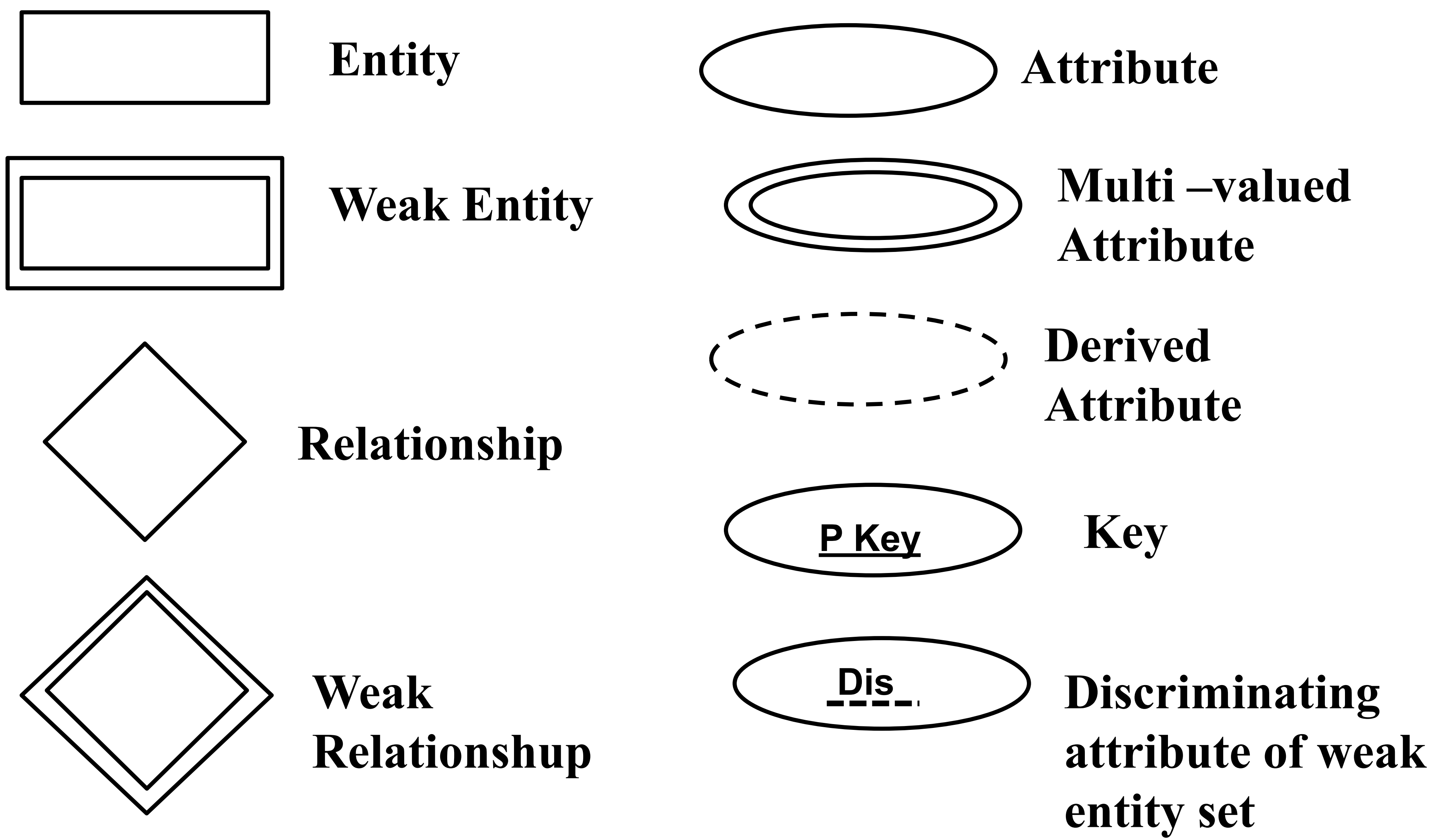


E-R Diagram (Symbol)



E-R Diagram

Construct an ER diagram for the university; the database has information about professors, courses and timetables. Time-table is important in the database iff course is present.

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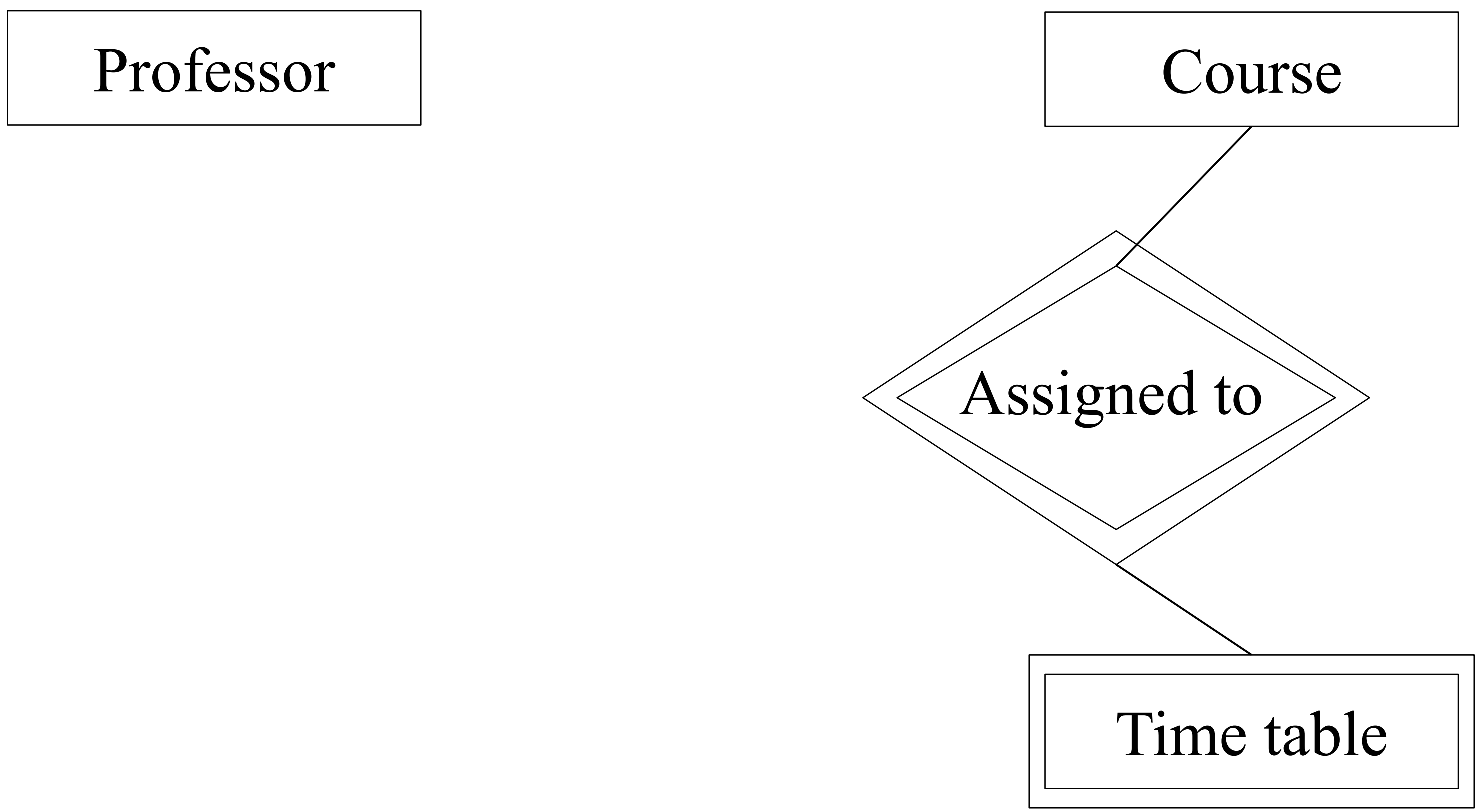

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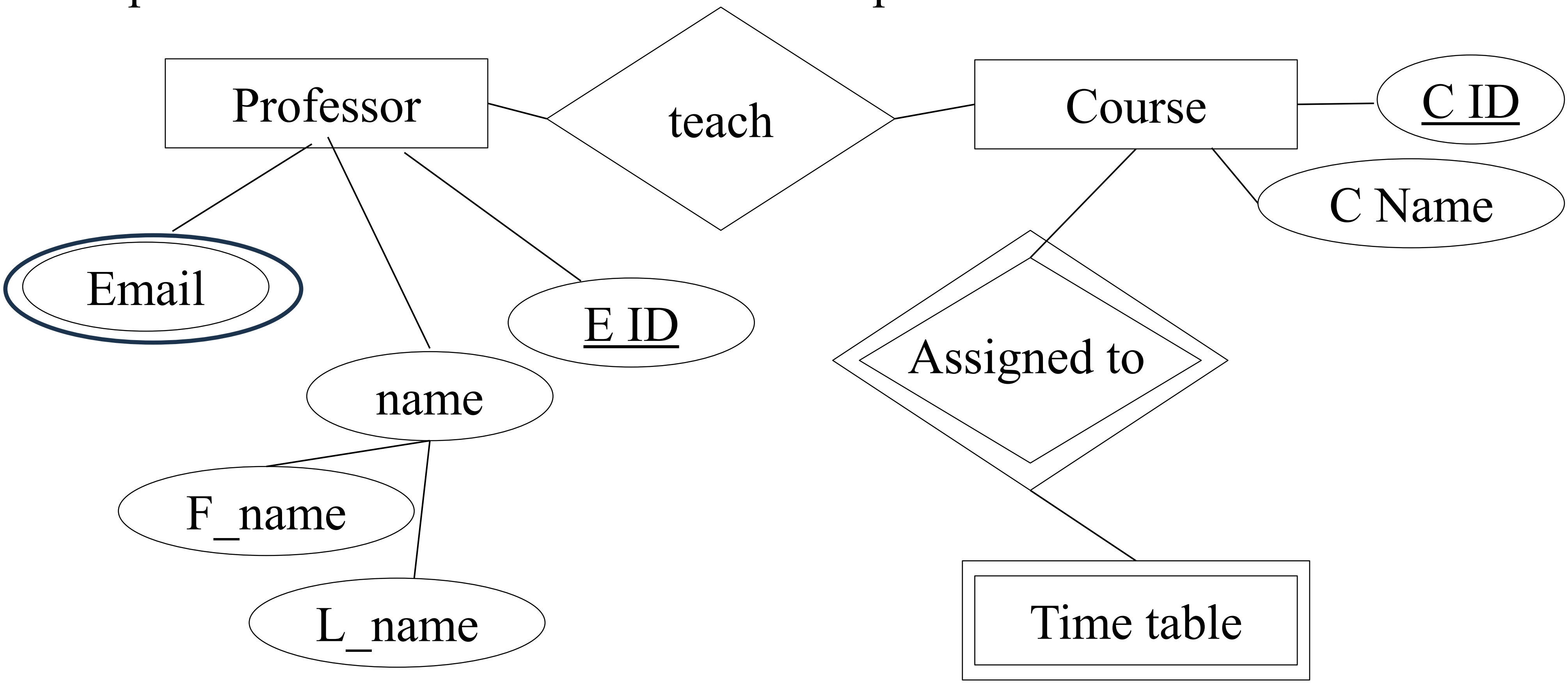

E-R Diagram

Construct an ER diagram for a car-insurance company whose customers own cars. Each car is associated with car accident / accidents.

E-R Diagram

Construct an ER diagram for a small retail shop system based on the following requirements:
The shop sells products. Customers shop at the store, and employees work at the store.

E-R Diagram

Construct an E-R diagram for a hospital with a set of **patients** and a set of **medical doctors**. Associate with each patient a log of the various **tests and examinations** conducted.

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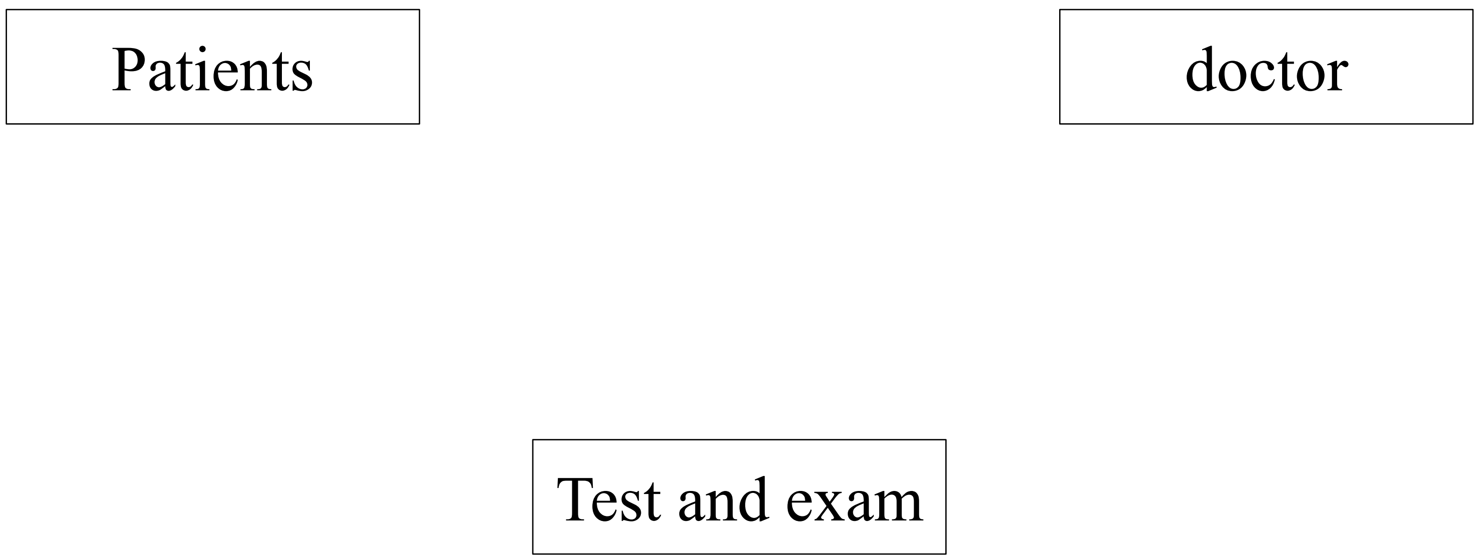

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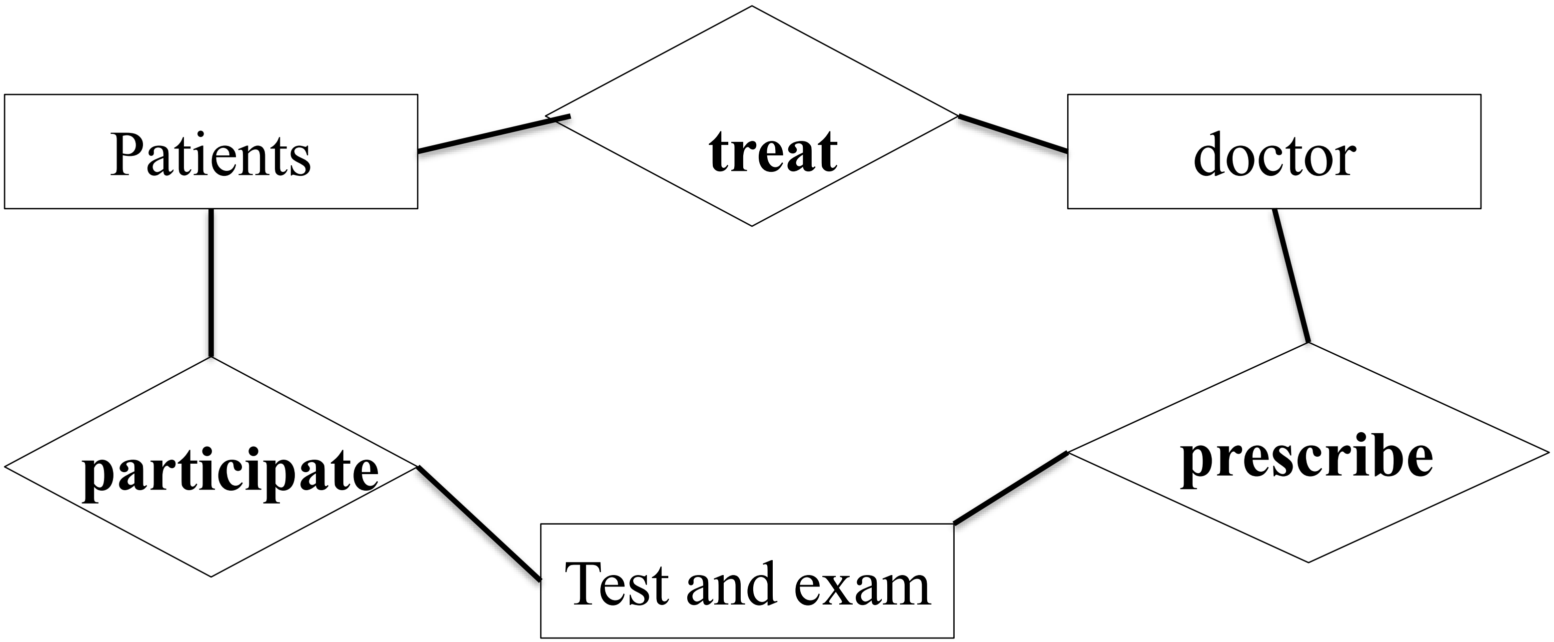

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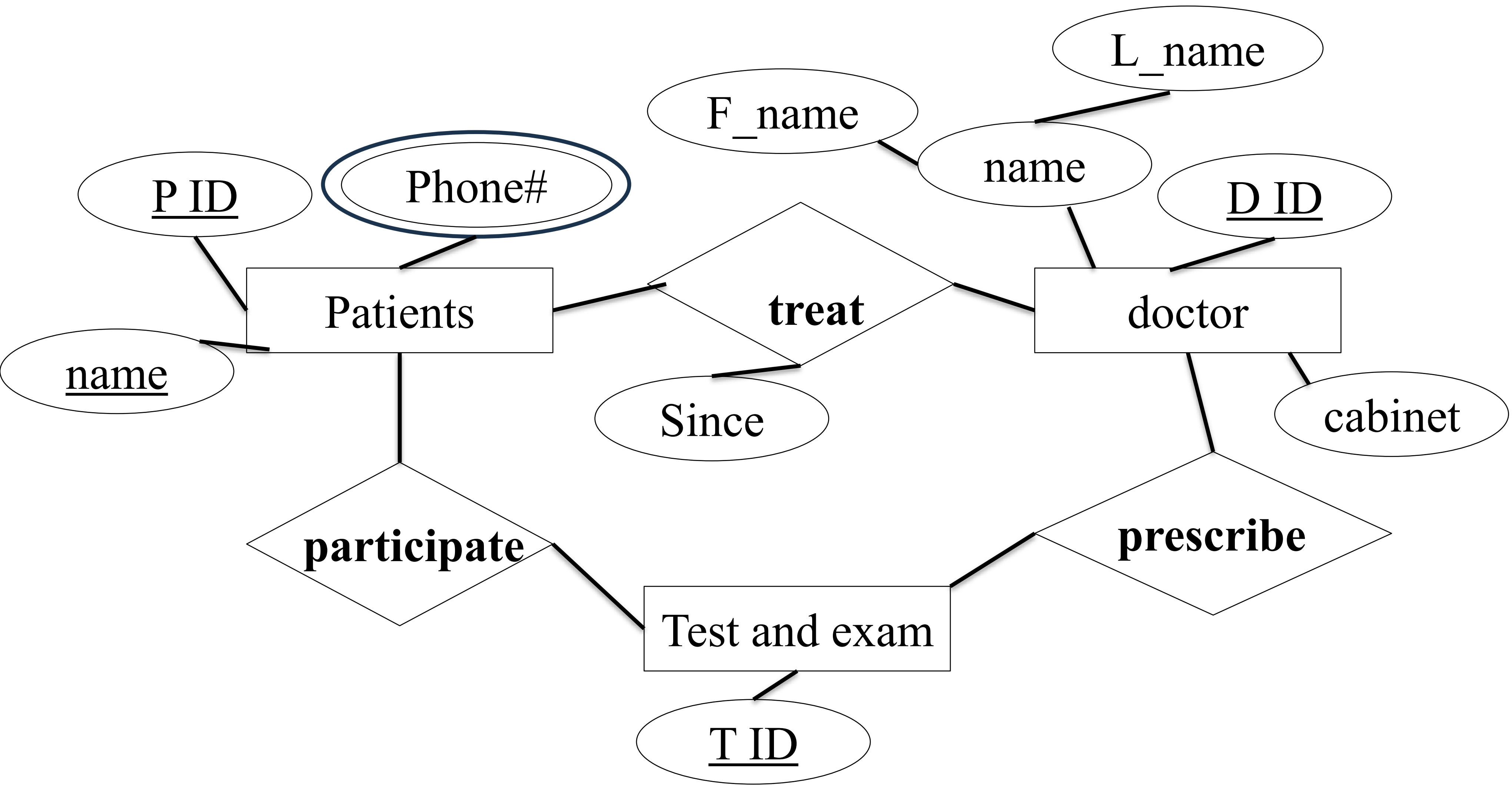
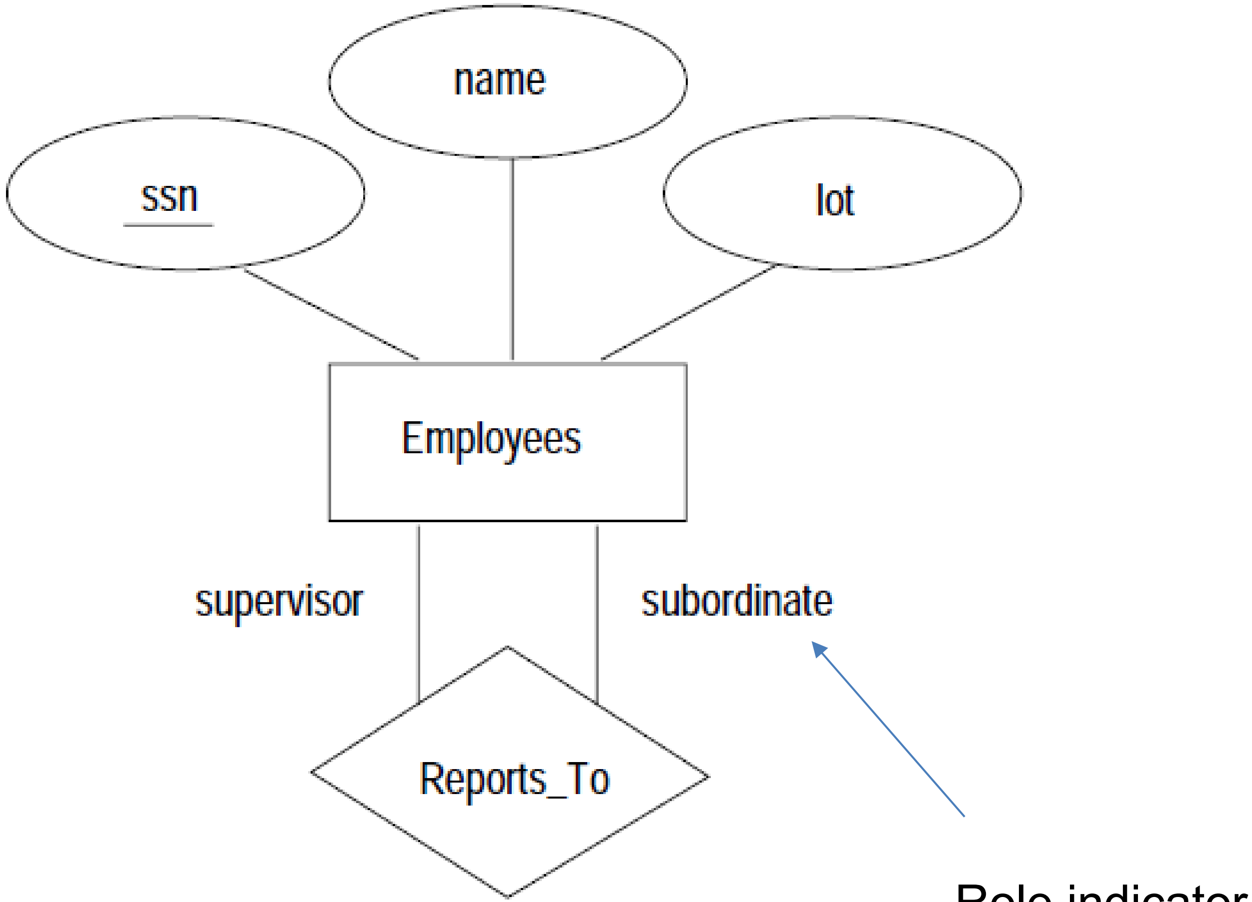


E-R Diagram
(Type of Relationship / Degree of Relationship)

- **Unary**
- **Binary**
- **Ternary**

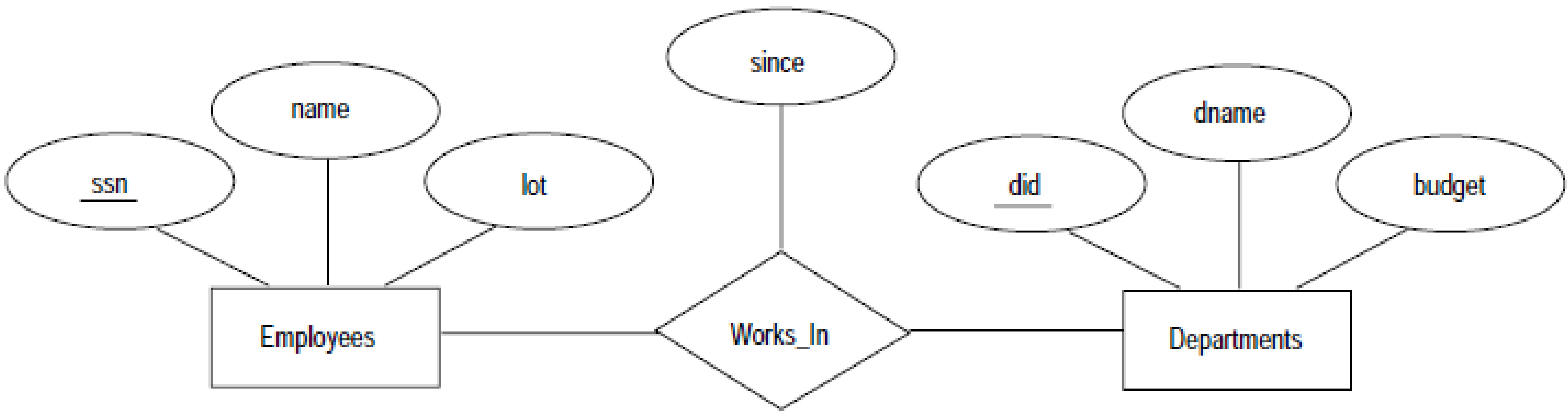
E-R Diagram
(Type of Relationship)

- **Unary:** Relationship **between entity of the same entity set**
- **Binary**
- **Ternary**



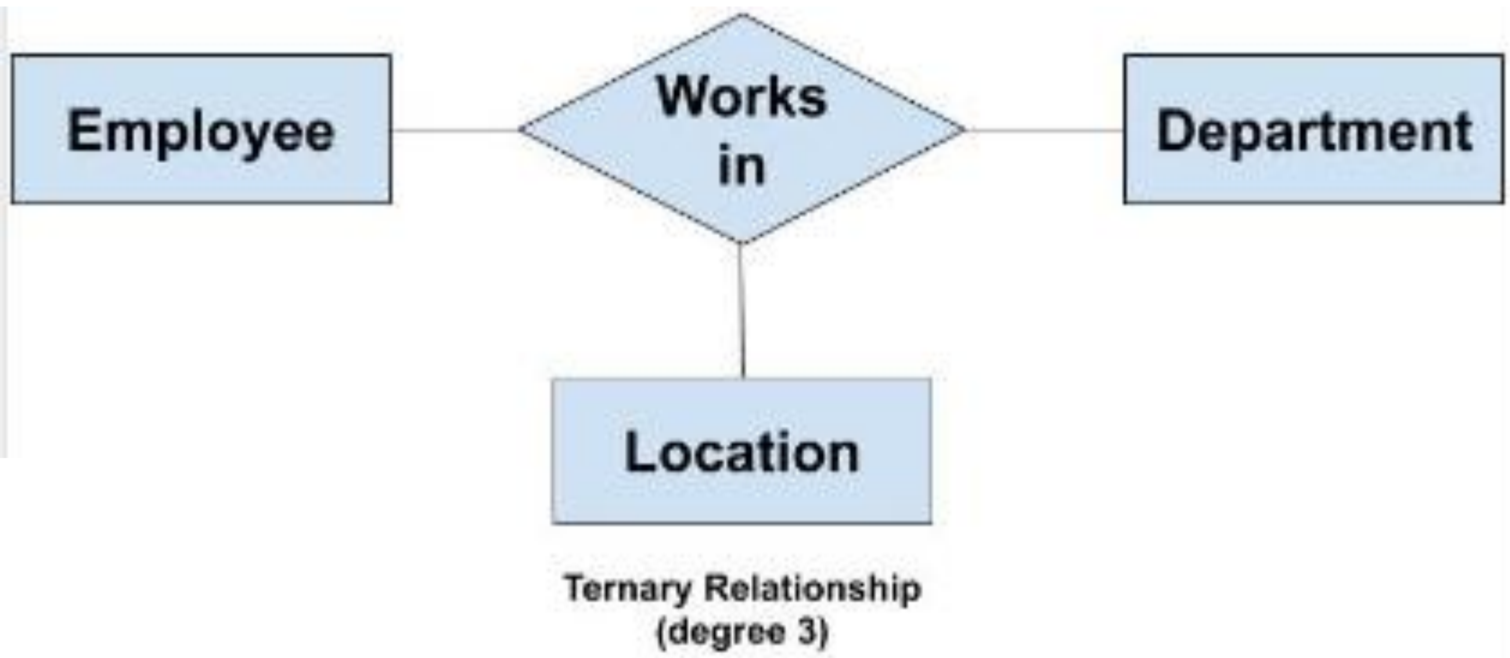
E-R Diagram
(Type of Relationship)

- **Unary:**
- **Binary:** Relationship **between entity of two entity set**
- **Ternary:**



E-R Diagram
(Type of Relationship)

- **Unary:**
- **Binary:**
- **Ternary:** Relationship **between entity of three entity set**



Mapping Cardinality

Mapping cardinalities, or cardinality ratios, express the number of entities to which another entity can be associated via a relationship set.

Mapping cardinalities are most <u>useful in describing binary relationship</u> sets, although they can contribute to the description of relationship sets that involve more than two entity sets.

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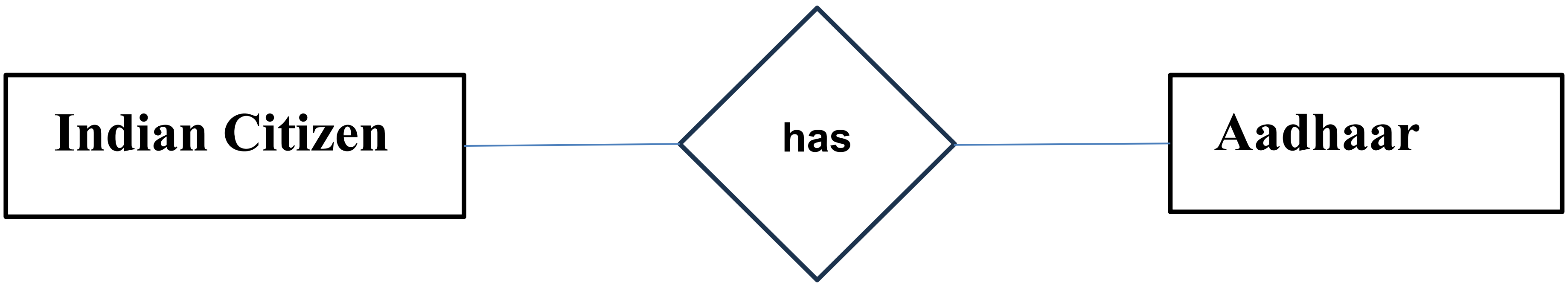
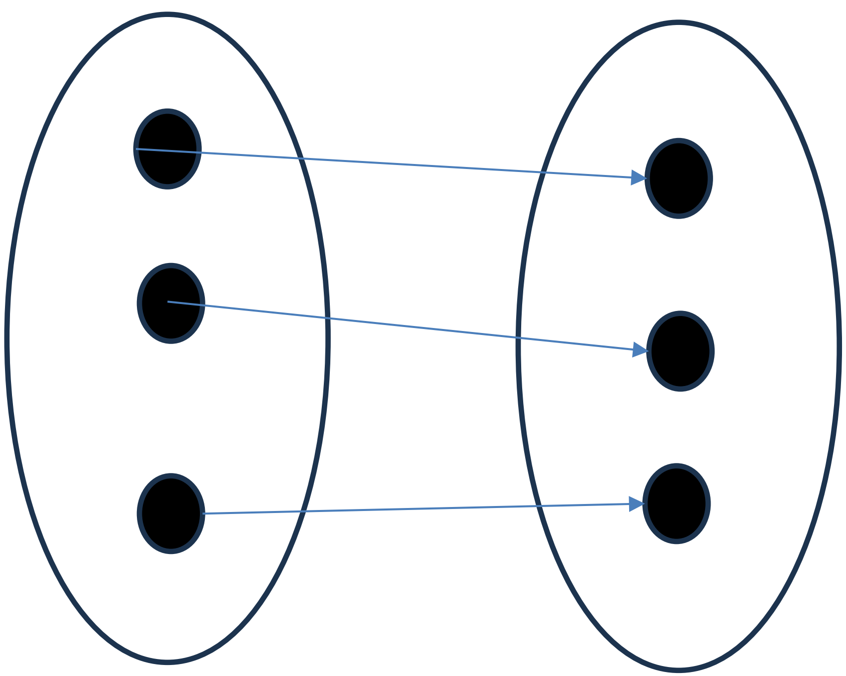
- One to One
- One to Many
- Many to One
- Many to Many

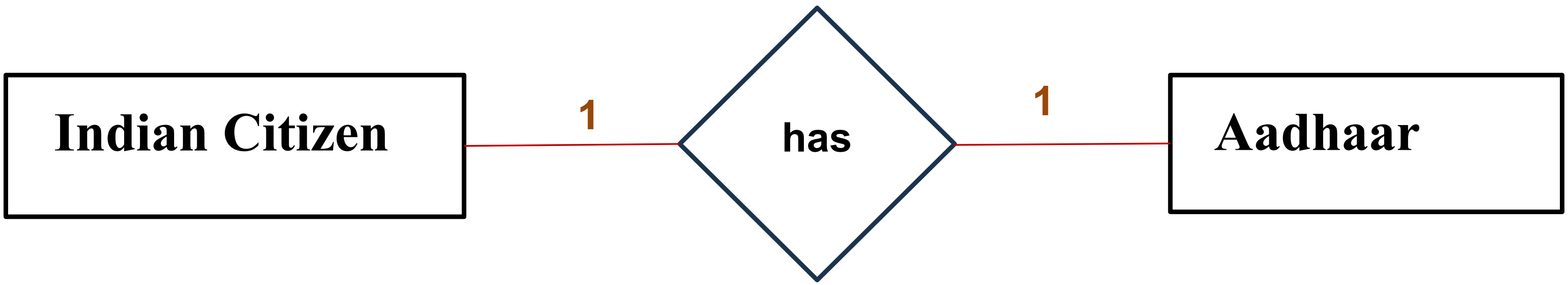
Mapping Cardinality
(one to one)





An entity in *A (entity-set)* is associated with *at most* one entity in *B (entity-set)*, and an entity in *B* is associated with *at most* one entity in *A*.

Mapping Cardinality
(one to one)



Mapping Cardinality
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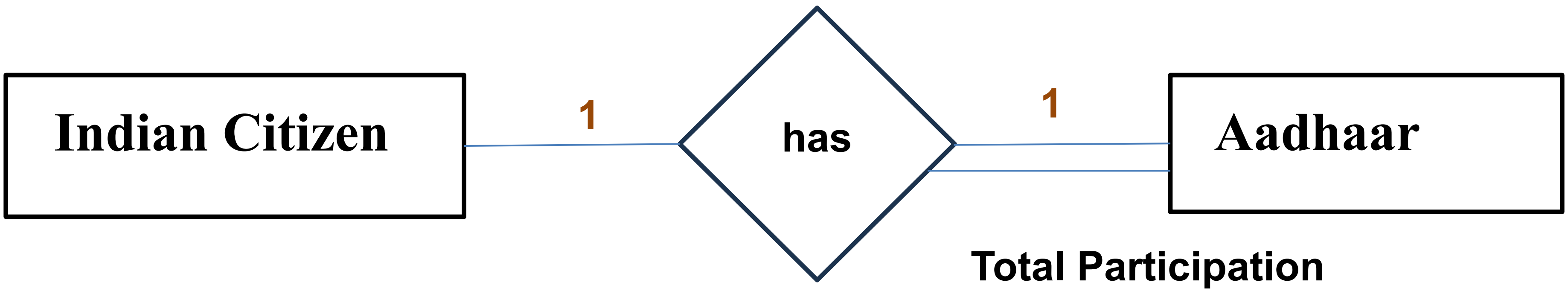
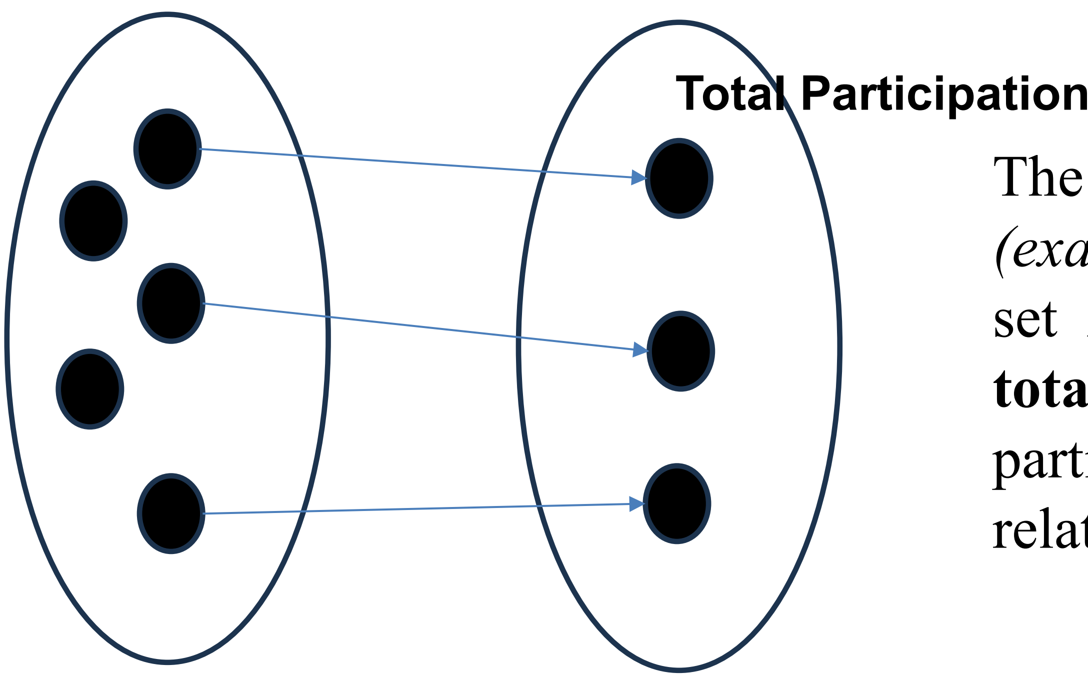
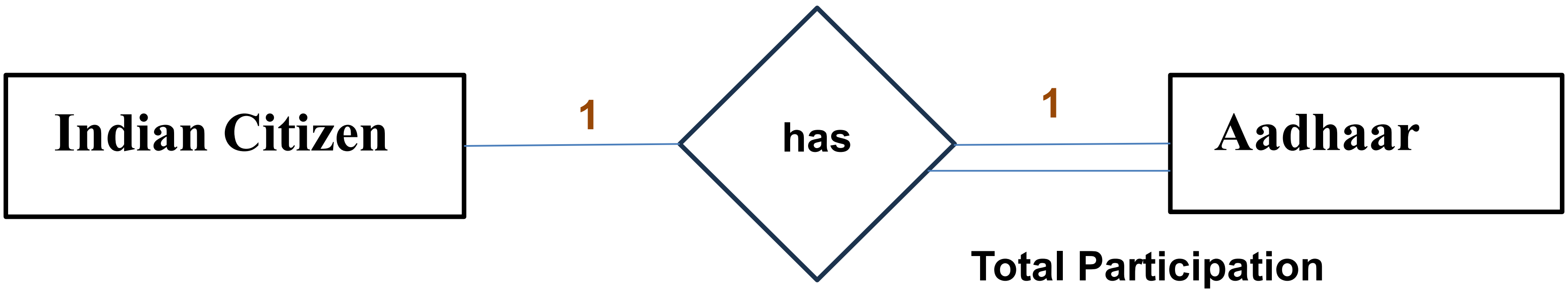
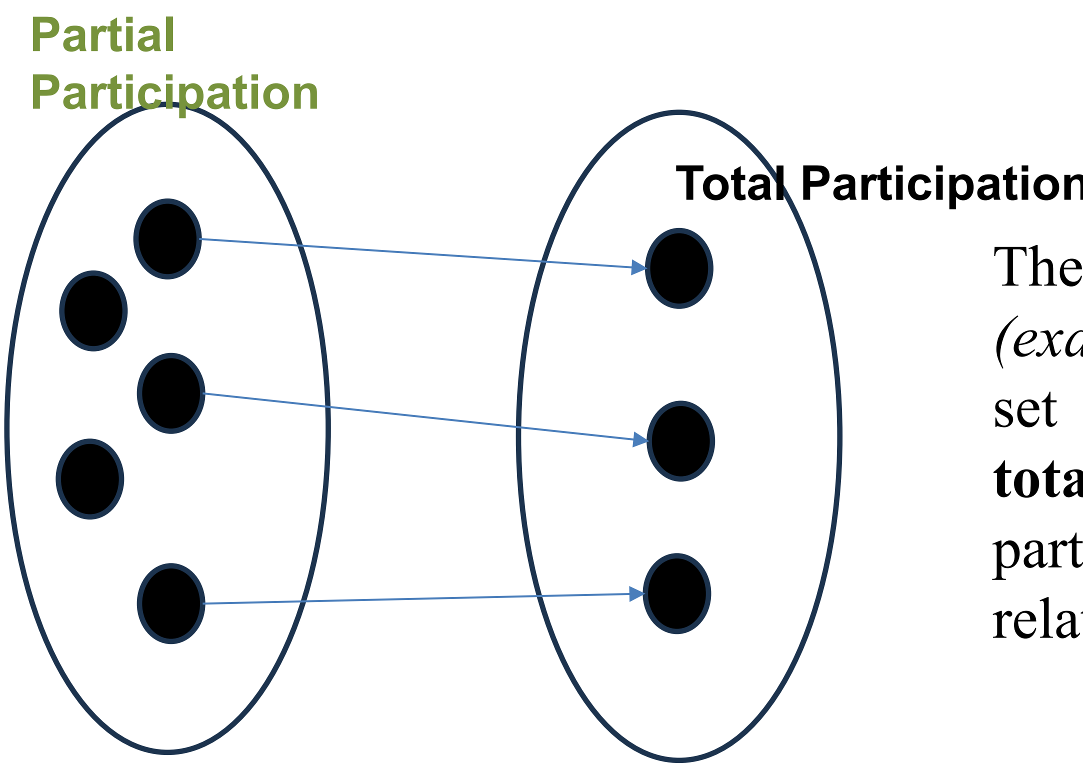




The participation of an entity set *E* *(exam: adhaar)* in a relationship set *R* *(exam: has)* is said to be **total** if every entity in *E* participates in at least one relationship in *R*.

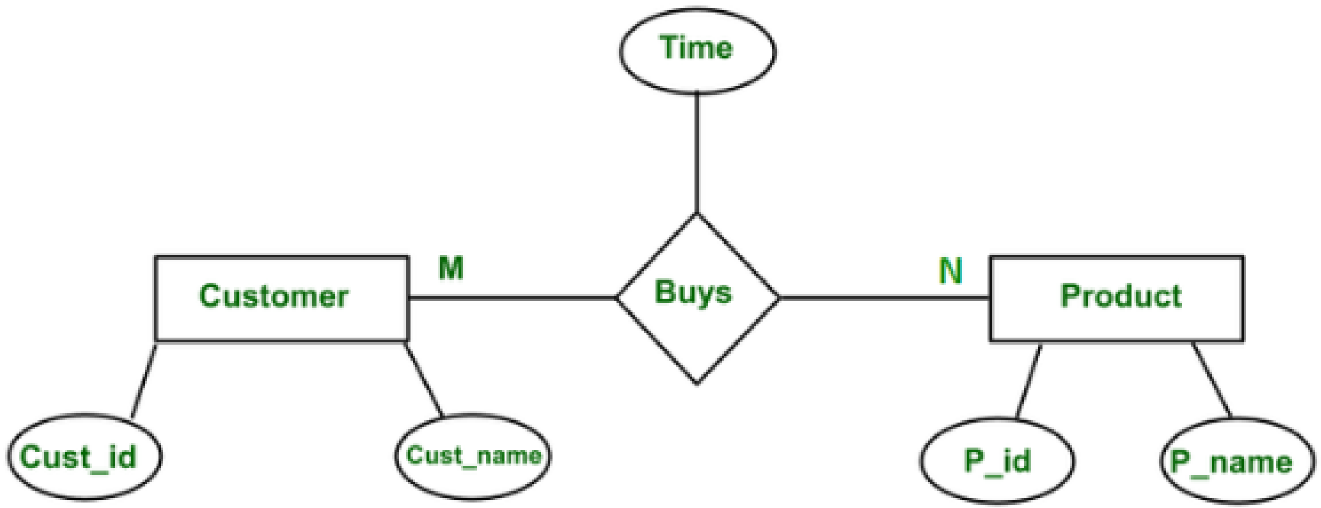
Mapping Cardinality
(one to one)





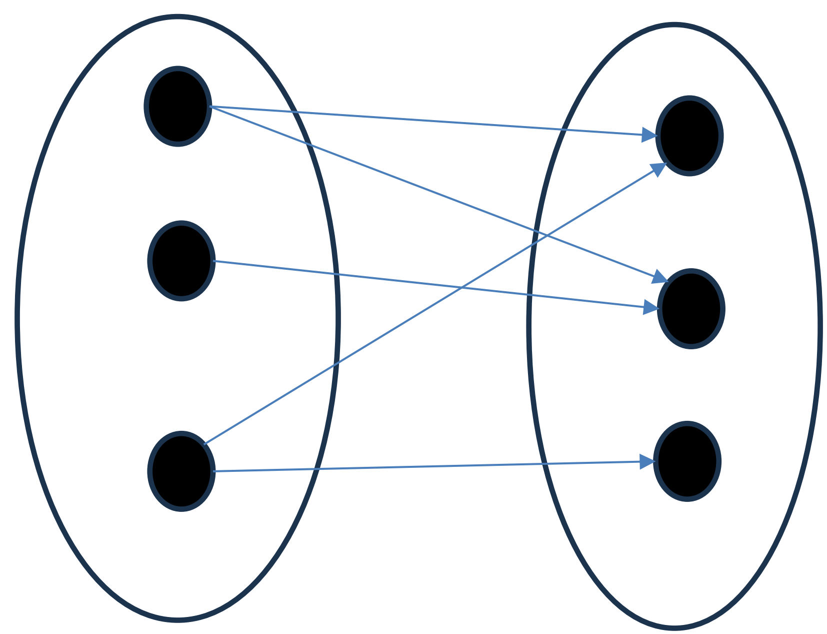
The participation of an entity set *E (exam: adhaar)* in a relationship set *R (exam: has)* is said to be **total** if every entity in *E* participates in at least one relationship in *R*.

Mapping Cardinality (Many to Many)



Example of Relationship in DBMS

An entity in A is associated with any number (zero or more) of entities in B, and an entity in B is associated with any number (zero or more) of entities in A



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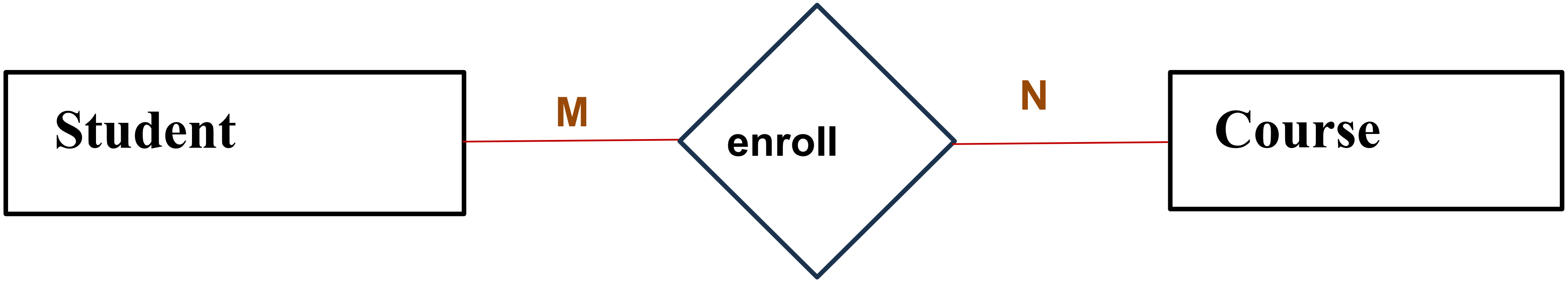
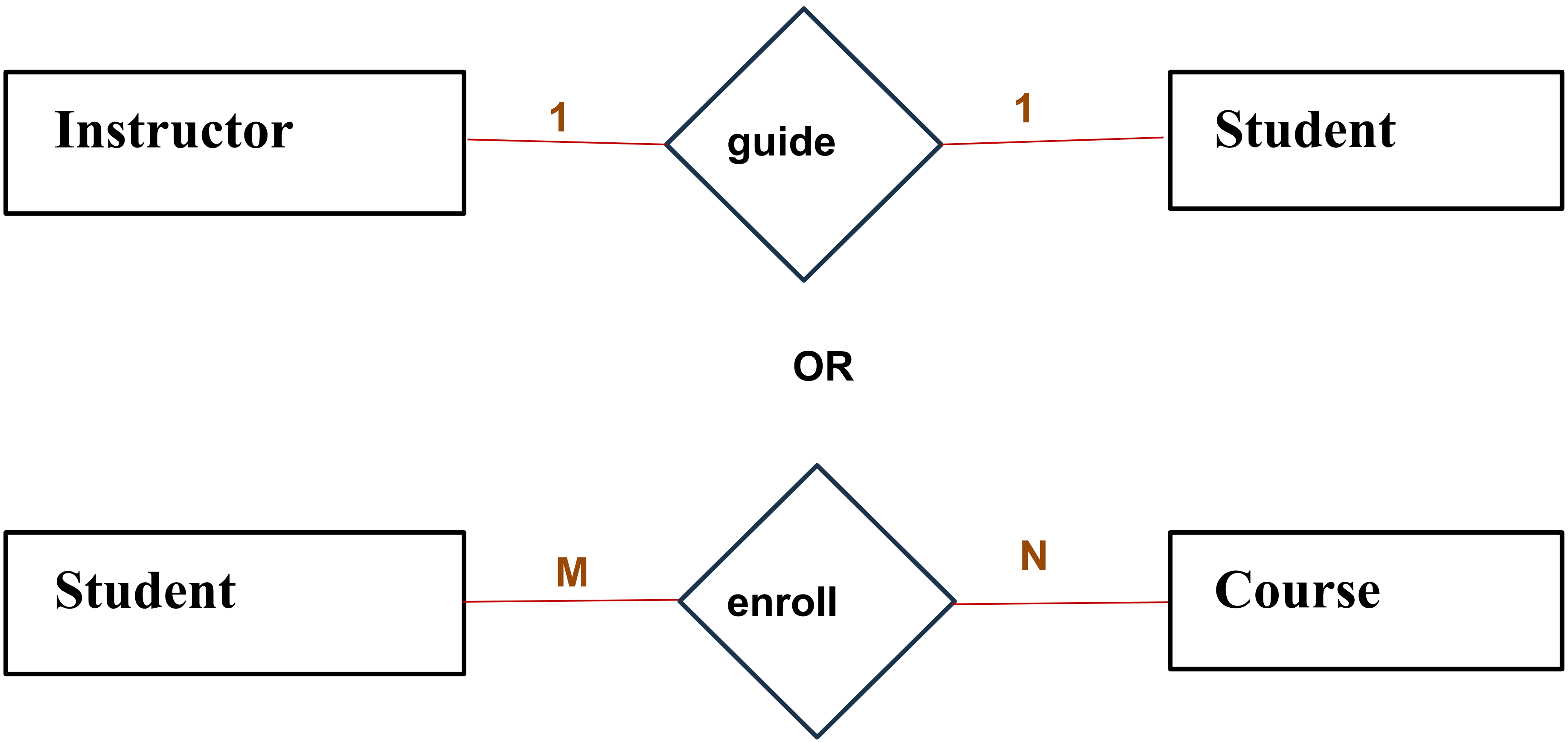


Mapping Cardinality
(Many to Many)



Mapping Cardinality
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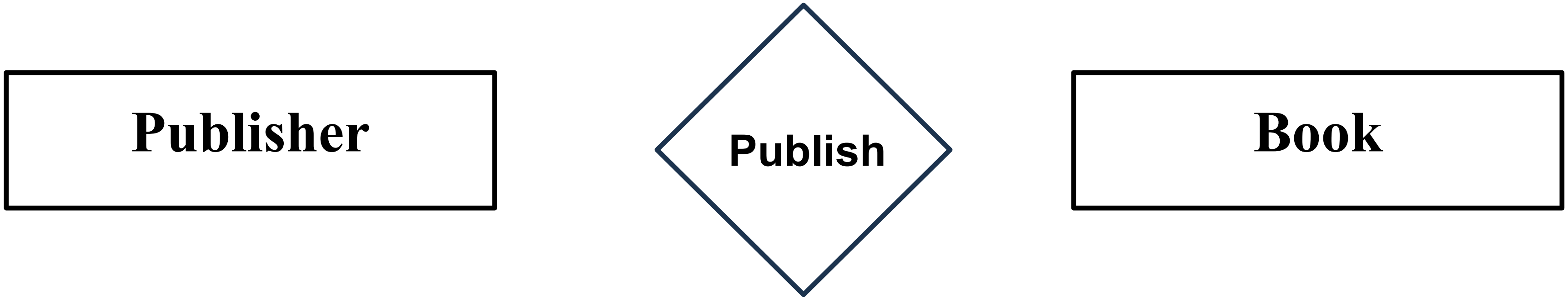

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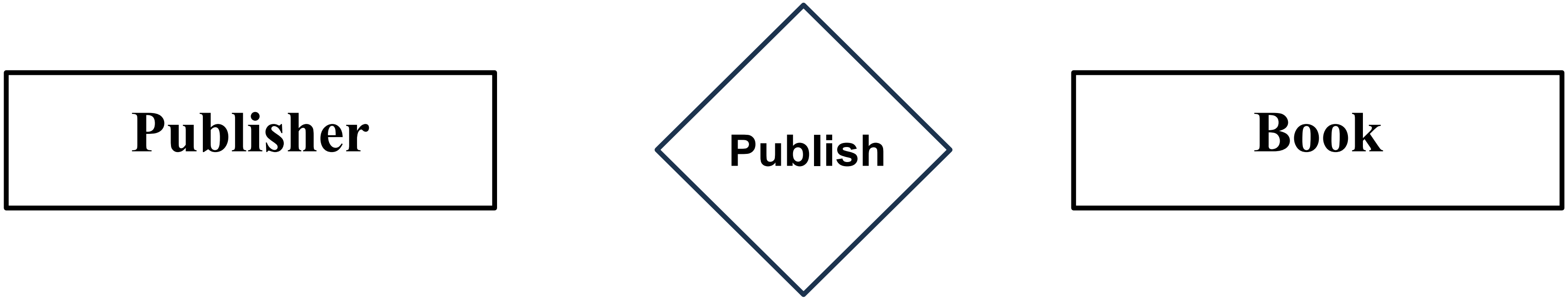
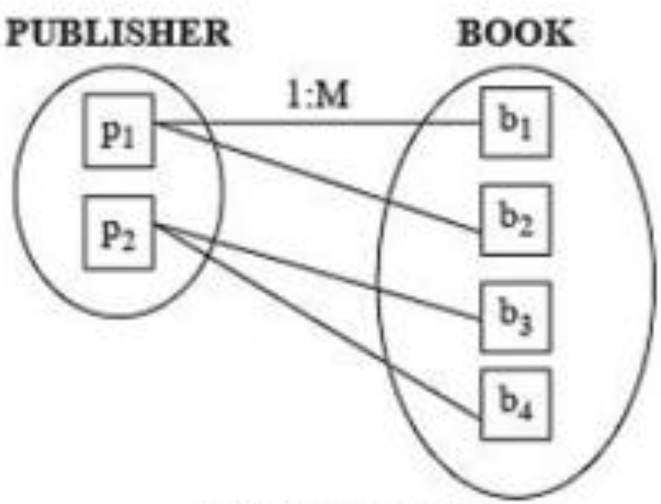
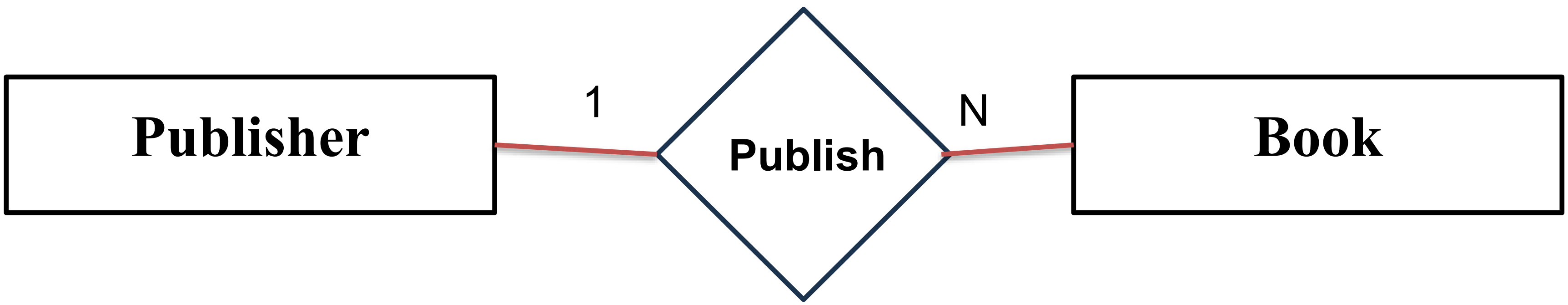


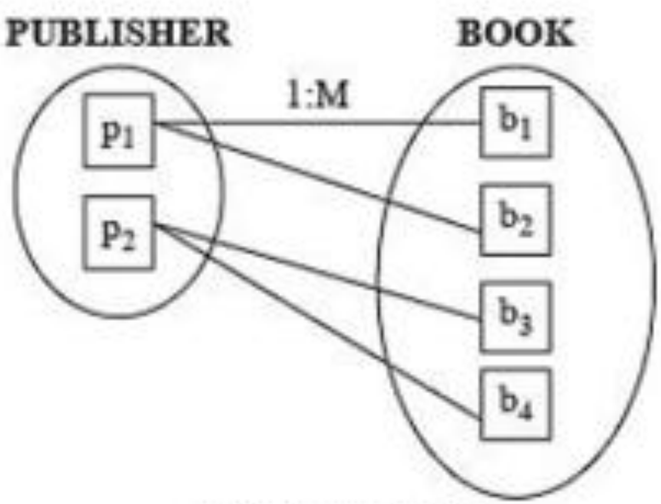


(b) One-to-many

One **Publisher** publishes many books.

Mapping Cardinality (One to Many)





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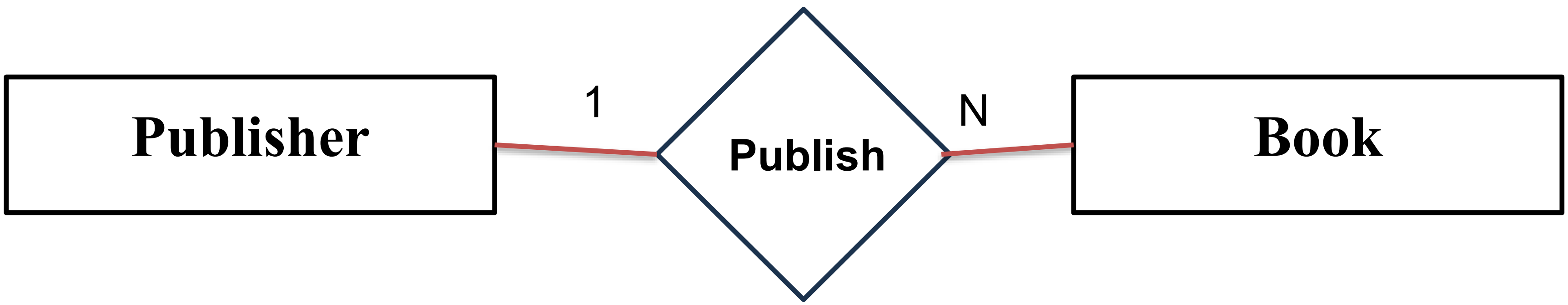

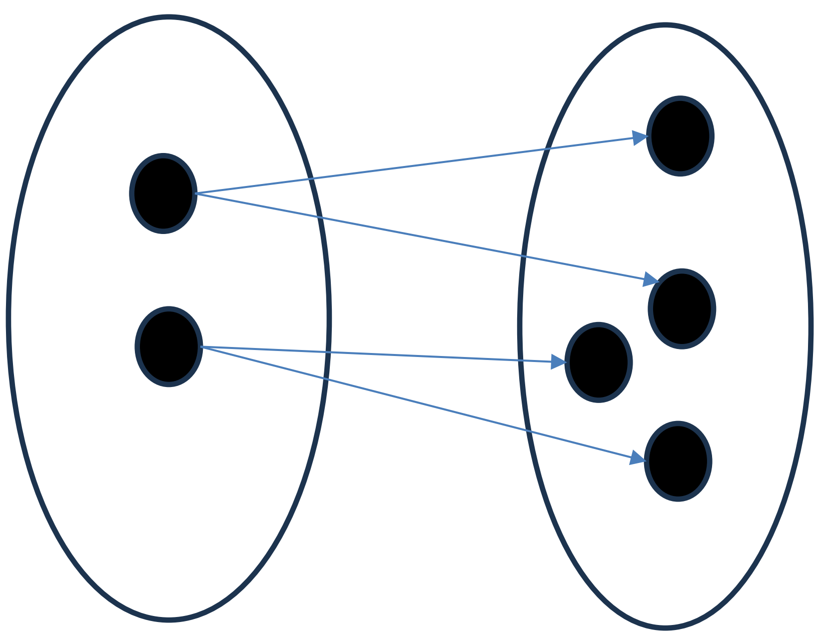

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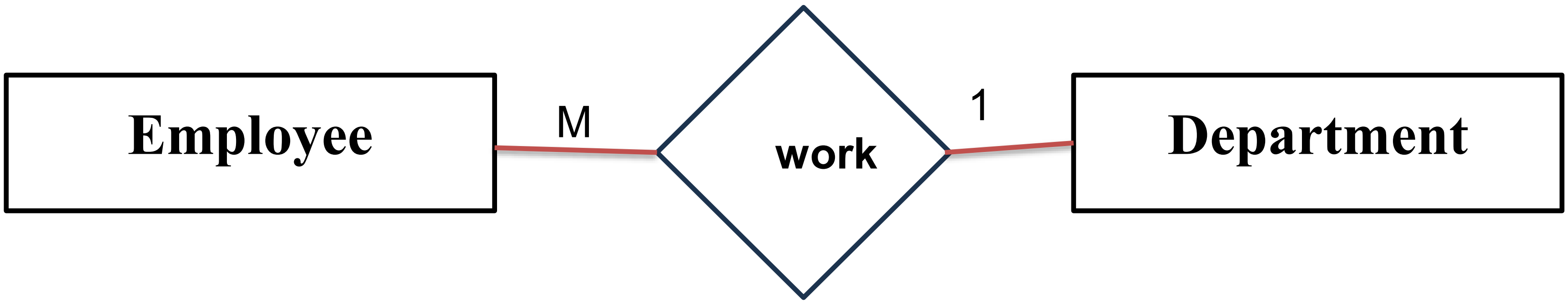
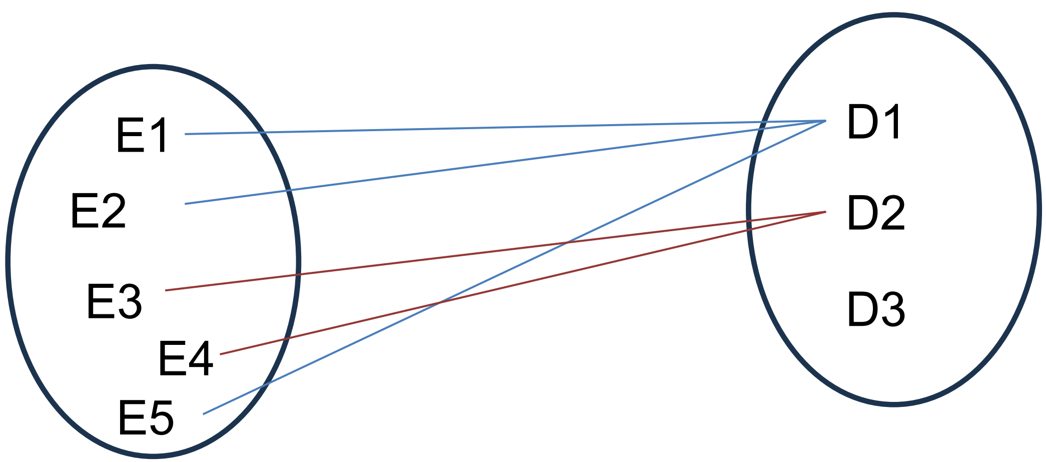
Mapping Cardinality (One to Many)



This indicates that an department may has many employe, but a employee work in at most one department



Mapping Cardinality
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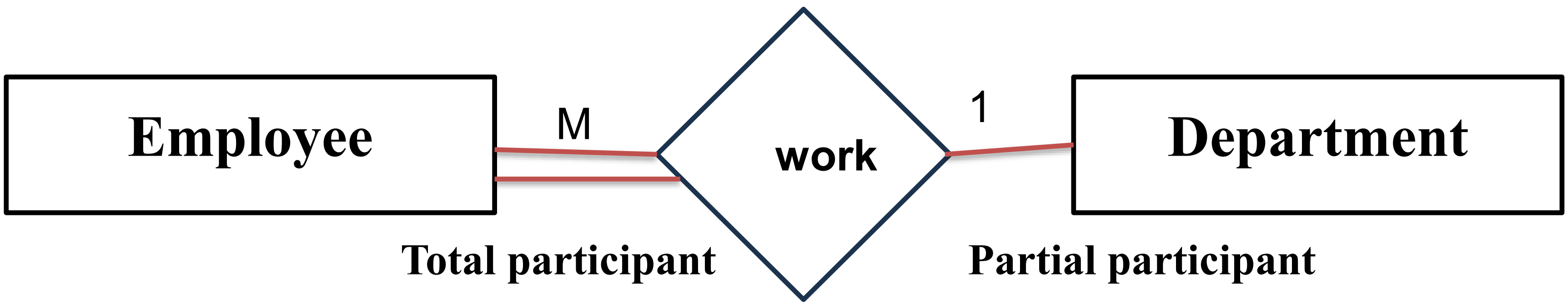

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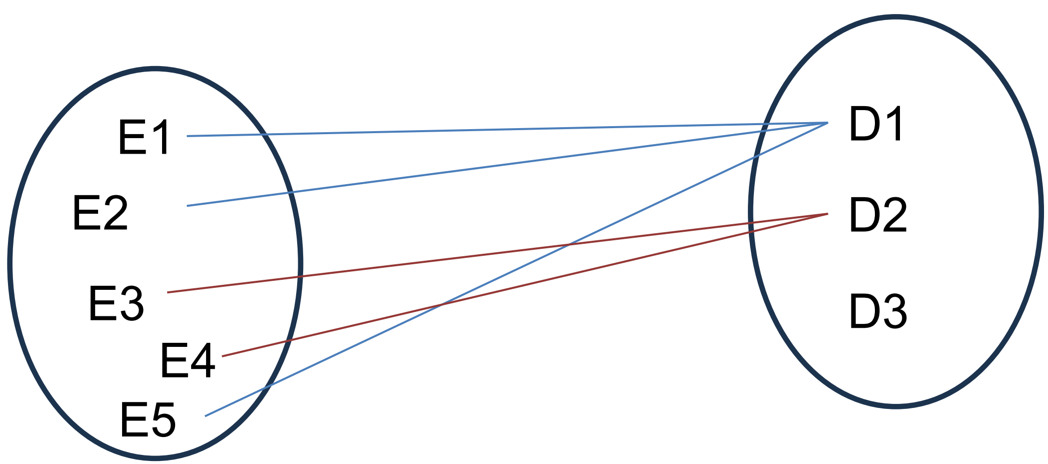
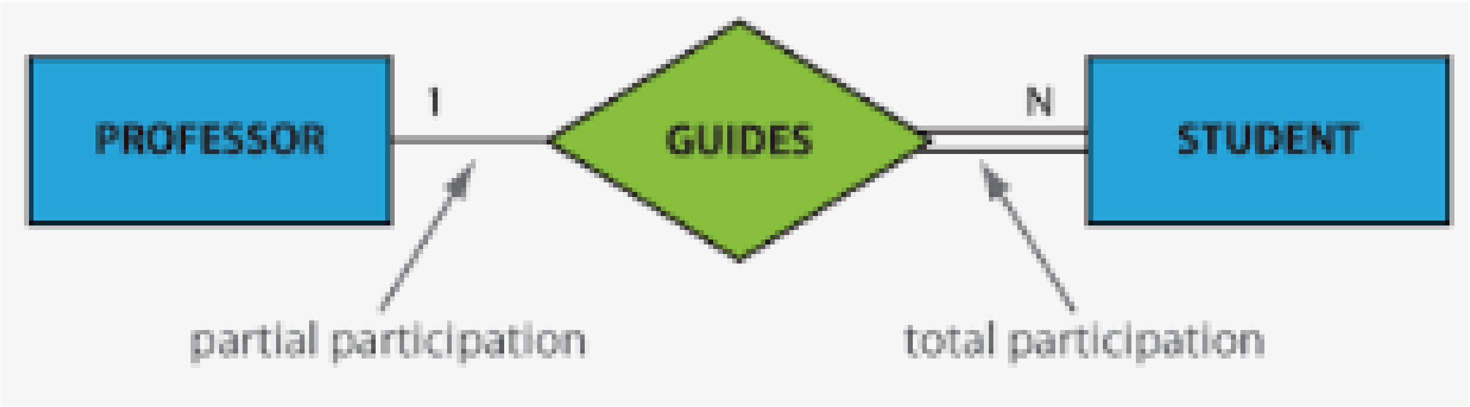
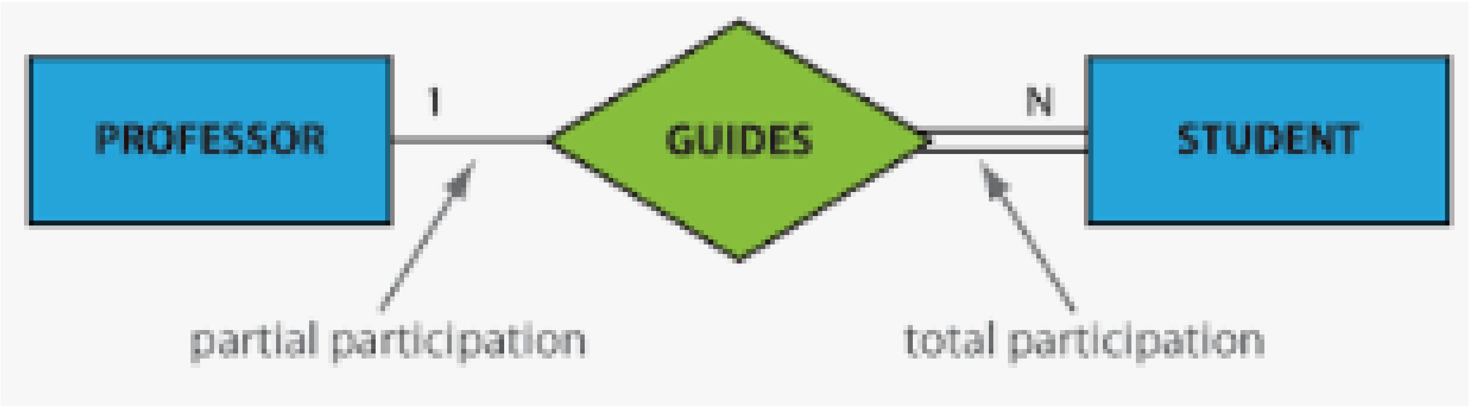


Mapping Cardinality (one to Many)



Mapping Cardinality (one to Many)

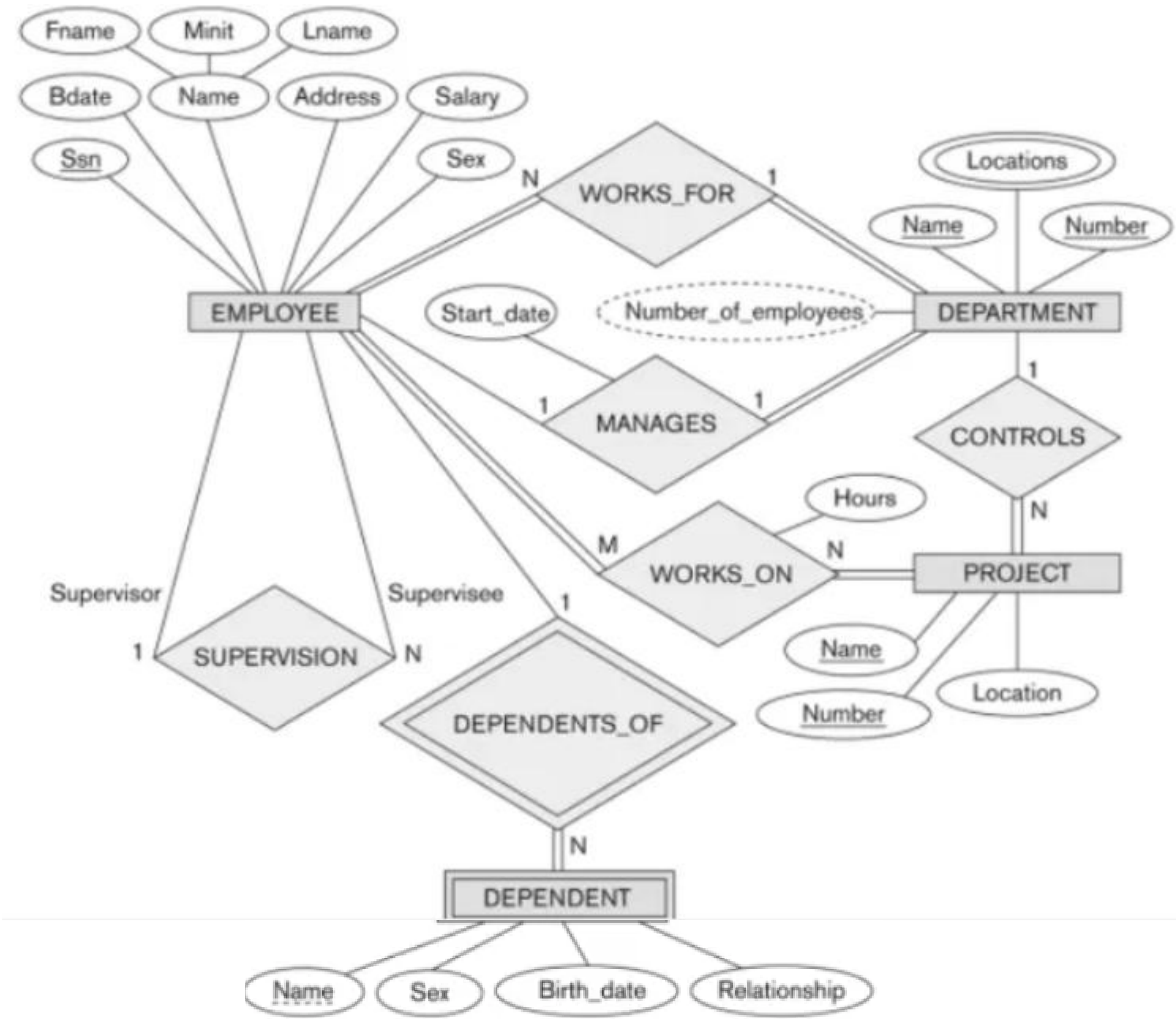


ERD interpretation

One professor can guide many students.

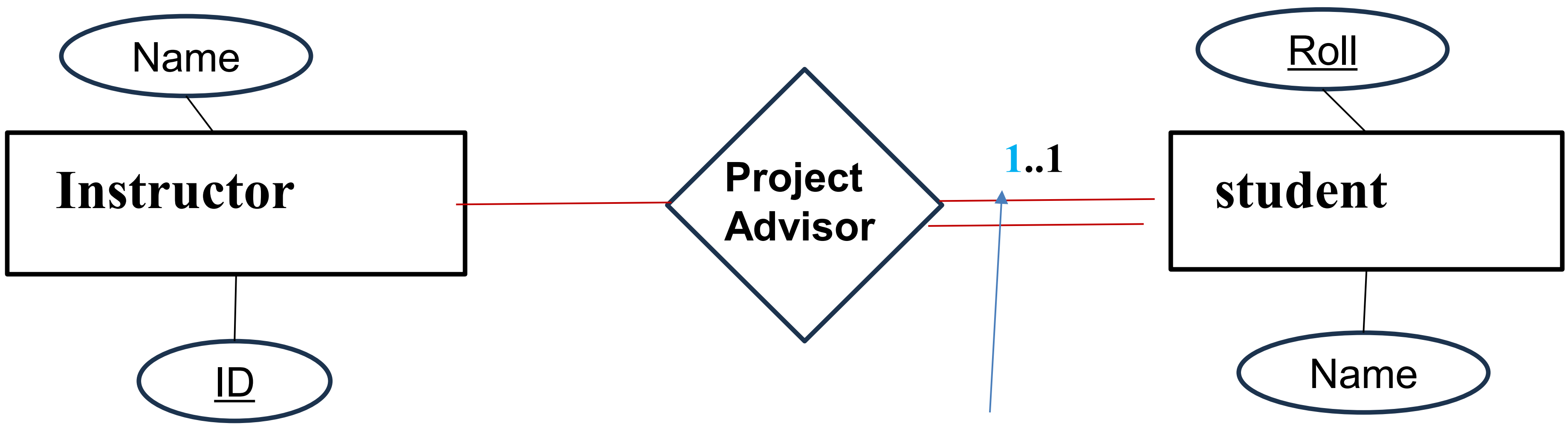
While it is not mandatory for a professor to guide every student, each student must be associated with a professor for guidance

Mapping Cardinality



Notation for Expression More Complex Cardinality

- A line may have an association minimum and maximum cardinality, shown in the form *l..h* , where *l* is minimum, and *h* is maximum cardinality.



Every student must have one instructor: minimum

Notation for Expression More Complex Cardinality

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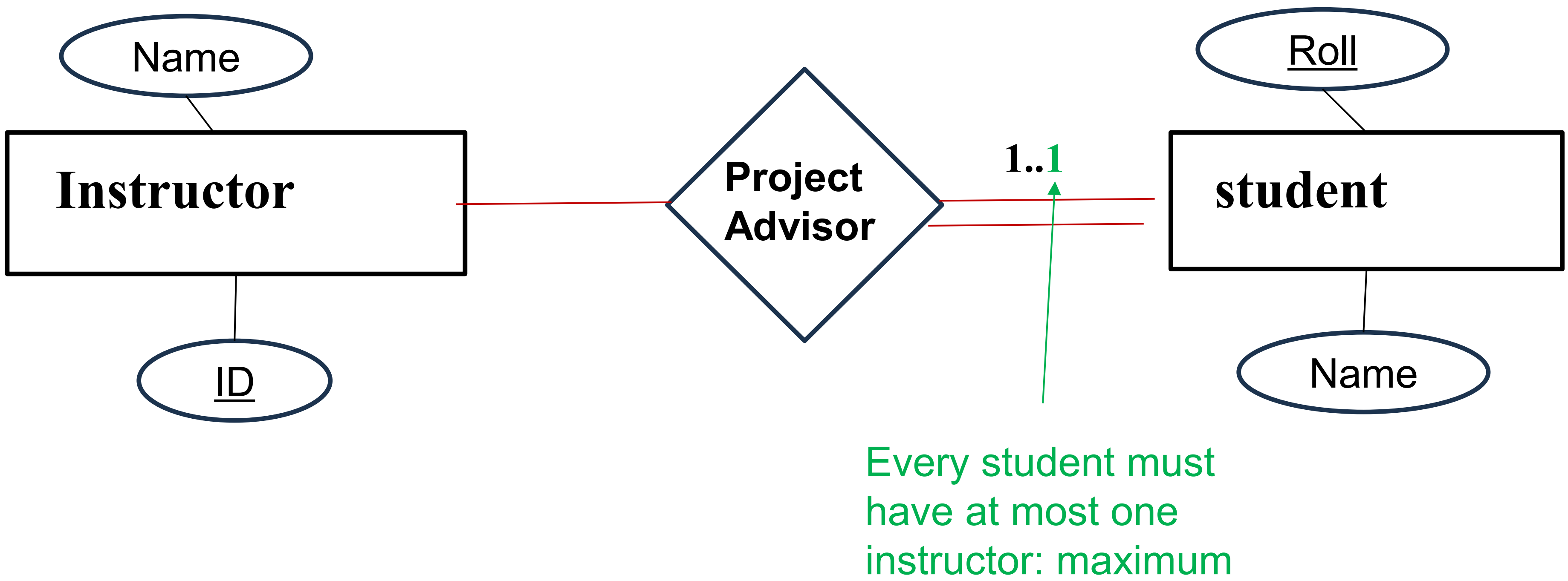

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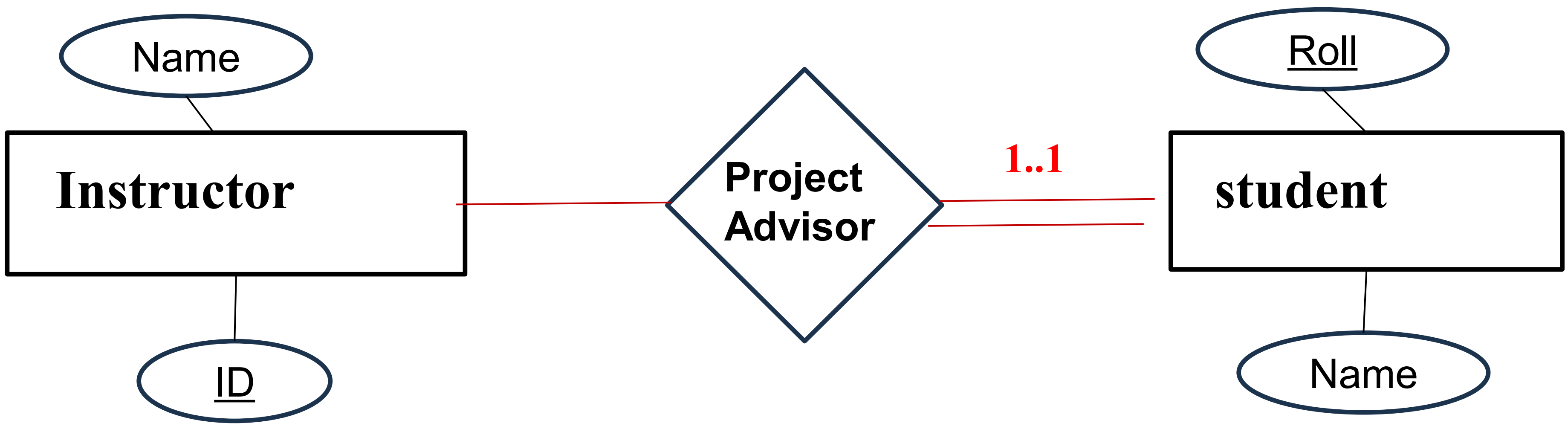

Every student must have one and only one instructor

Notation for Expression More Complex Cardinality

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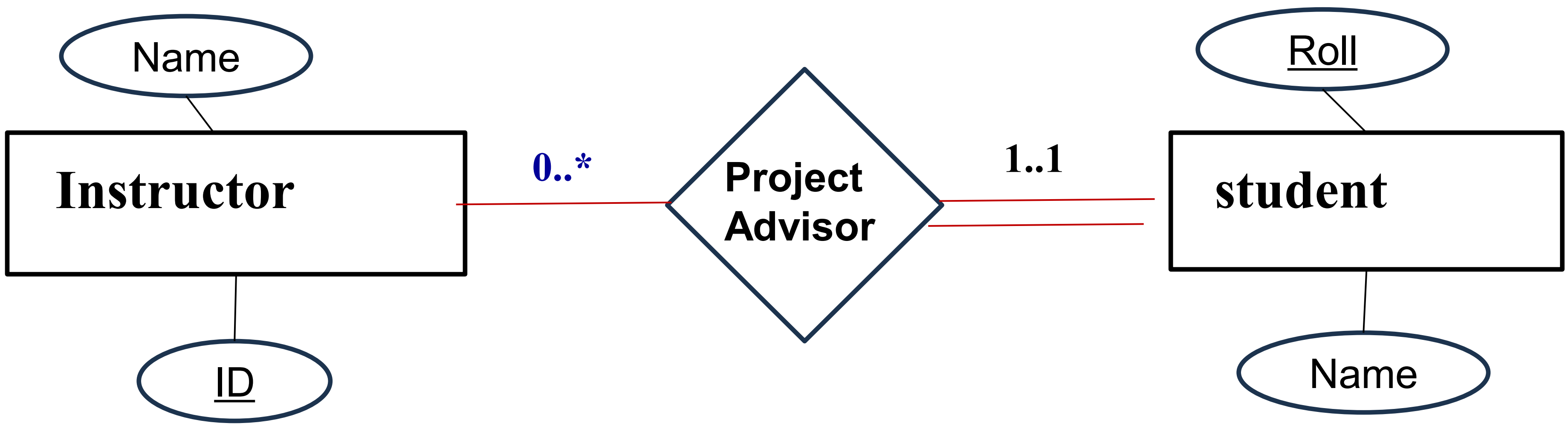

One Instructure may or may not guide a student, he/she can advise any number of student there is no upper limit

Notation for Expression More Complex Cardinality

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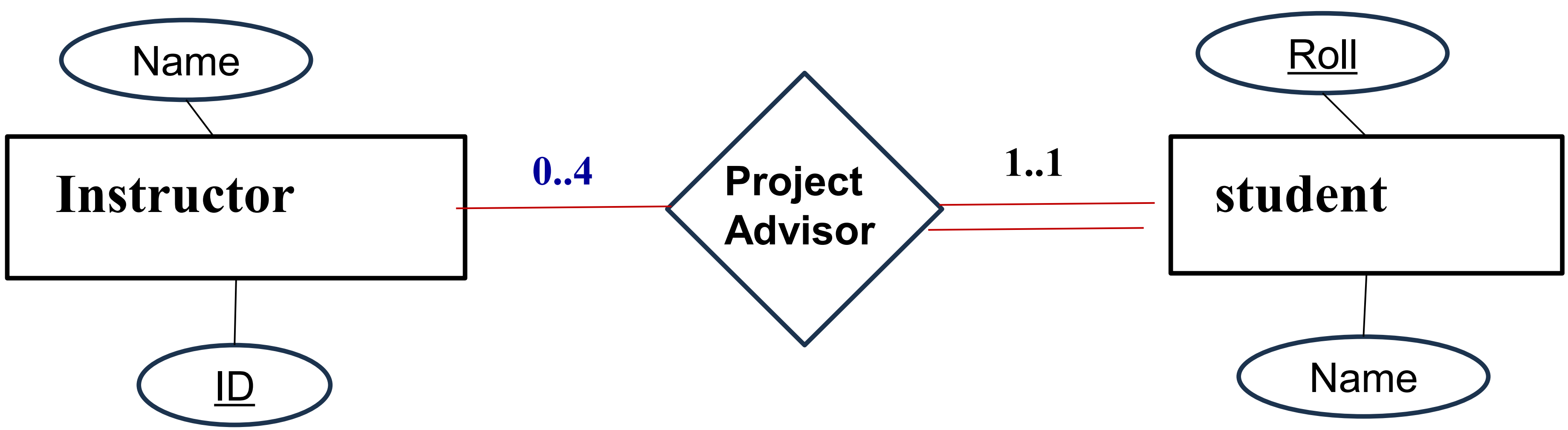

One Instructure may or may not guide a student, he/she can guide maximum (at most) 4 students

Notation for Expression More Complex Cardinality

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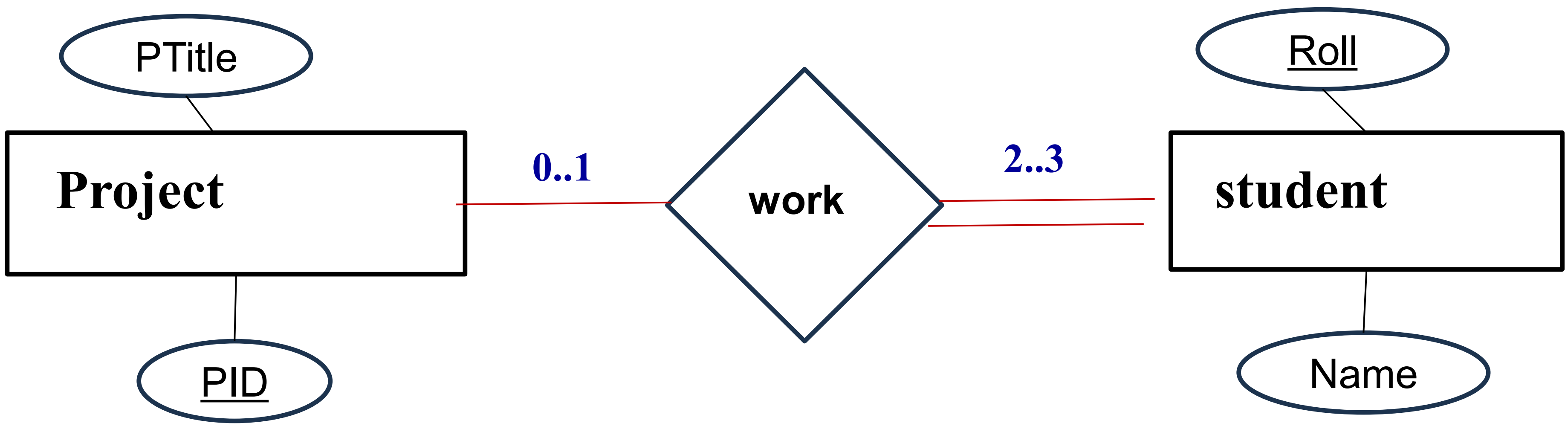
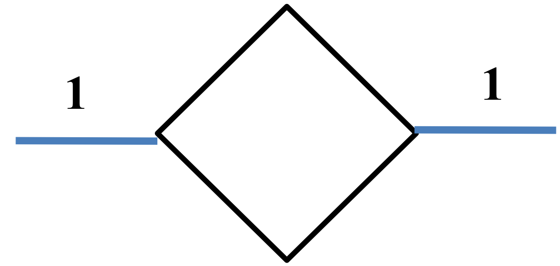


A project may or may not be undertaken by a student, and each project can only be associated with one student. A student must be involved in at least two projects and can participate in a maximum of three projects.

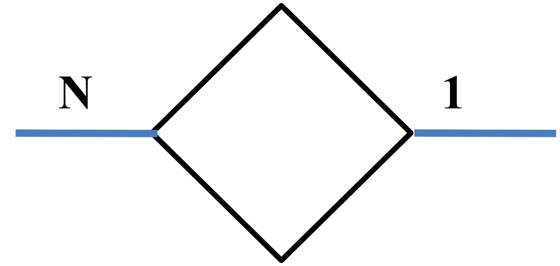
E-R Diagram (Symbol)

Cardinality

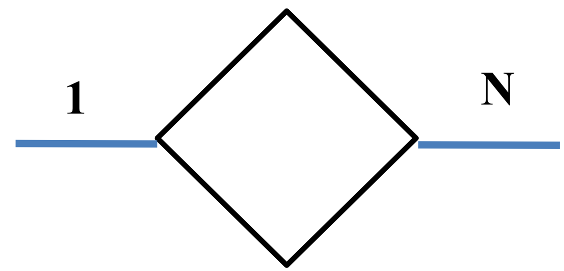
One to one



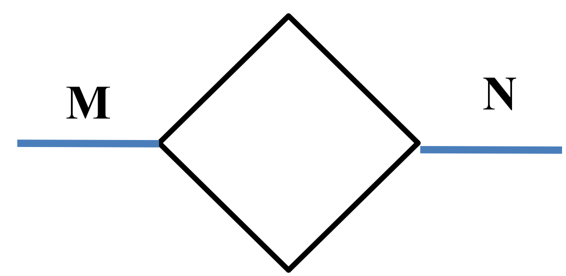
Many to one



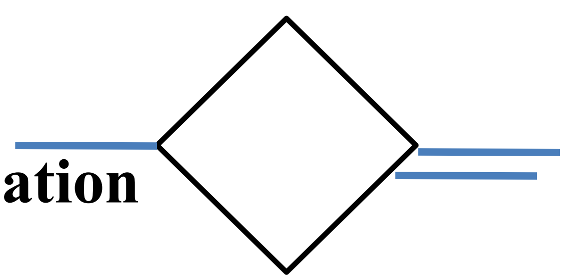
one to Many



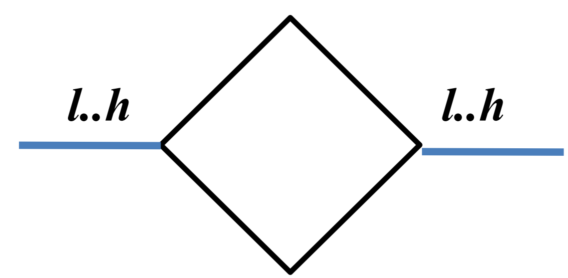
Many to Many



Total Participation



Complex Cardinality



E-R Diagram

Draw an Entity-Relationship (E-R) diagram for a university. The university has multiple departments, each of which can have numerous professors. One professor belong to one and only one department.

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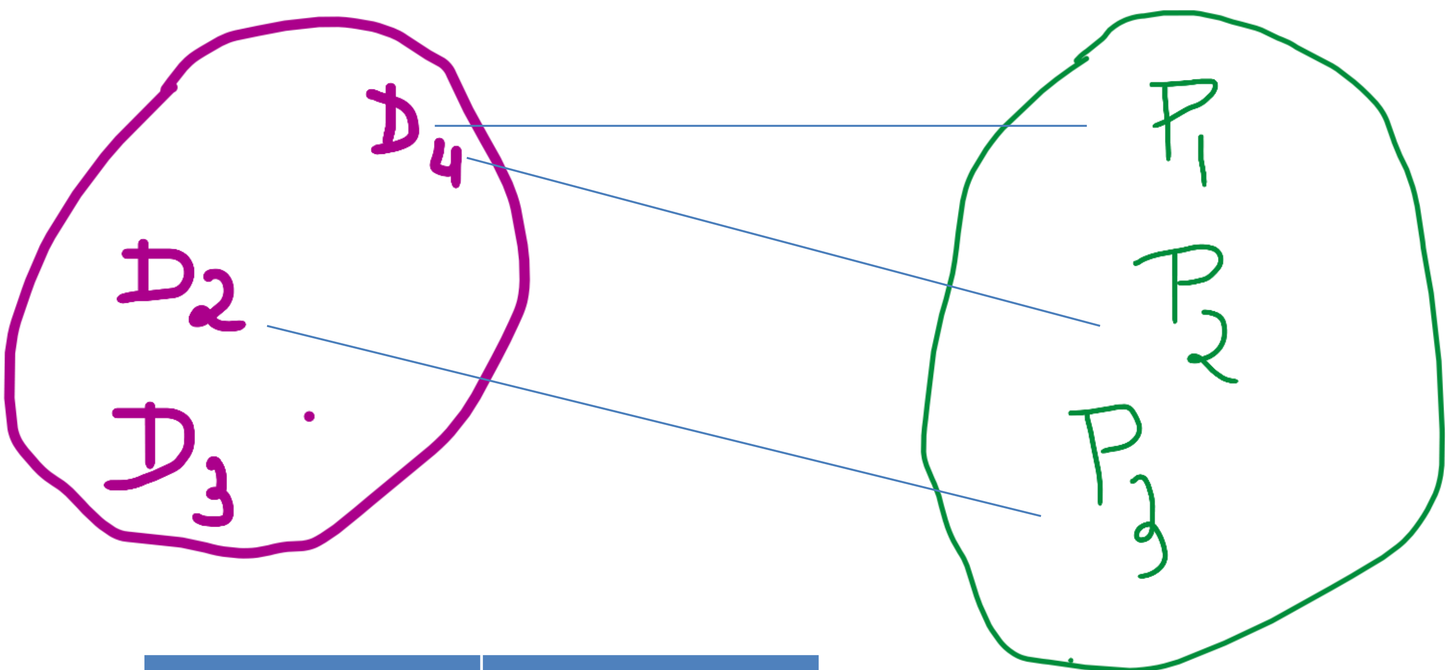
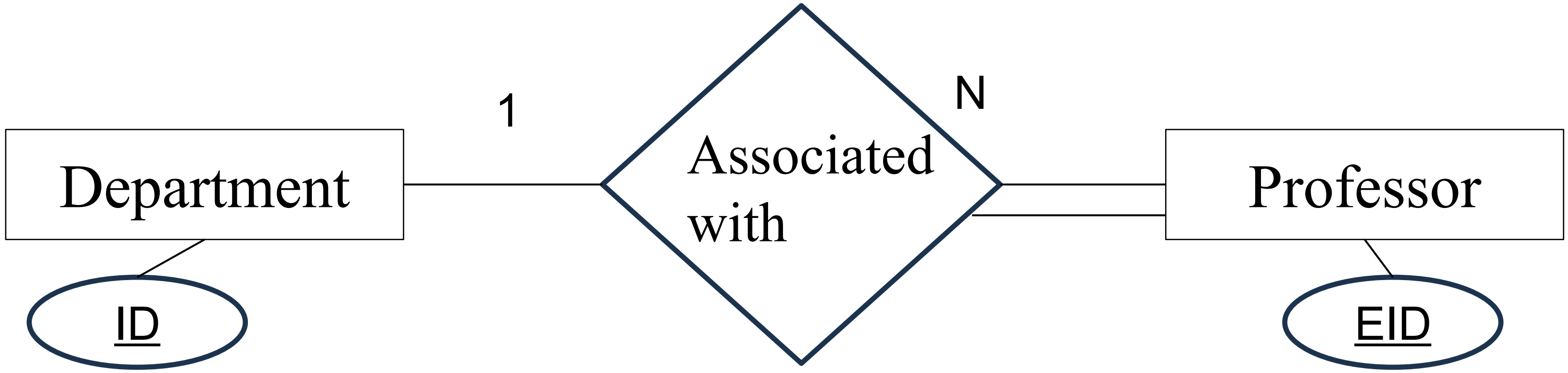


D_ID	P_ID
D4	P1
D4	P2
D2	P3

Many to One

E-R Diagram

Draw an Entity-Relationship (E-R) diagram for a university. There are multiple departments, each of which can have numerous professors. One professor belong to one and only one department.

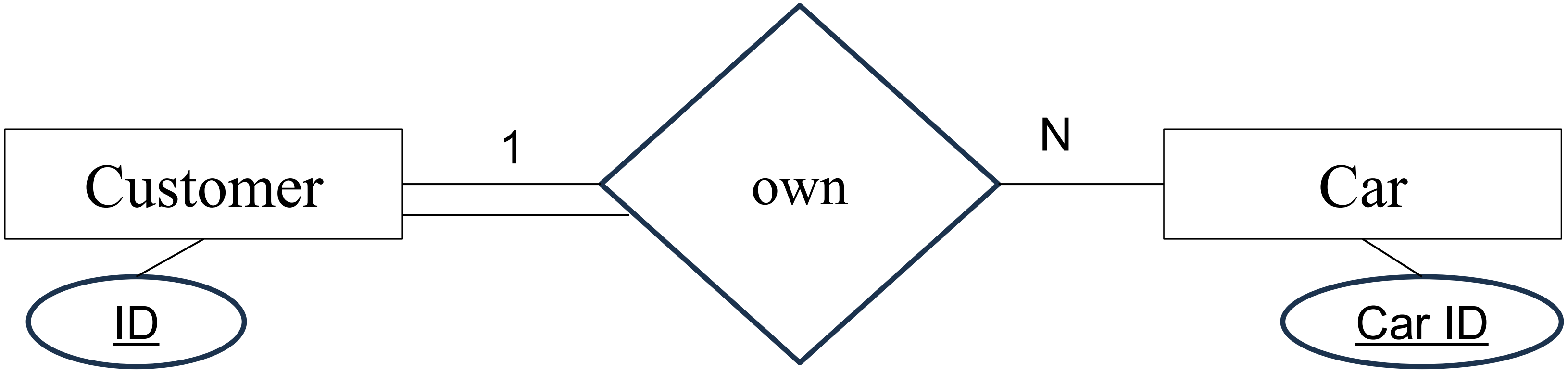


E-R Diagram

Construct an ER diagram for a car company (only sale car) whose customer own one or more cars each.

E-R Diagram

Construct an ER diagram for a car company (only sale car) whose customers own at least one car.



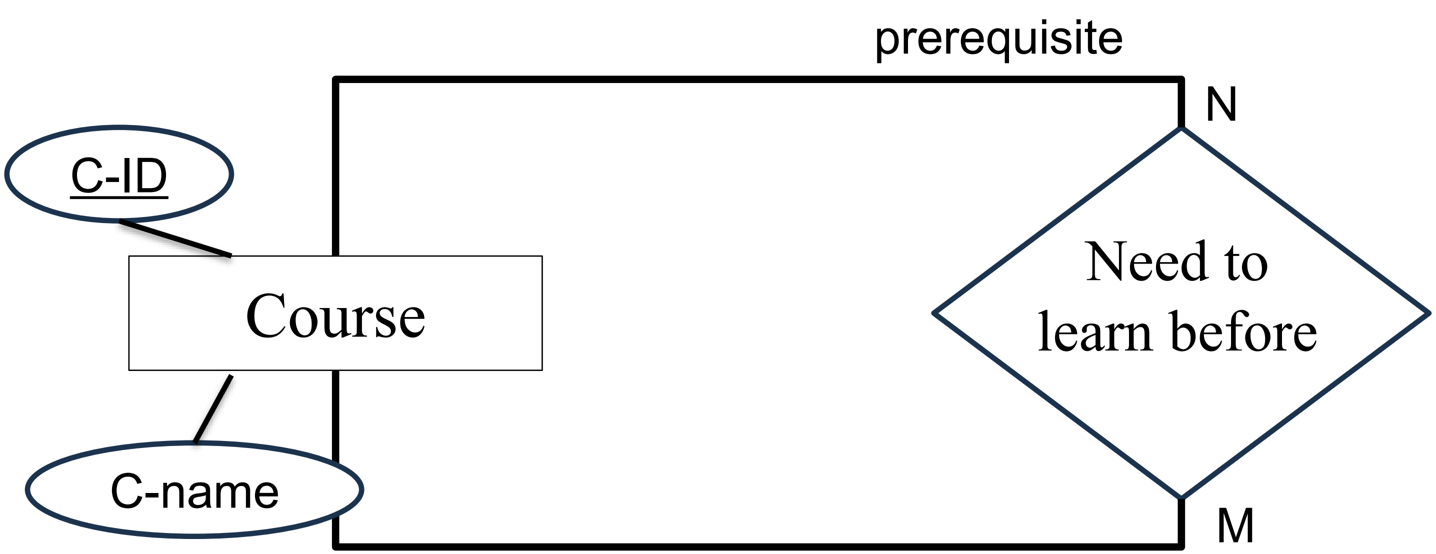
Assumption:
If a person purchases a car, only then are they considered a customer

E-R Diagram

Design an Entity-Relationship (E-R) diagram for a university course catalog system that includes courses and their prerequisite courses.

E-R Diagram

Design an Entity-Relationship (E-R) diagram for a university course catalog system that includes courses and their prerequisite courses.



Extended E-R Diagram feature

- Specialization
- Generalization
- Higher- and lower-level entity set
- Attribute Inheritance
- Aggregation

Extended E-R Diagram feature

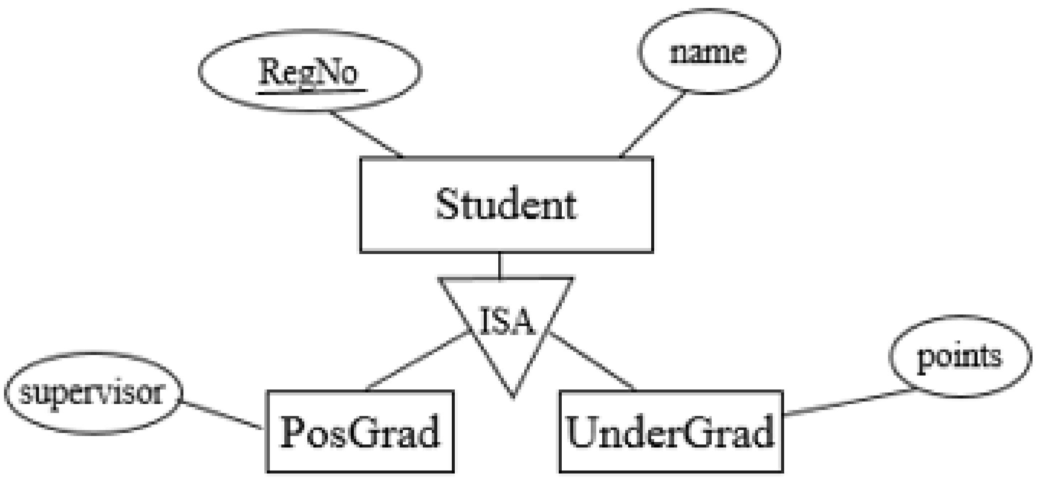
- **Specialization**

Specialization in the context of an Entity-Relationship (ER) model is a top-down approach where an entity is divided into sub-entities based on some distinguishing characteristics.

The process of designating subgrouping within an entity called specialization

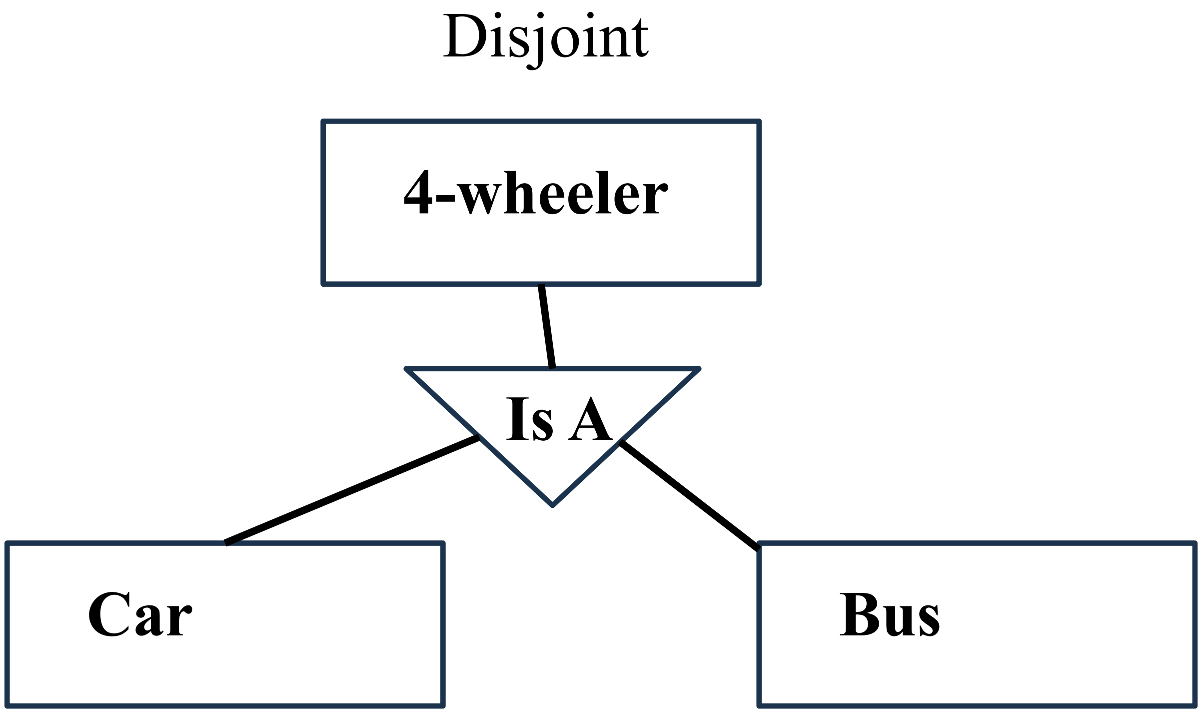
Superclass: An entity type that represents a general concept at a high level.

Subclass: An entity type that represents a specific concept at lower levels.



Extended E-R Diagram feature

- **Specialization**
  - Disjoint (mutually exclusive)
  - Overlapping



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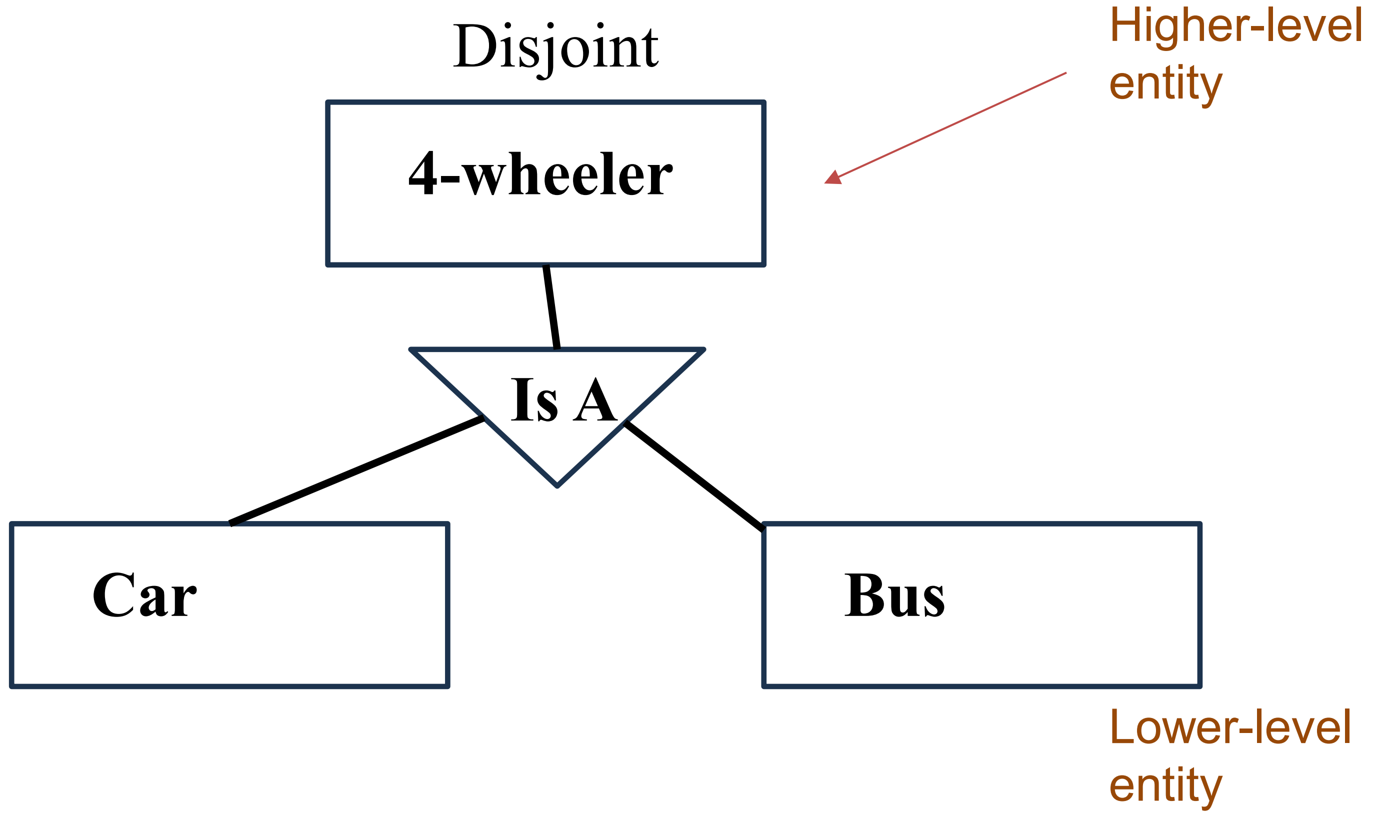

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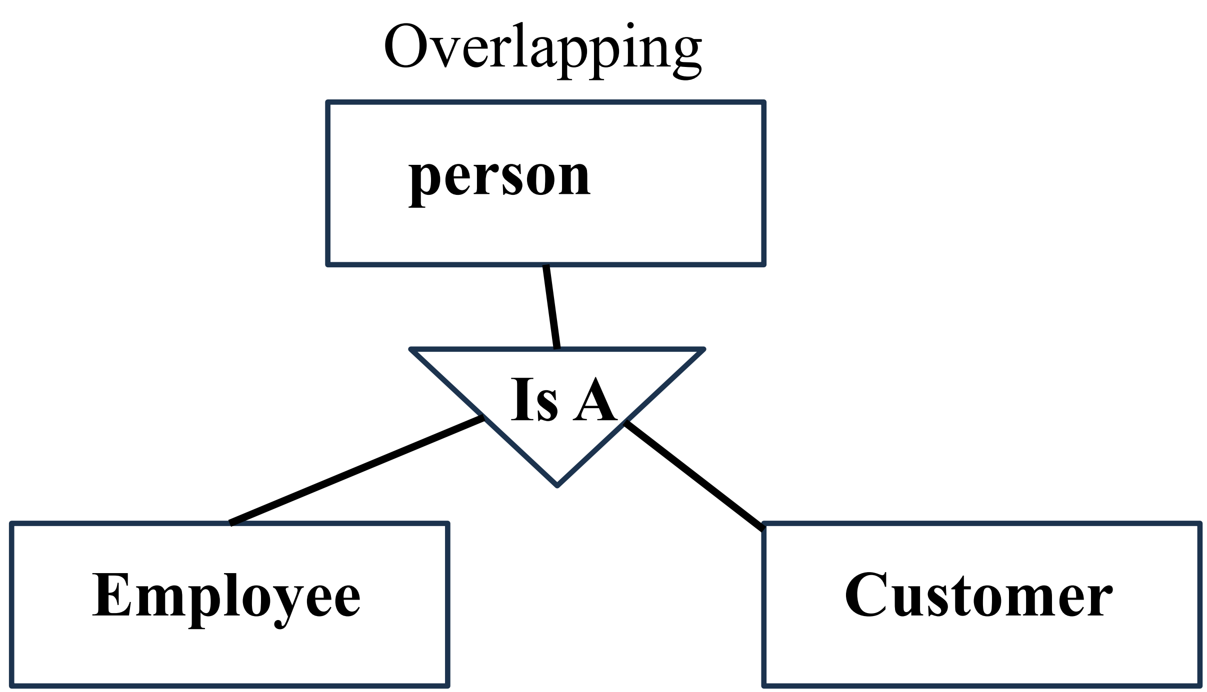
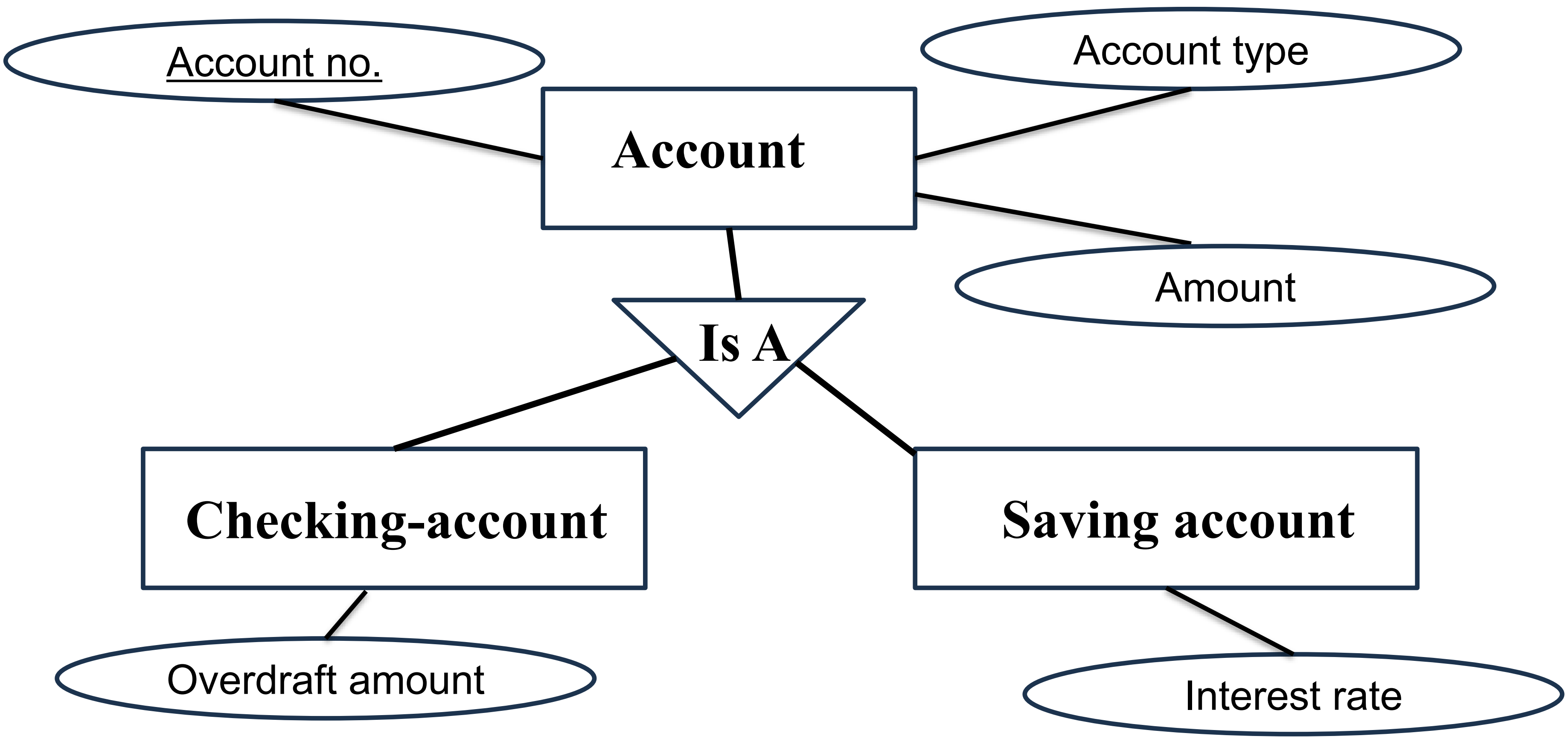


Extended E-R Diagram feature

- **Specialization**
  - Condition-defined
  - User-defined

Condition-defined

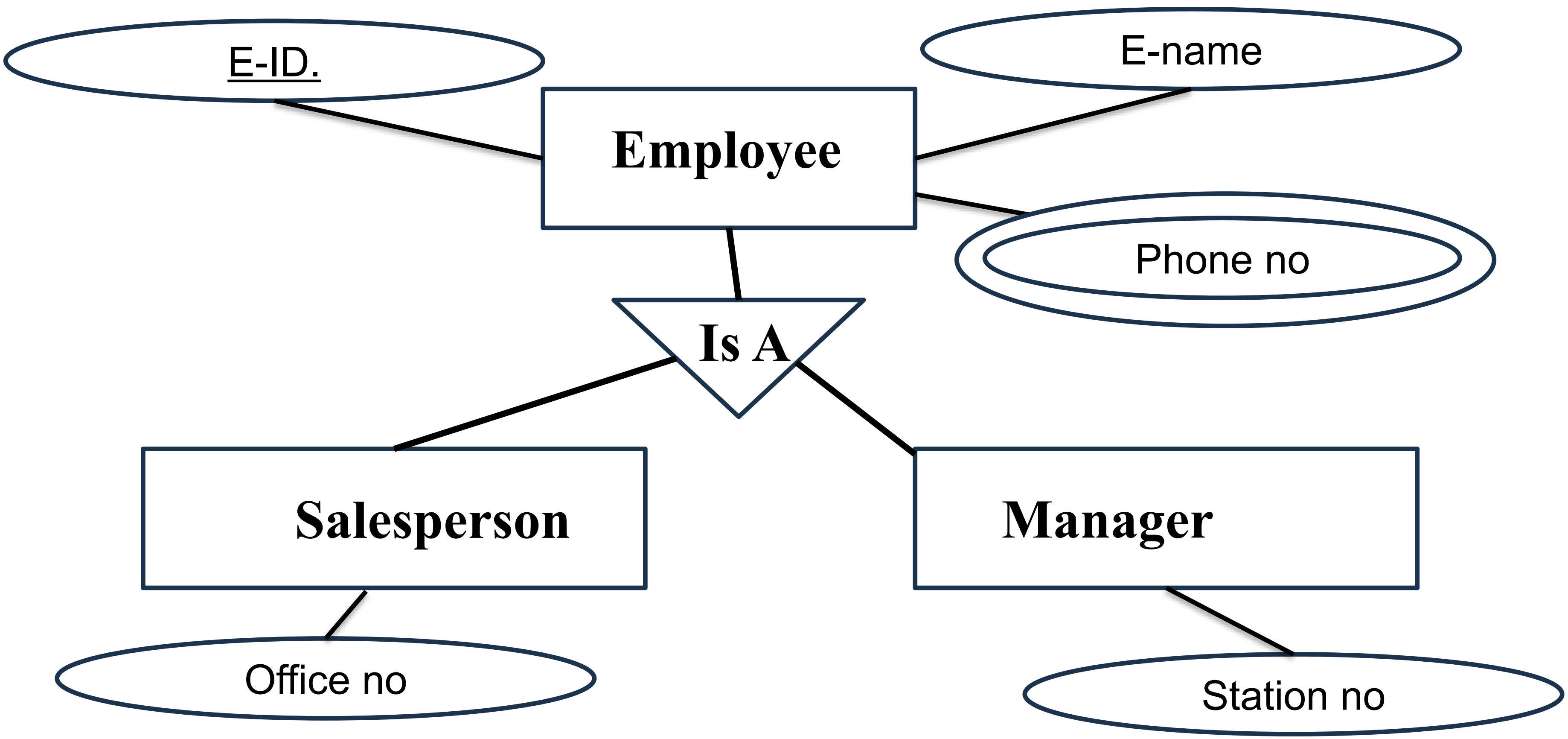


Extended E-R Diagram feature

- **Specialization**
  - Condition-defined
  - User-defined

User-defined: Based on the requirement or discretion of the user/system designer

The designer decides to create subclasses to better model, even if no strict attribute condition enforces it



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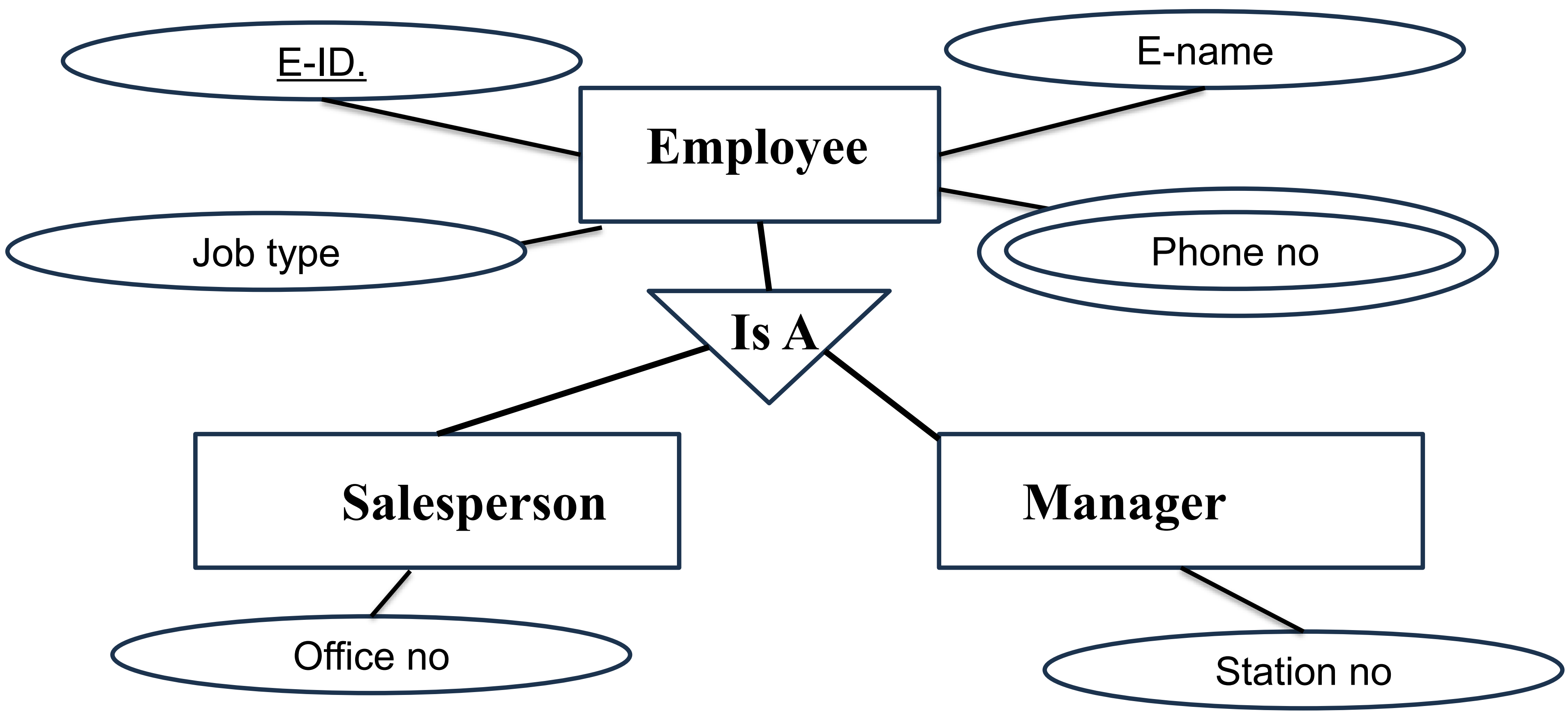
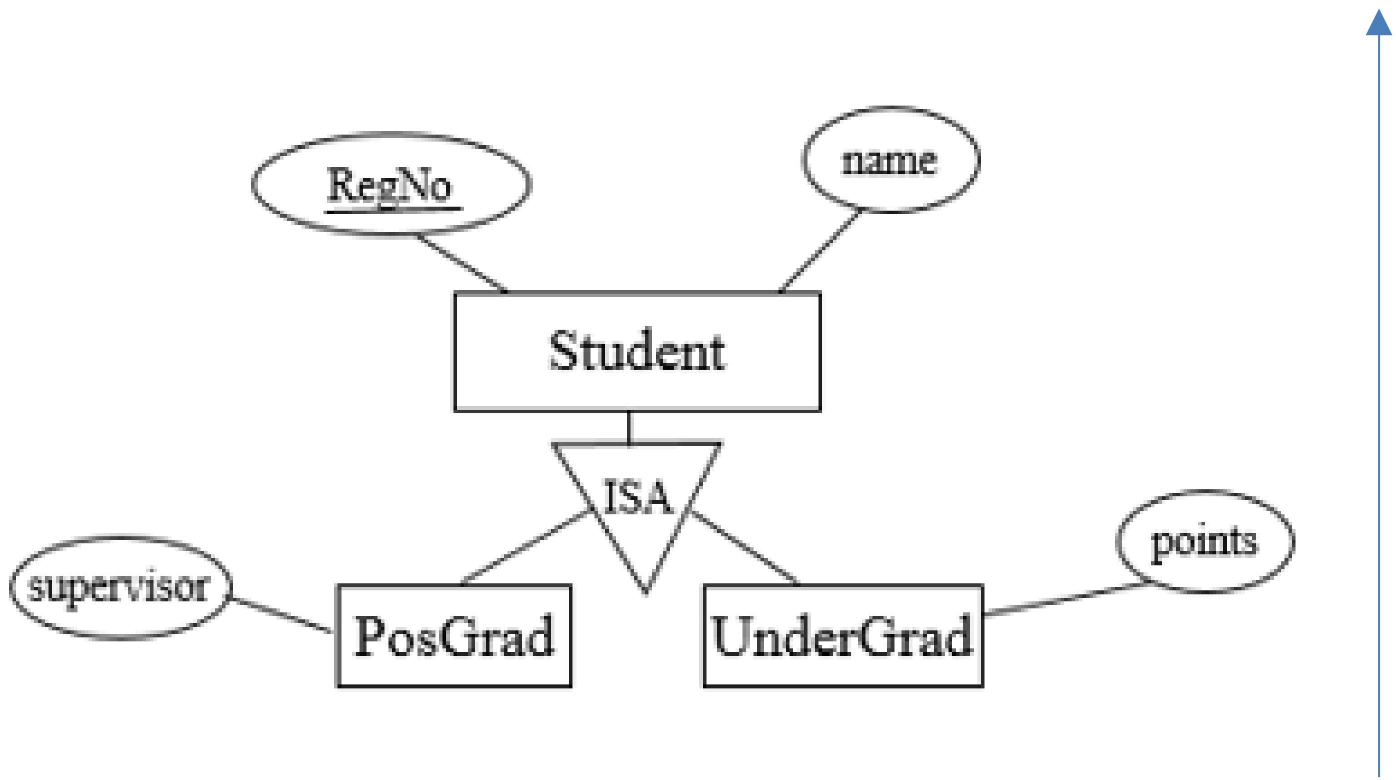


Extended E-R Diagram feature

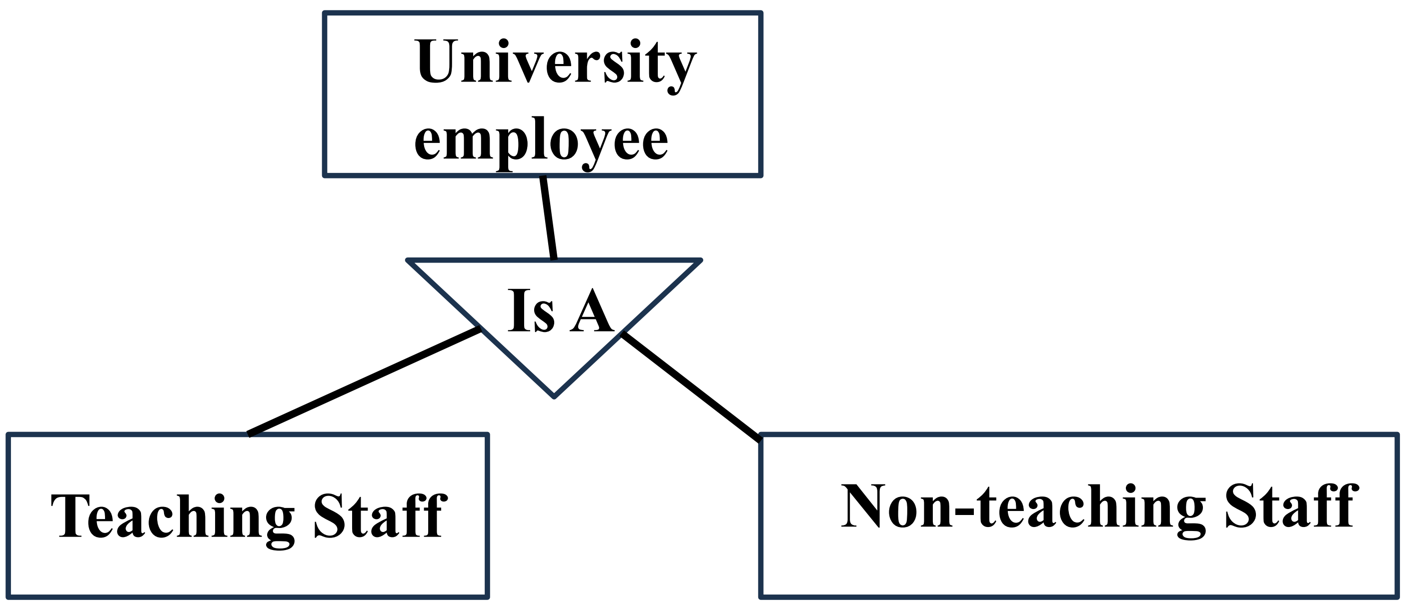
- **Generalization**

In ER modeling, generalization is the process of abstracting common characteristics from multiple entities to create a generalized, higher-level entity.



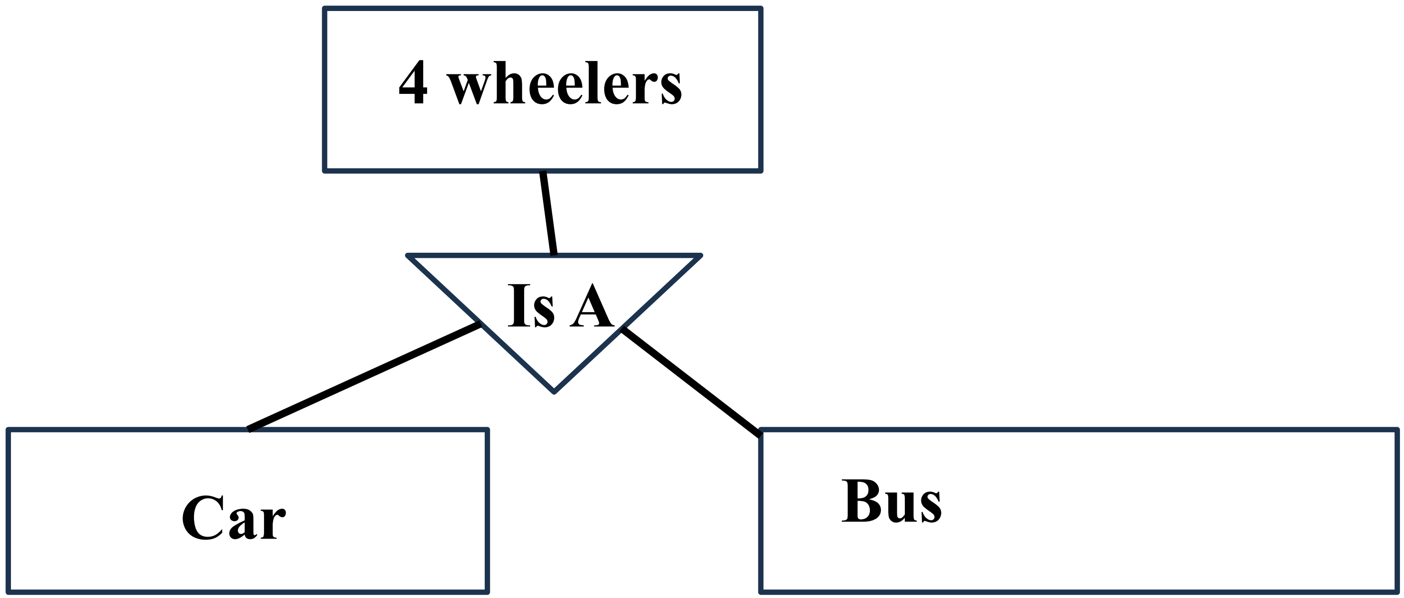
Extended E-R Diagram feature

- **Generalization and Specialization**
  - **Total generalization or total specialization:** each higher-level entity must belong to a lower-level entity-set



Extended E-R Diagram feature

- **Generalization and Specialization**
  - **Partial generalization or partial specialization:** Some higher level entity may not be belong to any lower-level entity



Extended E-R Diagram feature

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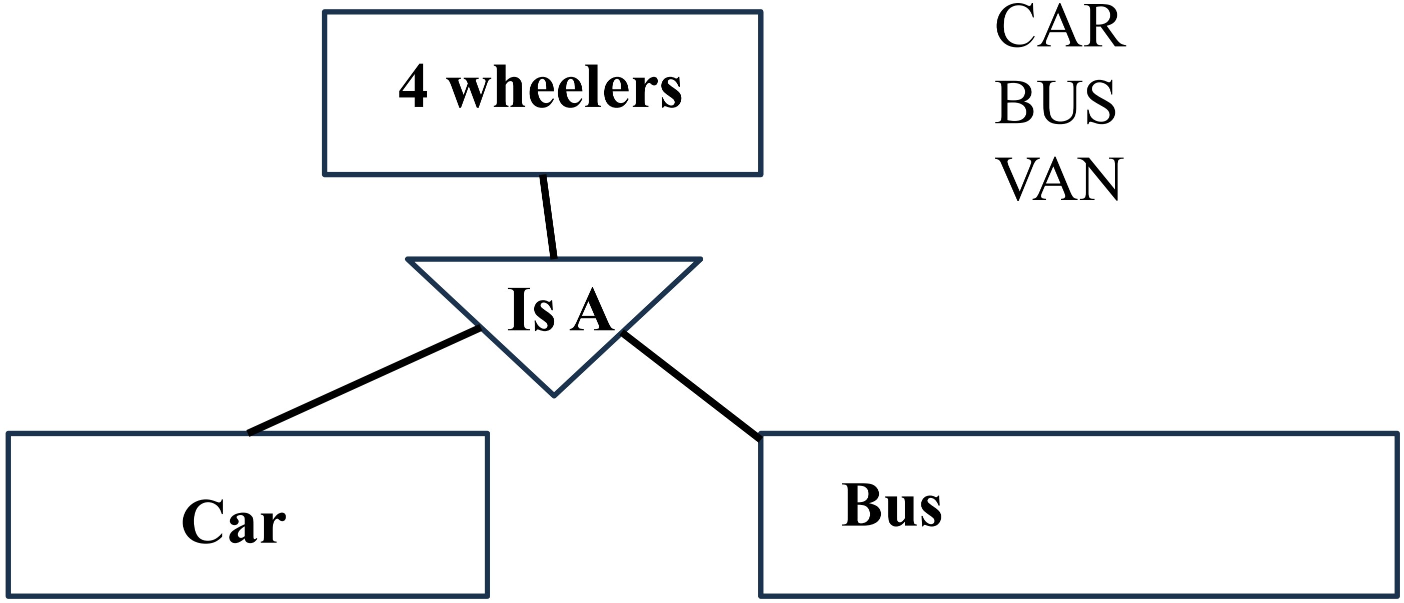
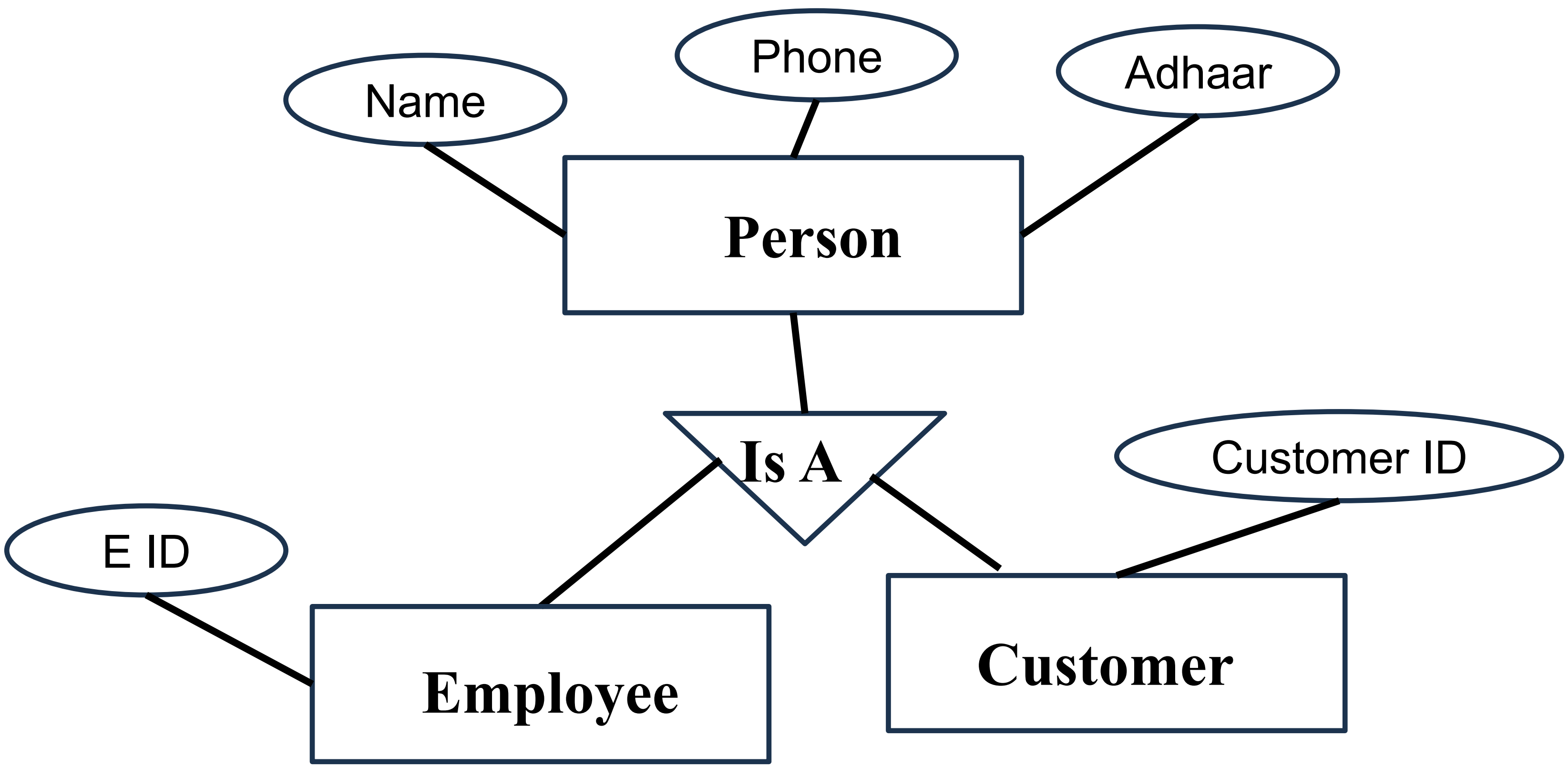


CAR
BUS
VAN

Extended E-R Diagram feature

Attribute Inheritance



Extended E-R Diagram feature

Draw an E-R diagram for an XYZ cricket league. There are many teams. Each team has a set of players. Players within a team can be categorized as active players or non-active players. A game is played between two teams (referred to as host team and guest team) and has a date and score.

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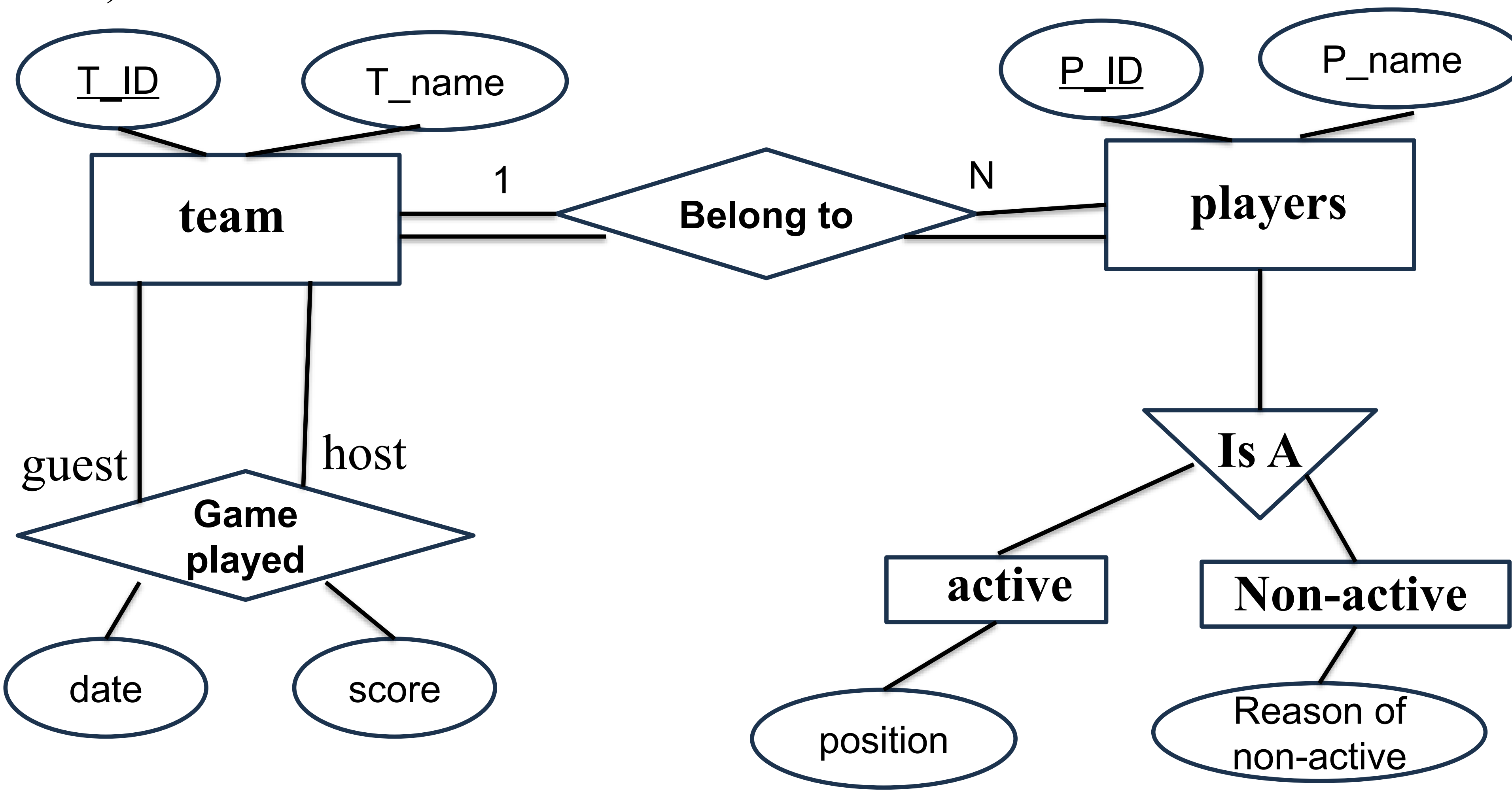

Extended E-R Diagram feature

Aggregation

Relationship-set participating in a relationship then aggregation is used

Extended E-R Diagram feature

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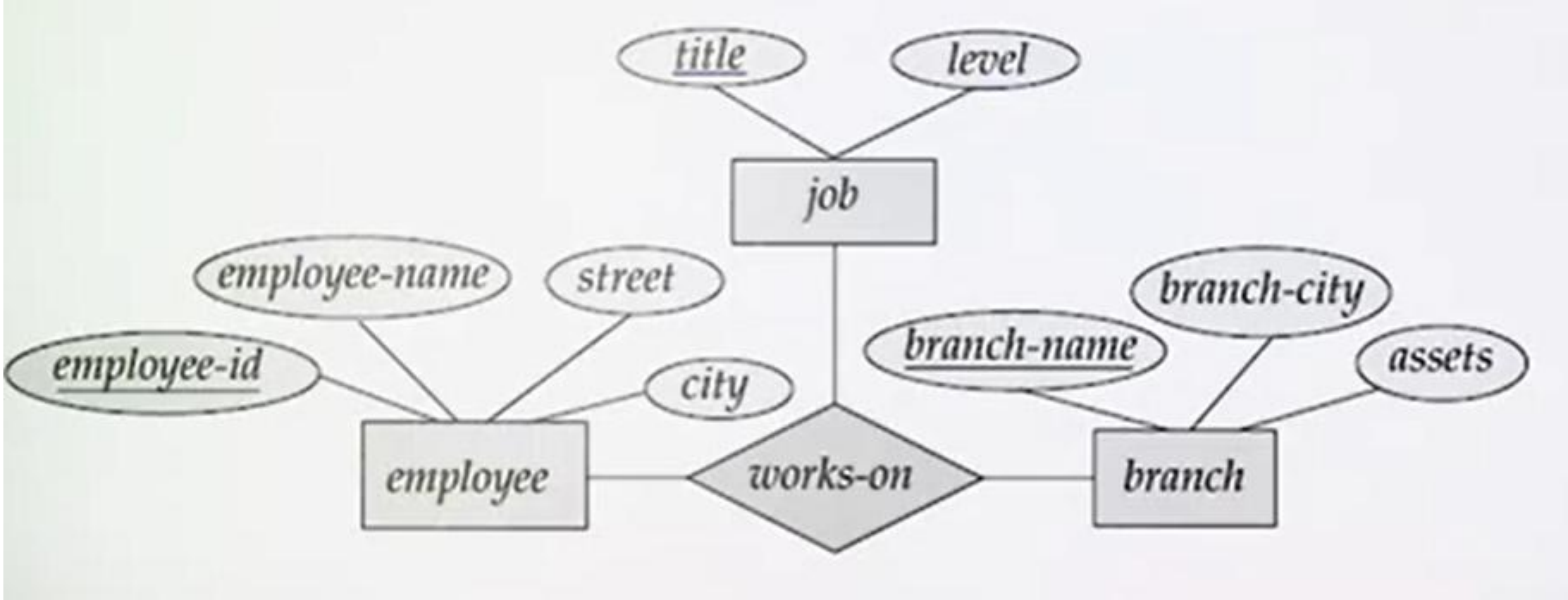
Relationship-set participating in a relationship then aggregation is used



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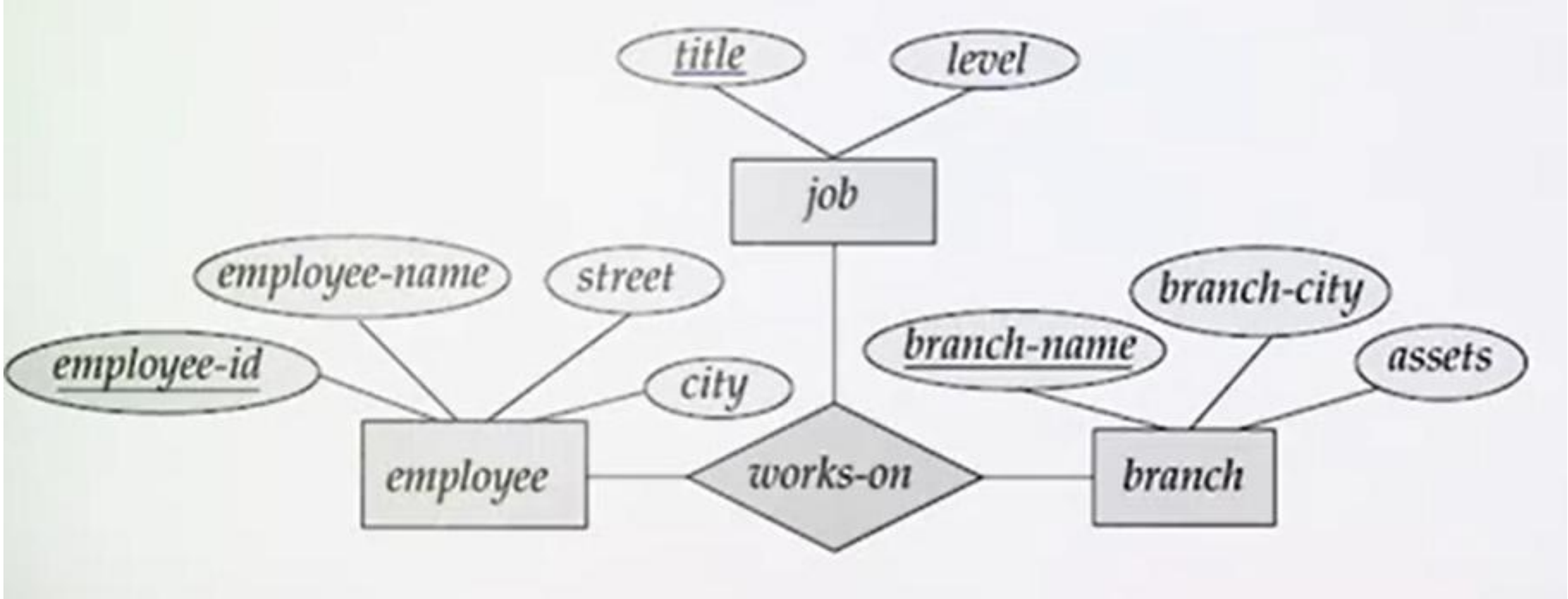

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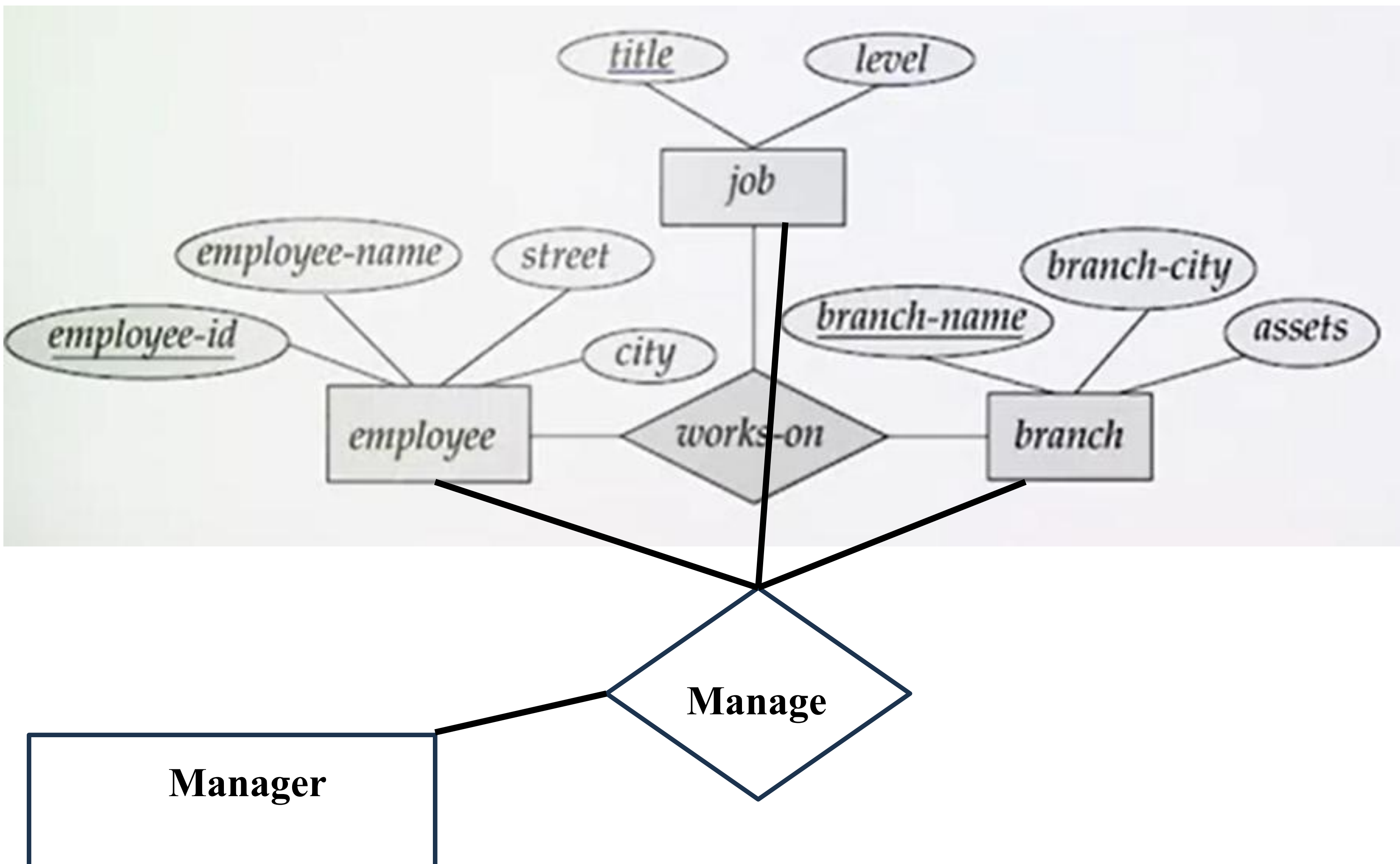

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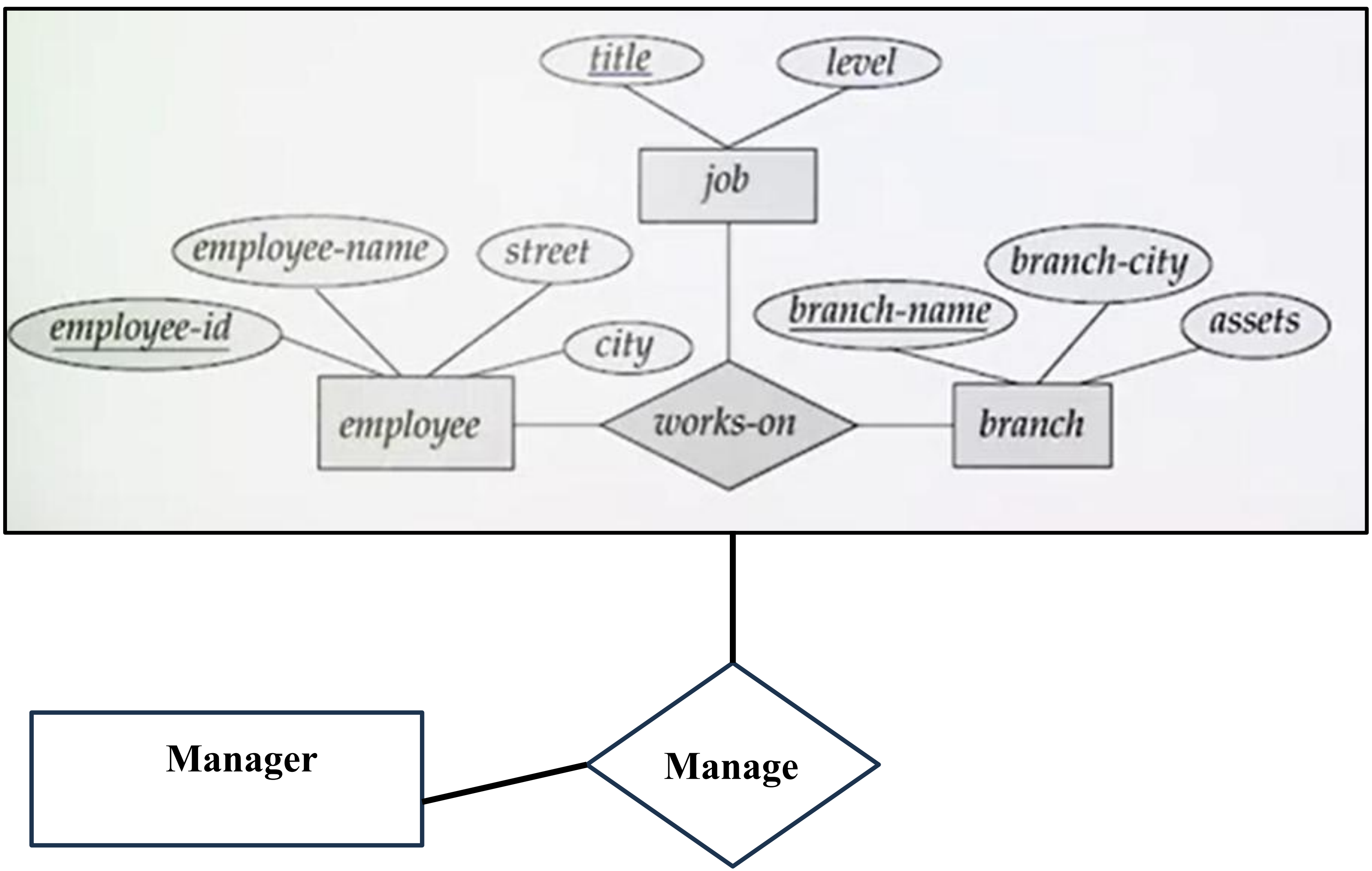
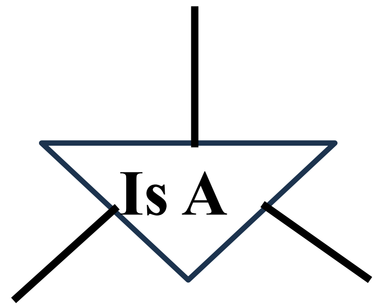
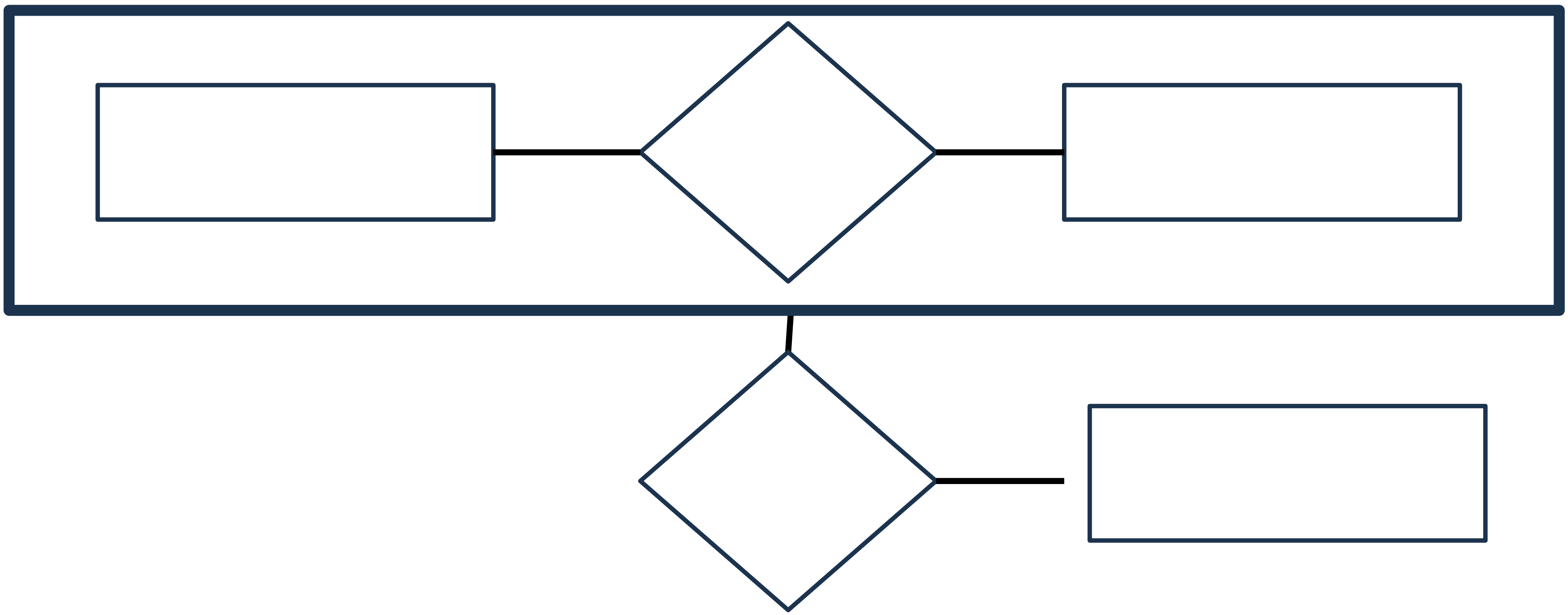


E-R Diagram (Symbol)

Specialization and Generalization and attribute inheritance



Aggregation



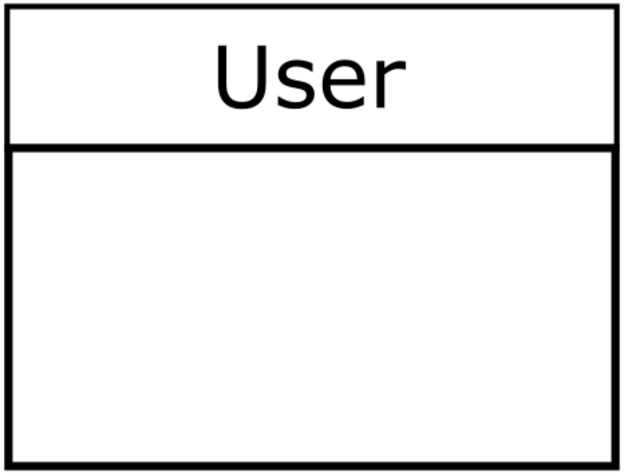
Different Type of E-R model

- **Chen Notation**
  Entity: Rectangle
  Attribute: Ovals connected to the entities
  Relationship: Represented by diamonds connected to the entities.
  Cardinality: 1:N; N:1, 1:1, M:N, *l..h*
  Total participation: *double line*

- **Crow's Foot Notation**
  Entity: Represented by rectangles.
  Attribute: Listed within the entity rectangles.
  Relationship: Lines with different symbols (crow's feet) to indicate cardinality.
  Cardinality:
    - A crow's foot symbol for "many"
    - A single line for "one"
    - A circle for optional
    - A perpendicular line for mandatory

Crow's Foot Notation

User is an entity-set



Crow's Foot Notation

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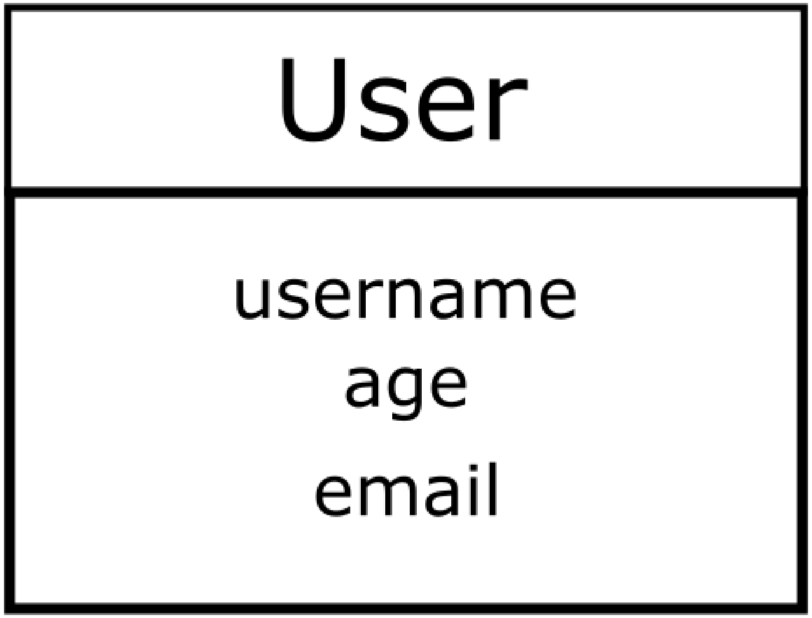
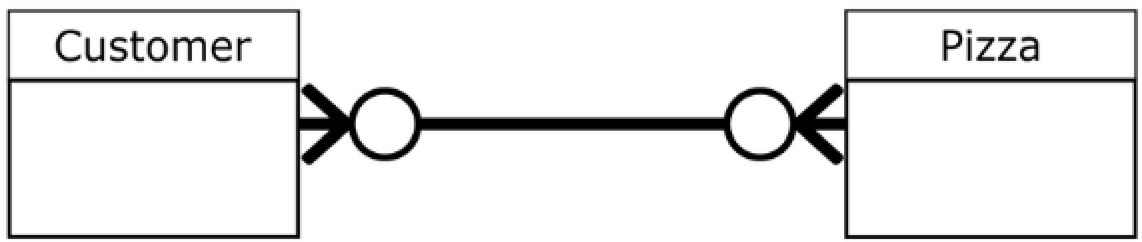


Crow's Foot Notation

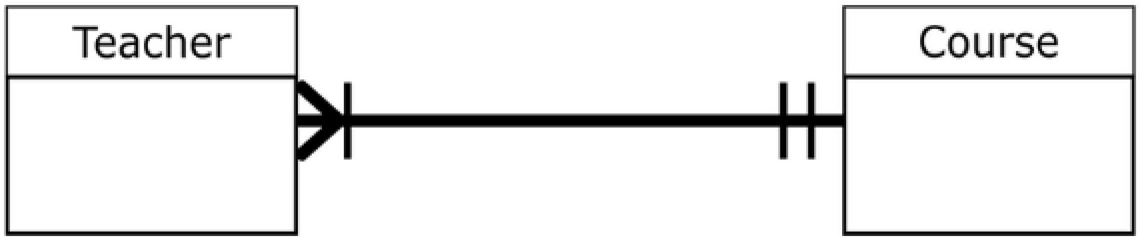
One course may buy many Pizzas, the type of same Pizza may be purchased by many customers.



Symbol	Meaning	Number
	One	N/A
	Many	N/A
	Mandatory-One	Exactly one
	Optional-One	Zero or one
	Mandatory-Many	One or More
	Optional-Many	Zero or more

Crow's Foot Notation

One course can be taught by one or many teachers



Unified Modeling Language (UML)

UML (Unified Modeling Language) notation is **commonly used in software engineering** and can also be adapted for **database modelling**, **specifically in ER (Entity-Relationship) diagrams**.

UML provides a standardized way to visualize the design of a system, and its notation can represent entities, attributes, relationships, and cardinality in database design.

Advantage:

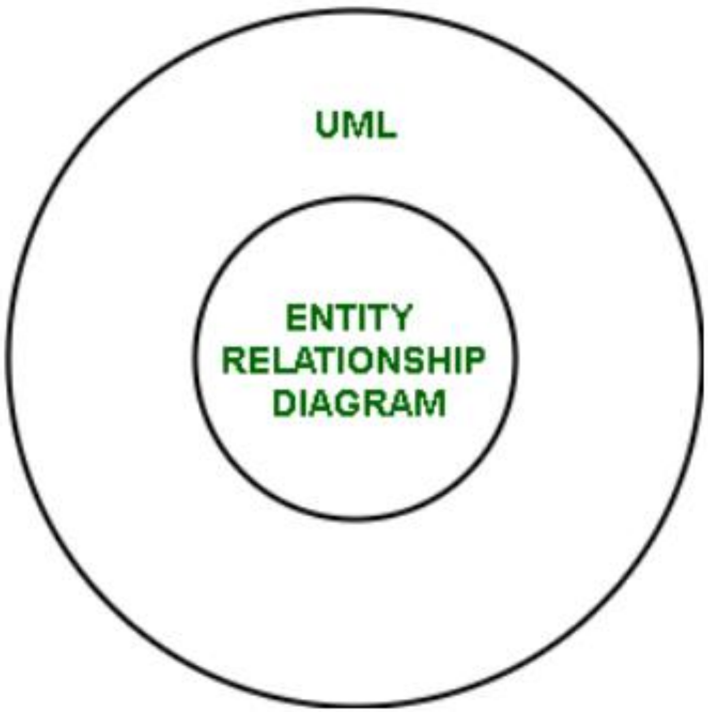
- UML is a standardized modeling language, which makes it widely understood and accepted in both software engineering and database design.
- UML is comprised of various diagrams, including class diagrams, sequence diagrams, activity diagrams, and so on, which are designed to provide specific support to particular activities in software development processes

Unified Modeling Language (UML)

Disadvantage:

- **Complexity:** UML can become unnecessarily complex for simple database models due to its broad scope.
- **Steep Learning Curve:** Each diagram in UML has been developed for a particular purpose, and it is important to understand them all while using UML

Unified Modeling Language (UML) And ER Diagram



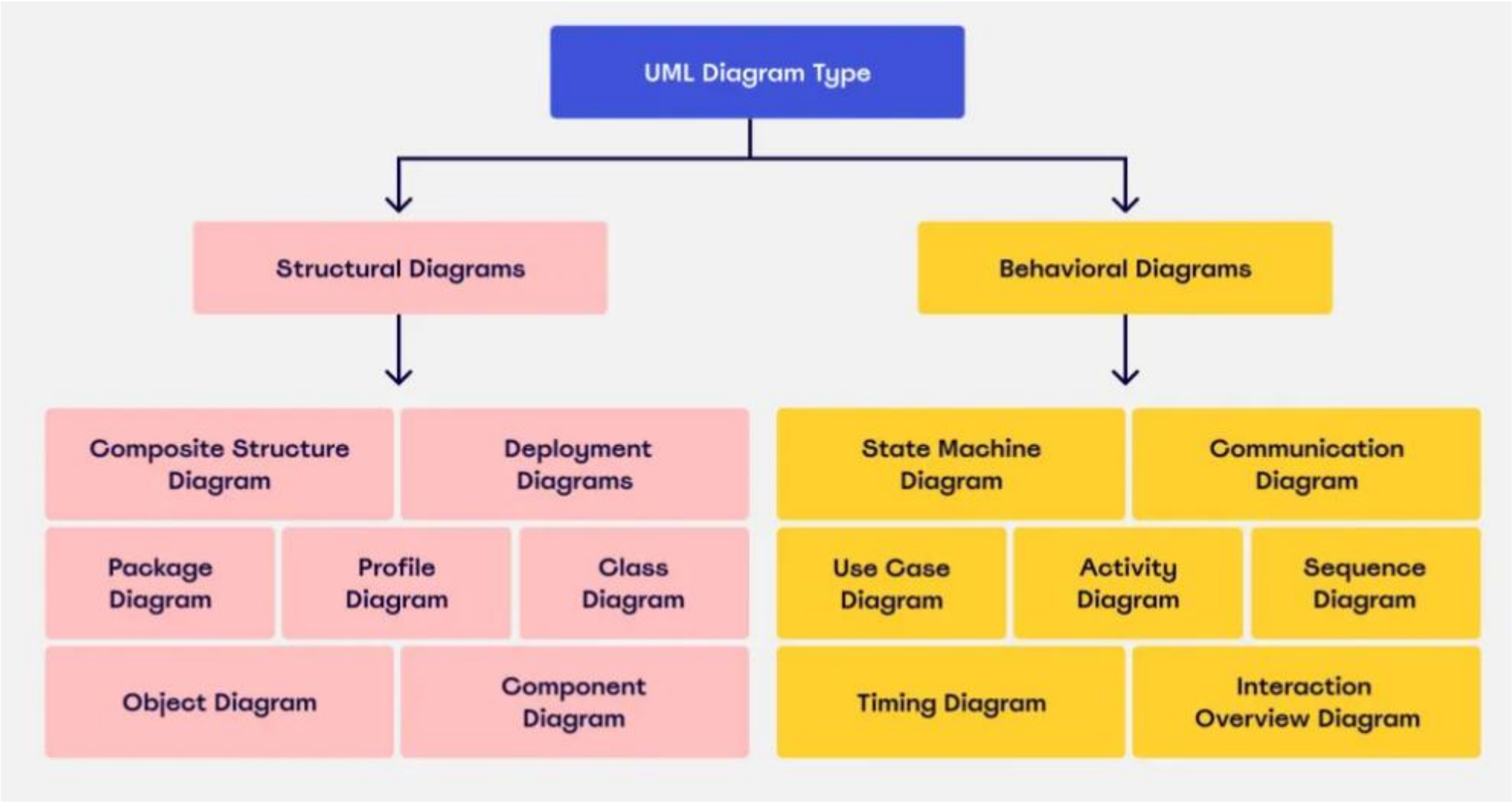
ER is the child of UML.

UML Class Diagrams are a **superset of ER diagrams**: they model more than just data (e.g., behavior/methods).

An **ER diagram can be mapped to a UML class diagram focusing only on the structure** (classes, attributes, associations).

Unified Modeling Language (UML)

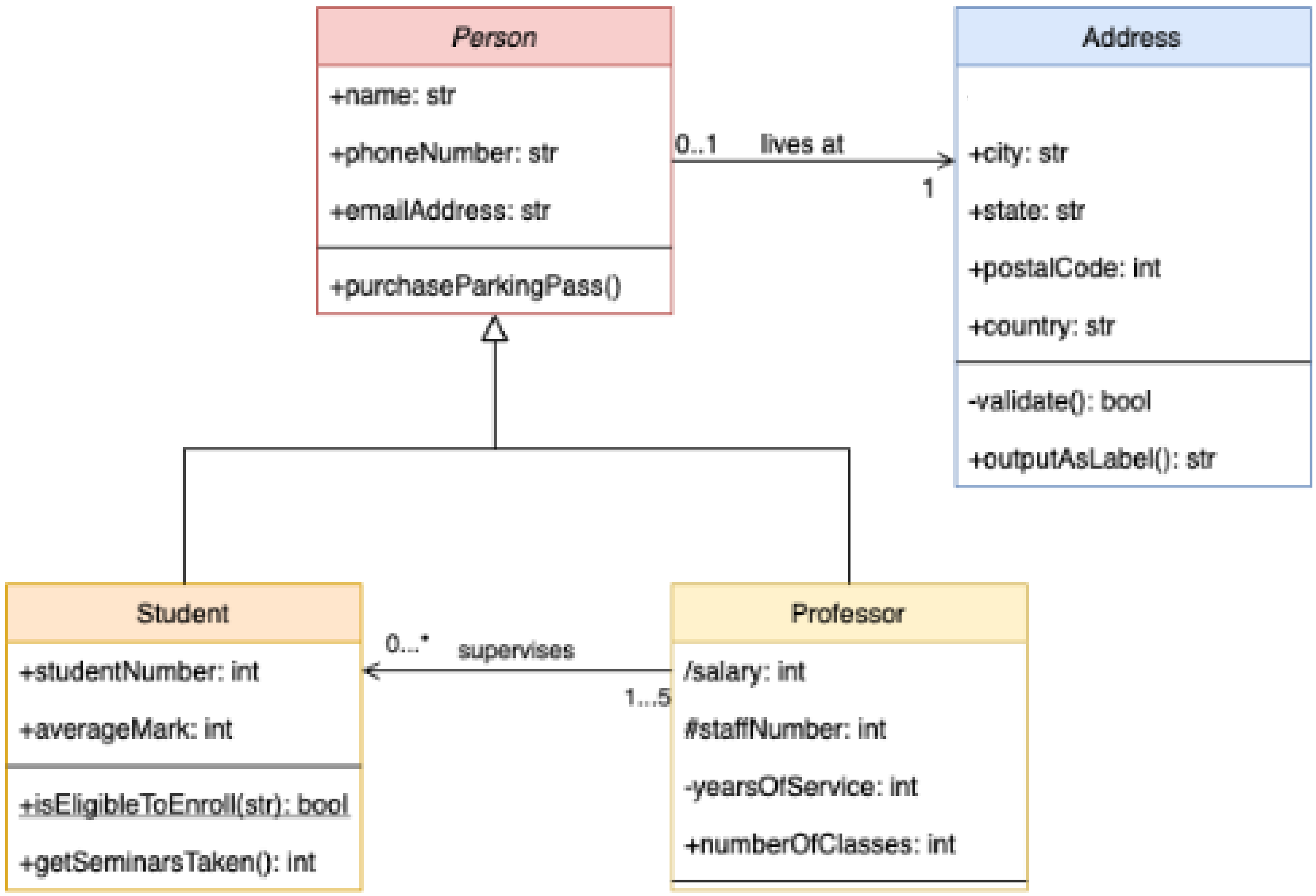
There are two main category



Unified Modeling Language (UML)

Structural diagrams represent the **static aspects of a system**, including its structure and the relationships between different elements.

Class Diagram



Unified Modeling Language (UML)

Class Diagram

Represented by rectangles divided into three sections:

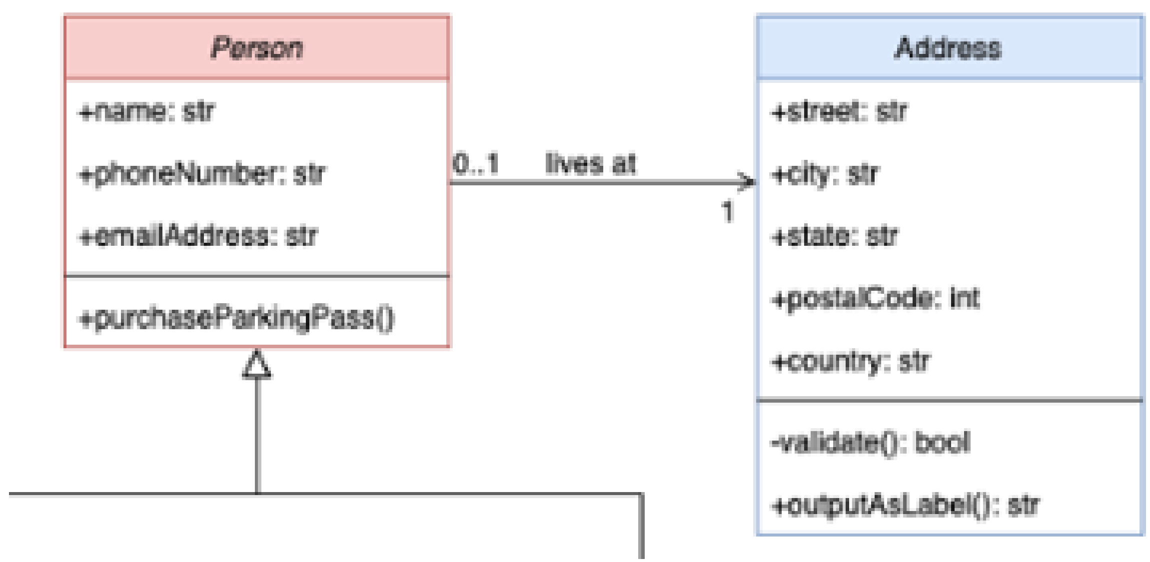
- **Top Section:** The name of the **class (entity)**.
- **Middle Section:** Attributes of the class.
- **Bottom Section:** Methods (not typically used in ER diagrams).

Unified Modeling Language (UML)

Class Diagram

Relationships (Associations):

- Represented by lines connecting classes (entities).
- Lines can be labeled to describe the nature of the relationship.
- Multiplicity (cardinality) is indicated at the ends of the lines.



Unified Modeling Language (UML)

ER Diagram → Class Diagram

ER Entity → UML Class

ER Relationship → UML Association

ER Attribute → UML Attribute

Extended E-R Diagram feature

Draw an E-R diagram for an XYZ cricket league. There are many teams. Each team has a set of players. Players within a team can be categorized as active players or non-active players. A game is played between two teams (referred to as host team and guest team) and has a date and score.

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